

Information Systems, Organizations, and Strategy



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Learning Objectives

After completing this chapter, you will be able to:

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| 3.1 Discuss technical and behavioral views of organizations. | 3.6 Discuss synergies, core competencies, and network economics. |
| 3.2 Discuss features of organizations that affect information systems. | 3.7 Understand how AI can be used in competitive strategy. |
| 3.3 Describe the impact of information systems on organizations. | 3.8 Describe the challenges posed by strategic information systems. |
| 3.4 Discuss Porter's competitive forces model. | 3.9 Understand how the information in this chapter can help your career. |
| 3.5 Discuss the value chain model. | |

Chapter Case Studies

- **Opening Case Study:** PepsiCo Tames the Doritos Crunch
- **Spotlight on Management Case Study:** Spotify's Podcasting Strategy Stumbles
- **Chapter Case Study:** AI Redesigns the Fashion Industry

Opening Case Study

PepsiCo Tames the Doritos Crunch

PepsiCo is a global leader in the food and beverage industry, noted for consumer favorites such as Pepsi Cola, Gatorade, Fritos, Ruffles potato chips, and Doritos. Although PepsiCo's business strategy emphasizes creating snacks or beverages attuned to changing consumer tastes and lifestyles, it also focuses on ways of making the entire experience of consuming its food products more enjoyable. How PepsiCo improved the customer experience for its highly popular Doritos-flavored tortilla chips is a case in point.

Doritos is one of PepsiCo's top-selling brands, and management believes it has huge upside growth potential worldwide. In 2023, Doritos was the leading tortilla chips brand in the United States, with more than 1.1 billion units sold. Doritos is also a top snacking choice for PC gamers. PepsiCo market researchers estimated that 85 percent of U.S. gamers consumed Doritos during the three-month period they studied, and it was the gamers' snack of choice. However, nearly a third of gamers reported that other people's crunching distracted them and impacted their performance. Gaming is a phenomenally popular pastime, with 3.25 million players worldwide, including roughly 65 percent of the U.S. population. This is a market that must be taken very seriously.

There was fierce consumer backlash when PepsiCo tried to modify the Doritos recipe to eliminate Doritos'

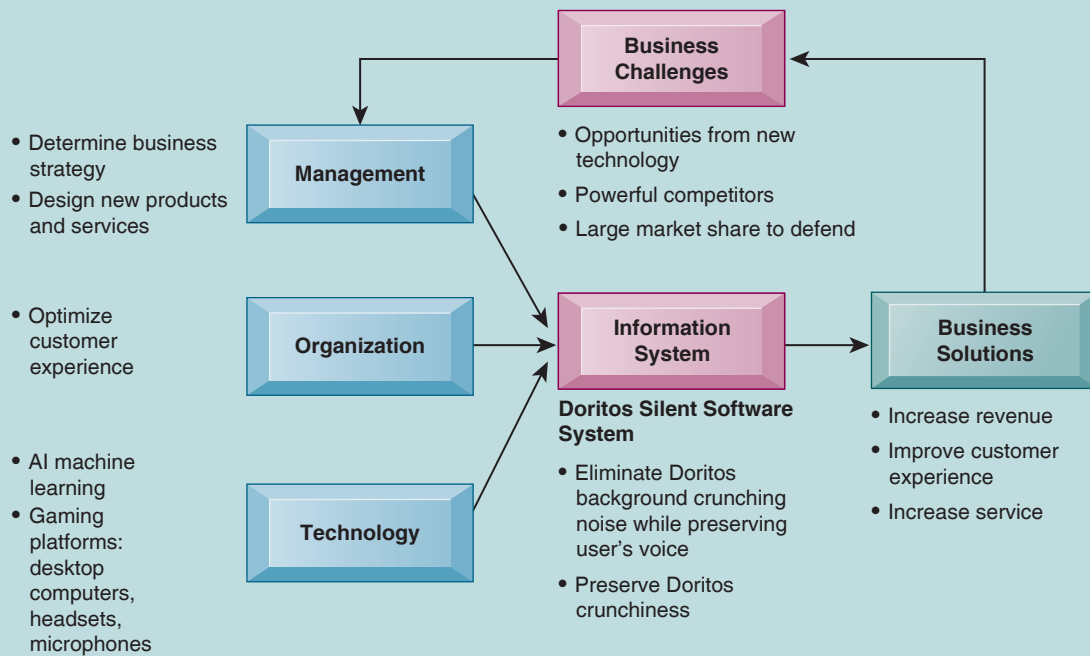
characteristic crunch sound. Doritos fans loved the crunch sound as much as Doritos' taste and did not want less crunchy chips. Could they have their crunch and eat it too without alienating those around them? Was this possible? Yes, thanks to artificial intelligence (AI) technology.

In November 2023, PepsiCo introduced Doritos Silent, a groundbreaking AI crunch-cancellation software tool that lets PC gamers enjoy their favorite chips without distracting other players. The crunch-cancellation software is able to retain the speaking voice of the person eating Doritos, while silencing crunching sounds so that they can't be heard by fellow game participants.

To create Doritos Silent, developers used AI machine learning (AI/ML) to analyze more than 5,000 different crunch sounds. Approximately 500 people were enlisted to crunch Doritos into a microphone to teach an AI model what Doritos crunching sounds like, so it could isolate it. Sounds of people talking were mixed with the crunch sounds so that the AI model could "learn" how to distinguish the sounds of crunching and voicing. Doritos Silent was trained exclusively on Doritos, but it works with other crunch sounds such as potato chips, crackers, raw vegetables, and other snacking foods, and is compatible with all types of desktop computers. PepsiCo has assured customers that no crunch will go undetected.



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SOURCES: PepsiCo.com, accessed August 17, 2024; Doritos.co.uk, accessed August 17, 2024; Sydney Page, "The Sound of Crunching Chips Is Annoying: Doritos Has Made a Silencer," *Washington Post*, November 23, 2023; Christopher Doering, "PepsiCo's Doritos Brand Creates Technology to 'Silence' Its Crunch," *Fooddive.com*, November 2, 2023; PepsiCo, "Doritos Silent Brings Crunch-canceling Tech to PC Gamers Around the World," *PepsiCo.com*, November 1, 2023.

PepsiCo's experience with improving its strategy with its Doritos brand illustrates some of the ways that information systems help businesses compete, as well as the challenges of finding the right business strategy and how to use technology in that strategy. The food and beverage industry is an extremely crowded and competitive playing field. Even though Doritos is the top tortilla chip snack in the United States, PepsiCo must continually be on the lookout for ways to improve Doritos and its customer experience to stay ahead of the pack. That includes meeting the needs of large market segments such as gamers, who love Doritos chips' flavor, but abhor the crunching sound that fellow gamers make when eating them.

The Opening Case Study diagram above calls attention to important points raised by this case and this

chapter. PepsiCo's business strategy focuses on creating snacks or beverages attuned to changing consumer tastes and lifestyles. It must not only focus on how its products taste, but also on the entire customer experience of using the products. PepsiCo used AI/ML to create Doritos Silent software that eliminates background crunching noises that gamers found annoying, while at the same time enabling PepsiCo to preserve Doritos' taste and texture. This helped PepsiCo maintain the popularity of Doritos among a very important market segment (gamers) and grow sales.

Case Study Questions

1. What competitive strategies is PepsiCo using for its Doritos brand?
2. Why is managing the customer experience important for those strategies?
3. What role did AI play in making Doritos more competitive?

CAREER BENEFIT

Developing Doritos Silent was a high-visibility project because there were such high expectations from and so much interest in AI. Developing a successful AI solution for a tricky and important problem would undoubtedly boost the careers of both IT and non-IT staff and give them a great deal of visibility throughout the organization. All of the people working on Doritos Silent would be more in demand internally and also at other organizations because of the AI expertise and experience they acquired.

3.1 Discuss Technical and Behavioral Views of Organizations

Information systems and organizations influence one another. Information systems are built by managers to serve the interests of the business firm. At the same time, the organization must be aware of and open to the influences of information systems to benefit from new technologies.

Examine Figure 3.1, which illustrates the complexity of the relationship between information technology (IT) and organizations. The relationship is influenced by many mediating factors, including the organization's structure, business processes, politics, culture, surrounding environment, and management decisions. You will need to understand how information systems can change social and work life in your firm. You will not be able to design new systems successfully or understand existing systems without understanding your own business organization.

As a manager, you will be the one to decide which systems will be built, what they will do, and how they will be implemented. You may not be able to anticipate all of the consequences of these decisions. Some of the changes that occur in business firms because of new IT investments cannot be foreseen and have results that may or may not meet your expectations. Who would have imagined 25 years ago, for instance, that email and instant messaging would become a dominant form of business communication and that some managers would be inundated with more than 200 email messages each day?

What Is an Organization?

Organization

A stable, formal social structure that takes resources from the environment and processes them to produce outputs.

An **organization** is a stable, formal social structure that takes resources from the environment and processes them to produce outputs. Examine Figure 3.2, which illustrates how an organization is defined from a technical perspective. This technical definition focuses on three elements of an organization. Capital and labor are primary production factors provided by the environment. The organization (the firm) transforms these inputs into products and services in a production function. The products and services are consumed by the environment in return for supplying inputs.

Figure 3.1 The Two-Way Relationship Between Organizations and Information Technology

This complex two-way relationship is mediated by many factors, not the least of which are the decisions made—or not made—by managers. Other factors mediating the relationship include the organizational culture, structure, politics, business processes, and environment.

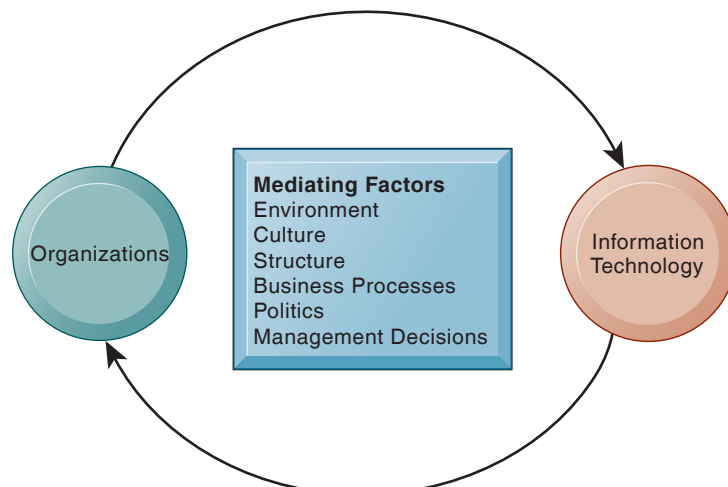
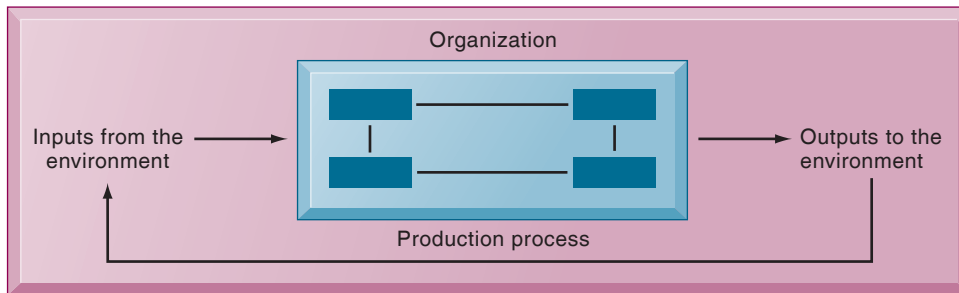


Figure 3.2 The Technical Microeconomic Definition of the Organization

In the microeconomic definition of organizations, capital and labor (the primary production factors provided by the environment) are transformed by the firm through the production process into products and services (outputs to the environment). The products and services are consumed by the environment, which supplies additional capital and labor as inputs in the feedback loop.



An organization is more stable than an informal group (such as a group of friends that meets every Friday for lunch) in terms of longevity and routineness. Organizations are formal legal entities with internal rules and procedures that must abide by laws. Organizations are also social structures because they are collections of social elements, much as a machine has a structure—a particular arrangement of valves, cams, shafts, and other parts.

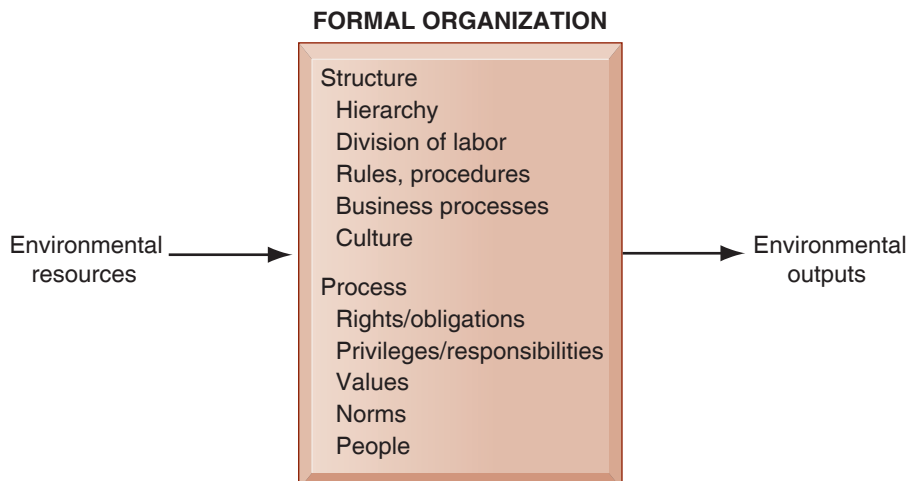
This definition of organizations is powerful and simple, but it is not very descriptive or even predictive of real-world organizations. A more realistic behavioral definition of an organization is a collection of rights, privileges, obligations, and responsibilities delicately balanced over a period of time through conflict and conflict resolution. Examine Figure 3.3, which illustrates the definition of an organization from a behavioral perspective.

In this behavioral view of the firm, people who work in organizations develop customary ways of working; they gain attachments to existing relationships; and they make arrangements with subordinates and superiors about how work will be done, the amount of work that will be done, and under what conditions work will be done. Most of these arrangements and feelings are not discussed in any formal rulebook.

How do these definitions of organizations relate to information systems technology? A technical view of organizations encourages us to focus on how inputs are combined to create outputs when technology changes are introduced into the company. The firm is seen as infi-

Figure 3.3 The Behavioral View of Organizations

The behavioral view of organizations emphasizes group relationships, values, and structures.



nately malleable, with capital and labor substituting for each other quite easily. But the more realistic behavioral definition of an organization suggests that building new information systems, or rebuilding old ones, involves much more than a technical rearrangement of machines or workers—that some information systems change the organizational balance of rights, privileges, obligations, responsibilities, and feelings that have been established over a long period of time.

Changing these elements can take a long time, be very disruptive, and require more resources to support training and learning. For instance, the length of time required to implement a new information system effectively is much longer than usually anticipated simply because there is a lag between implementing a technical system and teaching employees and managers how to use the system.

Technological change requires changes in who owns and controls information, who has the right to access and update that information, and who makes decisions about whom, when, and how. This more complex view forces us to look at the way work is designed and the procedures used to achieve outputs.

The technical and behavioral definitions of organizations are not contradictory. Indeed, they complement each other: The technical definition tells us how firms in competitive markets combine capital, labor, and information technology, whereas the behavioral model takes us inside the individual firm to see how that technology affects the organization's inner workings. Section 3.2 describes how each of these definitions of organizations can help explain the relationships between information systems and organizations.

3.2 Discuss Features of Organizations that Affect Information Systems

All modern organizations share certain characteristics. They are bureaucracies with clear-cut divisions of labor and specialization. Organizations arrange specialists in a hierarchy of authority in which everyone is accountable to someone, and authority is limited to specific actions governed by abstract rules or procedures. These rules create a system of impartial and universal decision making. Organizations try to hire and promote employees based on technical qualifications and professionalism (not personal connections). The organization is devoted to the principle of efficiency: maximizing output using limited inputs. Other features of organizations include their business processes, organizational culture, organizational politics, surrounding environments, structure, goals, constituencies, and leadership styles. All of these features affect the kinds of information systems used by organizations.

Routines and Business Processes

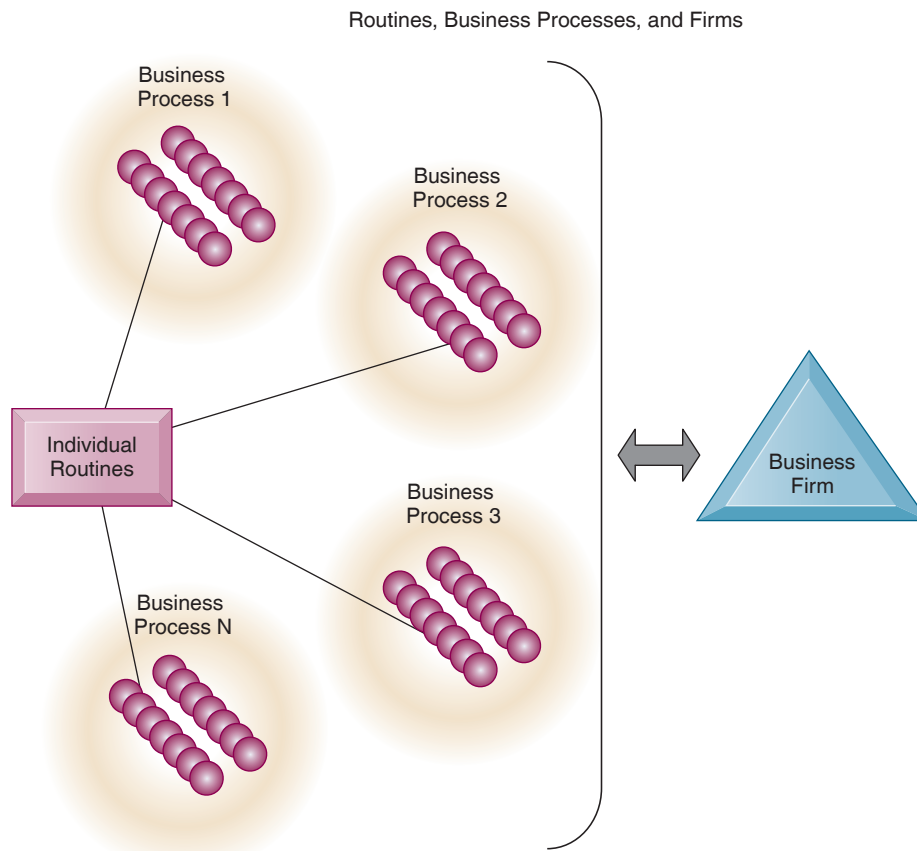
All organizations, including business firms, become very efficient over time because individuals in the firm develop routines for producing goods and services. **Routines**—sometimes called *standard operating procedures*—are precise rules, procedures, and practices that have been developed to cope with virtually all expected situations. As employees learn these routines, they become highly productive and efficient, and the firm is able to reduce its costs over time as efficiency increases. For instance, when you visit a doctor's office, receptionists have a well-developed set of routines for gathering basic information from you, nurses have a different set of routines for preparing you for an interview with a doctor, and the doctor has a well-developed set of routines for diagnosing you. *Business processes*, which we introduced in Chapters 1 and 2, are collections of such routines. Examine Figure 3.4, which shows that a firm is a collection of these business processes.

Routines

Precise rules, procedures, and practices that have been developed to cope with expected situations. Sometimes called standard operating procedures.

Figure 3.4 Routines, Business Processes, and Firms

All organizations are composed of individual routines and behaviors, a collection of which make up a business process. A collection of business processes makes up the business firm. New information system applications require that individual routines and business processes change to achieve high levels of organizational performance.



Organizational Politics

People in organizations occupy different positions with different specialties, concerns, and perspectives. As a result, they naturally have divergent viewpoints about how resources, rewards, and punishments should be distributed. These differences matter to both managers and employees, and they result in political struggles for resources, competition, and conflict within every organization. Political resistance is one of the great difficulties of bringing about organizational change—especially the development of new information systems. Virtually all large information systems investments by a firm that bring about significant changes in strategy, business objectives, business processes, and procedures become politically charged events. Managers who know how to work with the politics of an organization will be more successful than less-skilled managers in implementing new information systems.

Organizational Culture

All organizations have bedrock, unassailable, unquestioned (by the members) assumptions that define their goals and products. Organizational culture encompasses this set of assumptions about what products the organization should produce and how, where, and for whom it should produce them. Generally, these cultural assumptions are taken totally for granted and are rarely publicly announced or discussed. Business processes—the actual way business firms produce value—are usually ensconced in the organization's culture.

You can see organizational culture at work by looking around your university or college. Some bedrock assumptions of university life are that professors know more than students, the reason students attend college is to learn, and classes follow a regular schedule. Organizational culture is a powerful unifying force that restrains political conflict and promotes common understanding, agreement on procedures, and common practices. If we all share the same basic cultural assumptions, agreement on other matters is more likely.

At the same time, organizational culture is a powerful restraint on change, especially technological change. Most organizations will do almost anything to avoid making changes in basic assumptions. Any technological change that threatens commonly held cultural assumptions usually meets a great deal of resistance. However, there are times when the only sensible way for a firm to move forward is to employ a new technology that directly opposes an existing organizational culture. When this occurs, the technology is often stalled while the culture slowly adjusts.

Organizational Environments

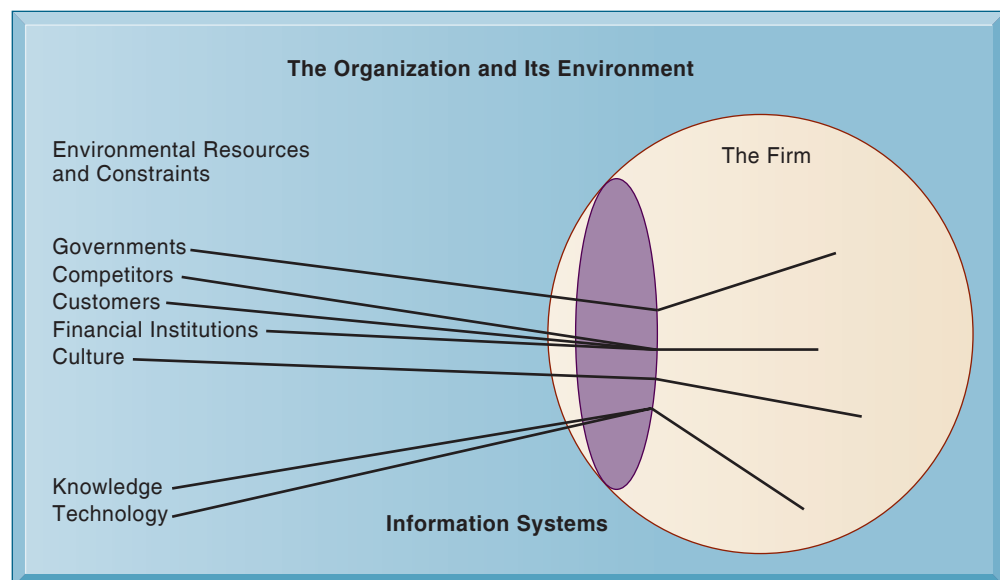
Organizations reside in environments from which they draw resources and to which they supply goods and services. Organizations and environments have a reciprocal relationship. On the one hand, organizations are open to and dependent on the social and physical environment that surrounds them. Without financial and human resources—people willing to work reliably and consistently for a set wage or revenue from customers—organizations could not exist. Organizations must respond to legislative and other requirements imposed by government as well as the actions of customers and competitors. On the other hand, organizations can influence their environments. For example, business firms sometimes form alliances with other businesses to influence the political process.

Examine Figure 3.5, which illustrates the role of information systems in helping organizations perceive changes in their environments and also in helping organizations act on their environments. Information systems are key instruments for *environmental scanning*, helping managers identify external changes that might require an organizational response.

Environments generally change much faster than organizations. New technologies, new products, and changing public tastes and values (many of which result in new government regulations) put strains on any organization's culture, politics, and people. Most organizations are unable to quickly adapt to a rapidly changing environment. Inertia built into an organization's

Figure 3.5 Environments and Organizations Have a Reciprocal Relationship

Environments shape what organizations can do, but organizations can influence their environments and decide to change environments altogether. Information technology plays a critical role in helping organizations perceive environmental change and in helping organizations act on their environment.



standard operating procedures, the political conflict raised by changes to the existing order, and the threat to closely held cultural values inhibit organizations from making significant changes. Young firms typically lack resources to sustain even short periods of troubled times. It is not surprising that less than 50 percent of the *Fortune* 500 companies in 2000 still exist today.

Disruptive Technologies: Riding the Wave

Sometimes, a technology and resulting business innovation come along to radically change the business landscape and environment. These innovations are loosely called “disruptive” (Christensen, 2003; Christensen, Raynor, and McDonald, 2015). What makes a technology disruptive? **Disruptive technologies** are those that substitute for and significantly reduce the market share of existing products, services, and business models. The car substituted for the horse-drawn carriage, the word processor for typewriters, streaming music services for portable CD players, and digital photography for process film photography. Table 3.1 describes just a few disruptive technologies from the past.

In some cases, entire industries are put out of business. In other cases, disruptive technologies expand the market, usually with less functionality and much less cost than existing products. Eventually, they turn into low-cost competitors for whatever was sold before. Disk drives are an example: Small hard disk drives used in PCs extended the market for disk drives by offering cheap digital storage for small files.

Some firms are able to create these technologies and ride the wave to profits; others learn quickly and adapt their business; still others are obliterated because their products, services, and business models become obsolete. They may be very efficient at doing what no longer needs to be done! There are also cases where no firms benefit, and all the gains go to consumers (firms fail to capture any profits). Moreover, not all change or technology is disruptive. Managers of existing businesses often do make the right decisions and find ways to continue competing. Disruptive technologies are tricky. Firms that invent disruptive technologies as “first movers” do not always benefit if they lack the resources to exploit the technology or fail to see the opportunity. The MITS Altair 8800 is widely regarded as the first PC, but its inventors did not take advantage of their first mover status. Second movers, so-called fast followers, such as IBM and Microsoft, reaped the rewards. Citibank’s ATMs revolutionized retail banking, but they were copied by other banks. Now all banks use ATMs, with the benefits going mostly to the consumers.

Disruptive technologies

Technologies that substitute for and significantly reduce the market share of existing products, services, and business models.

Table 3.1 Disruptive Technologies: Winners and Losers

Technology	Description	Winners and Losers
Microprocessor chips (1971)	Thousands and eventually millions of transistors on a silicon chip	Microprocessor firms win (Intel, Texas Instruments), while transistor firms (GE) decline.
Personal computers (1975)	Small, inexpensive, but fully functional desktop computers	PC manufacturers (HP, Apple, IBM) and chip manufacturers prosper (Intel), while mainframe (IBM) and minicomputer (DEC) firms lose.
World Wide Web (1989)	A global database of digital files and “pages” instantly available	Owners of online content and news benefit, while traditional publishers (newspapers, magazines, and broadcast television) lose.
Internet music, video, TV services (1998)	Repositories of downloadable music, video, and TV broadcasts on the web	Owners of Internet platforms, telecommunications providers owning Internet backbone (ATT, Verizon), and local Internet service providers win, while content owners and physical retailers (Tower Records, Blockbuster) lose.
Software as web service	Using the Internet to provide remote access to online software	Online software services companies (Salesforce.com) win, while traditional “boxed” software companies (Microsoft, SAP, Oracle) lose or adapt.
Artificial intelligence	Using machine learning and generative AI to make decision making more rapid and accurate and to create new knowledge and content	Organizations providing AI technology and services such as NVIDIA and organizations with AI expertise win, while providers and users of less advanced technologies, along with book, film, music, and other content creators and providers, lose.

Table 3.2 Organizational Structures

Organizational Type	Description	Examples
Entrepreneurial structure	Young, small firm in a fast-changing environment. It has a simple structure and is managed by an entrepreneur serving as its single chief executive officer.	Small start-up business
Machine bureaucracy	Large bureaucracy existing in a slowly changing environment, producing standard products. It is dominated by a centralized management team and centralized decision making.	Midsize manufacturing firm
Divisionalized bureaucracy	Combination of multiple machine bureaucracies, each producing a different product or service, all topped by one central headquarters.	<i>Fortune</i> 500 firms, such as General Motors
Professional bureaucracy	Knowledge-based organization where goods and services depend on the expertise and knowledge of professionals. Dominated by department heads with weak centralized authority.	Law firms, school systems, hospitals
Adhocracy	Task force organization that must respond to rapidly changing environments. Consists of large groups of specialists organized into short-lived multidisciplinary teams and has weak central management.	Consulting firms, such as the RAND Corporation

Organizational Structure

All organizations have a structure or shape. Mintzberg's classification, described in Table 3.2, identifies five basic kinds of organizational structure (Mintzberg, 1971).

The kind of information systems you find in a business firm—and the nature of problems with these systems—often reflects the type of organizational structure. For instance, in a professional bureaucracy such as a hospital, it is not unusual to find parallel patient record systems operated by the administration, another by doctors, and another by other professional staff such as nurses and social workers. In small entrepreneurial firms, you will often find poorly designed systems developed in a rush that quickly outgrow their usefulness. In huge multidivisional firms operating in hundreds of locations, you will frequently find there is not a single integrating information system, but instead each locale or each division has its own set of information systems.

Other Organizational Features

Organizations have goals and use different means to achieve them. Some organizations have coercive goals (e.g., prisons); others have utilitarian goals (e.g., businesses). Still others have normative goals (universities, religious groups). Organizations also serve different groups or have different constituencies, some primarily benefiting their members, others benefiting clients, stockholders, or the public. The nature of leadership differs greatly from one organization to another—some organizations may be more democratic or authoritarian than others. Another way organizations differ is by the tasks they perform and the technology they use. Some organizations perform primarily routine tasks that can be reduced to formal rules that require little judgment (such as manufacturing auto parts), whereas others (such as consulting firms) work primarily with nonroutine tasks.

3.3 Describe the Impact of Information Systems on Organizations

Information systems have become integral, online, interactive tools deeply involved in the minute-to-minute operations and decision making of large organizations. Over the past decade, information systems have fundamentally altered the economics of organizations and greatly increased the possibilities for organizing work. Theories and concepts from economics and sociology help us understand the changes brought about by IT.

Economic Impacts

From the point of view of economics, IT changes both the relative costs of capital and the costs of information. Information systems technology can be viewed as a factor of production that can be substituted for traditional capital and labor. As the cost of information technology decreases, it is substituted for labor, which historically has been a rising cost. Hence, information technology should result in a decline in the number of middle managers and clerical workers as information technology substitutes for their labor.

As the cost of information technology decreases, it also substitutes for other forms of capital, such as buildings and machinery, which remain relatively expensive. Hence, over time, we should expect managers to increase their investments in IT because of its declining cost relative to other capital investments.

IT also affects the cost and quality of information and changes the economics of information. Information technology helps firms contract in size because it can reduce **transaction costs**—the costs incurred when a firm buys on the marketplace what it cannot make itself. According to **transaction cost theory**, firms and individuals seek to economize on transaction costs, much as they do on production costs. Using markets is expensive because of costs such as locating and communicating with distant suppliers, monitoring contract compliance, buying insurance, obtaining information on products, and so forth (Coase, 1937; Williamson, 1985). Traditionally, firms have tried to reduce transaction costs through vertical integration, by getting bigger, hiring more employees, and buying their own suppliers and distributors, as both General Motors and Ford used to do.

Information technology, especially the use of networks, can help firms lower the cost of market participation (transaction costs), making it worthwhile for firms to contract with external suppliers instead of using internal sources. As a result, firms can shrink in size (numbers of employees) because it is far less expensive to outsource work to a competitive marketplace rather than hire employees.

For instance, by using computer links to external suppliers, automakers such as Ford, Toyota, and Honda can achieve economies by obtaining more than 70 percent of their parts from those external suppliers. Information systems make it possible for companies such as Apple to outsource assembly of iPhones to contract manufacturers such as Foxconn instead of making their products themselves.

As transaction costs decrease, firm size (the number of employees) should shrink because it becomes easier and cheaper for the firm to contract for the purchase of goods and services in the marketplace rather than to make the product or offer the service itself. Firm size can stay constant or contract even as the company increases its revenues. For example, when Eastman Chemical Company split off from Kodak in 1994, it had \$3.3 billion in revenue and 24,000 full-time employees. In 2023, it generated \$9.2 billion in revenue with only 14,000 employees.

Information technology can also reduce internal management costs. According to **agency theory**, the firm is viewed as a “nexus of contracts” among self-interested individuals rather than as a unified, profit-maximizing entity (Jensen and Meckling, 1992). A principal (owner) employs “agents” (employees) to perform work on the principal’s behalf. However, agents need constant supervision and management; otherwise, they will tend to pursue their own interests rather than those of the owners. As firms grow in size and scope, agency costs or coordination costs rise because owners must expend more and more effort supervising and managing employees.

Information technology, by reducing the costs of acquiring and analyzing information, permits organizations to reduce agency costs because it becomes easier for managers to oversee a greater number of employees. By reducing overall management costs, information technology enables firms to increase revenues while shrinking the number of middle managers and clerical workers. We have seen examples in earlier chapters where information technology expanded the power and scope of small organizations by enabling them to perform coordinated activities such as processing orders or keeping track of inventory with very few clerks and managers.

Because IT reduces both agency and transaction costs for firms, we should expect firm size to shrink over time as more capital is invested in IT. Firms should have fewer managers, and we expect to see revenue per employee increase over time.

Transaction costs

Costs of participating in a market; for example when a firm buys in the marketplace what it cannot make itself, or a consumer expends time and resources.

Transaction cost theory

Economic theory that firms and individuals seek to economize on transaction costs. Traditionally this was done vertical integration, with firms becoming larger, but today information technology enables firms to do so while decreasing the number of employees.

Agency theory

Economic theory that views the firm as a nexus of contracts among self-interested individuals who must be supervised and managed.

Organizational and Behavioral Impacts

Theories based in the sociology of complex organizations also provide some understanding of how and why firms change with the implementation of new IT applications.

IT Flattens Organizations

Large bureaucratic organizations, which primarily developed before the computer age, are often inefficient, slow to change, and less competitive than newly created organizations. Some of these large organizations have downsized, reducing the number of employees and the number of levels in their organizational hierarchies.

Behavioral researchers have theorized that information technology facilitates flattening of hierarchies by broadening the distribution of information to empower lower-level employees and increase management efficiency. Examine Figure 3.6, which illustrates this organizational flattening. IT pushes decision-making rights lower in the organization because lower-level employees receive the information they need to make decisions without supervision. (This empowerment is also possible because of higher educational levels among the workforce, which give employees more capabilities to make intelligent decisions.) Because managers now receive so much more accurate information on time, they become much faster at making decisions, so fewer managers are required. Management costs decline as a percentage of revenues, and the hierarchy becomes much more efficient.

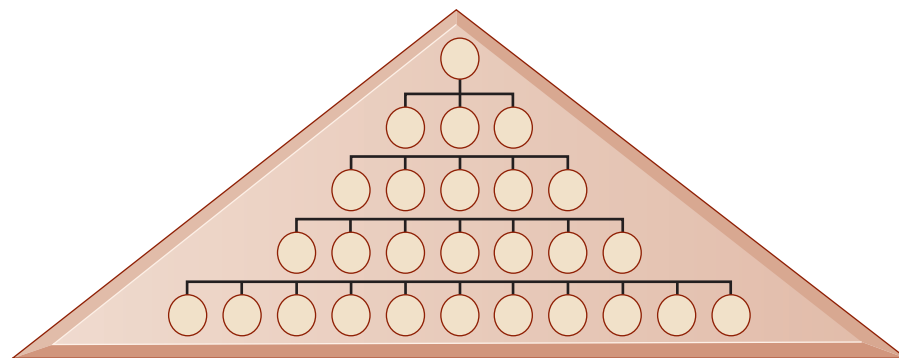
These changes mean that the management span of control has also been broadened, enabling high-level managers to manage and control more workers spread over greater distances. Many companies have eliminated thousands of middle managers as a result of these changes.

Understanding Organizational Resistance to Change

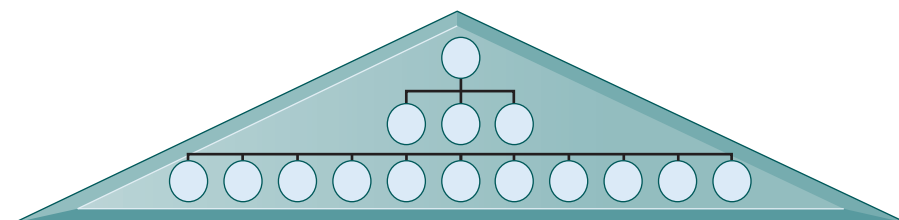
Information systems inevitably become bound up in organizational politics because they influence access to a key resource—namely, information. Information systems can affect who does what to whom, when, where, and how in an organization. Many new information systems

Figure 3.6 Flattening Organizations

Information systems can reduce the number of levels in an organization by providing managers with information to supervise larger numbers of workers and by giving lower-level employees more decision-making authority.



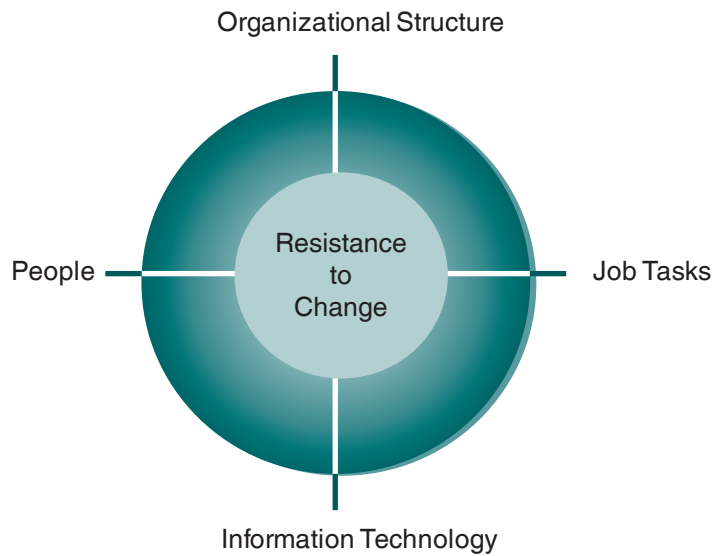
A traditional hierarchical organization with many levels of management



An organization that has been “flattened” by removing layers of management

Figure 3.7 Organizational Resistance to Information System Innovations

Implementing information systems has consequences for task arrangements, structures, and people. According to this model, to implement change, all four components must be changed simultaneously.



require changes in personal routines that can be painful for those involved and require retraining and additional effort that may or may not be compensated. Because information systems potentially change an organization's structure, culture, business processes, and strategy, there is often considerable resistance to them when they are introduced.

There are several ways to visualize organizational resistance. Research on organizational resistance to innovation suggests that four factors are paramount: the nature of the IT innovation, the organization's structure, the culture of people in the organization, and the tasks affected by the innovation. Examine Figure 3.7, which illustrates the interaction of these four factors. Here, changes in technology are absorbed, interpreted, deflected, and defeated by organizational task arrangements, structures, and people. In this model, the only way to bring about change is to change the technology, tasks, structure, and people simultaneously. Other authors have spoken about the need to "unfreeze" organizations before introducing an innovation, quickly implementing it, and "refreezing" or institutionalizing the change (Kolb and Frohman, 1970).

Because organizational resistance to change is so powerful, many information technology investments flounder and do not increase productivity. Indeed, research on project implementation failures demonstrates that the most common reason for the failure of large projects to reach their objectives is not the failure of the technology, but organizational and political resistance to change. Chapter 14 treats this issue in detail. Therefore, as a manager involved in future IT investments, your ability to work with people and organizations is just as important as your technical awareness and knowledge.

Implications for the Design and Understanding of Information Systems

To deliver genuine benefits, information systems must be built with a clear understanding of the organization in which they will be used. In our experience, the central organizational factors to consider when planning a new system are the following:

- The environment in which the organization must function
- The structure of the organization: hierarchy, specialization, routines, and business processes
- The organization's culture and politics
- The type of organization and its style of leadership

- The principal interest groups affected by the system and the attitudes of workers who will be using the system
- The kinds of tasks, decisions, and business processes that the information system is designed to assist

3.4 Discuss Porter's Competitive Forces Model

In almost every industry you examine, you will find that some firms do better than most others. There's almost always a standout firm. In the automotive industry, Toyota is considered a superior performer. In pure online retail, Amazon is the leader; in offline retail, Walmart, the largest retailer on earth, is the leader. In web search, Google is considered the leader.

Firms that "do better" than others are said to have a competitive advantage over others: They either have access to special resources that others do not, or they are able to use commonly available resources more efficiently—usually because of superior knowledge and information assets. In any event, they do better in terms of revenue growth, profitability, or productivity growth (efficiency), all of which ultimately in the long run translate into higher stock market valuations than their competitors.

But why do some firms do better than others, and how do they achieve competitive advantage? How can you analyze a business and identify its strategic advantages? How can you develop a strategic advantage for your own business? And how do information systems contribute to strategic advantages? One answer to that question is Michael Porter's competitive forces model.

Porter's Competitive Forces Model

Arguably, the most widely used model for understanding competitive advantage is Michael Porter's **competitive forces model**. Examine Figure 3.8, which illustrates how this model provides a general view of the firm, its competitors, and the firm's environment. Earlier in this chapter, we described the importance of a firm's environment and the dependence of firms on environments. Porter's model is all about the firm's general business environment. In this model, five competitive forces shape the fate of the firm.

Competitive forces model

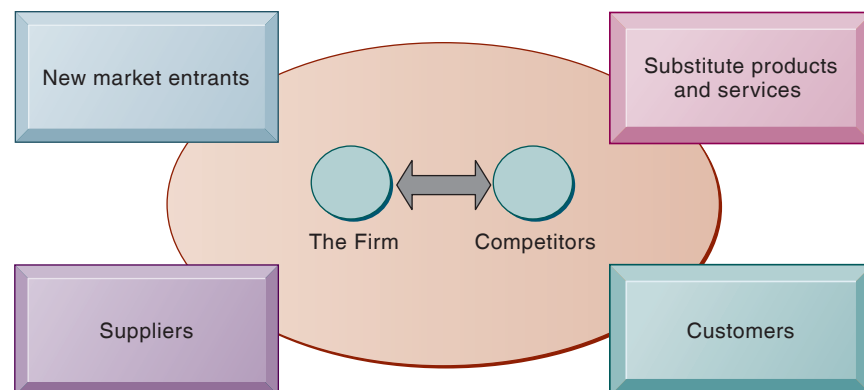
Most widely used model for understanding competitive advantage. Views the strategic position of the firm as being determined not only by competition with its traditional direct competitors, but also four other forces in the firm's environment: new market entrants, substitute products and services, customers, and suppliers.

Traditional Competitors

All firms share market space with other competitors who are continuously devising new, more-efficient ways to produce by introducing new products and services, and attempting to attract customers by developing their brands and imposing switching costs on their customers.

Figure 3.8 Porter's Competitive Forces Model

In Porter's competitive forces model, firm strategy and the strategic position of the firm are determined not only by competition with its traditional direct competitors, but also by four other forces in the firm's environment: new market entrants, substitute products and services, customers, and suppliers.



New Market Entrants

In a free economy with mobile labor and financial resources, new companies are always entering the marketplace. In some industries, there are very low barriers to entry, whereas in other industries, entry is very difficult. For instance, it is fairly easy to start a pizza business or just about any small retail business, but it is much more expensive and difficult to enter the computer chip business, which has very high capital costs and requires significant expertise and knowledge that are hard to obtain. New companies have several possible advantages: They are not locked into old plants and equipment, they often hire younger workers who are less expensive and perhaps more innovative, they are not encumbered by old worn-out brand names, and they are “more hungry” (more highly motivated) than traditional occupants of an industry. These advantages are also their weaknesses: They depend on outside financing for new plants and equipment, which can be expensive; they have a less-experienced workforce; and they have little brand recognition.

Substitute Products and Services

In just about every industry, there are substitutes that your customers might use if your prices become too high. New technologies create new substitutes all the time. Electric batteries can substitute for gasoline or diesel in cars and trucks, and wind, solar, and hydro power for industrial electricity generation. Likewise, Internet and wireless telephone service can substitute for traditional telephone service. And, of course, streaming television services that allow you to stream television and movies have become a substitute for traditional broadcast and pay cable TV services. The more substitute products and services in your industry, the less you can control pricing and the lower your profit margins.

Customers

A profitable company depends, in large measure, on its ability to attract and retain customers (while denying them to competitors) and charge high prices. The power of customers grows if they can easily switch to a competitor’s products and services, or if they can force a business and its competitors to compete on price alone in a transparent marketplace where there is little **product differentiation** (a competitive strategy for creating brand loyalty by developing new and unique products and services that are not easily duplicated by competitors), and all prices are known instantly (such as on the Internet). For instance, in the used college textbook market on the Internet, students (customers) can find multiple suppliers of just about any current college textbook. In this case, online customers have extraordinary power over used-book firms.

Product differentiation

Competitive strategy for creating brand loyalty by developing new and unique products and services that are not easily duplicated by competitors.

Suppliers

The market power of suppliers can have a significant impact on firm profits, especially when the firm cannot raise prices as fast as suppliers can. The more alternative suppliers a firm has, the greater control it can exercise over suppliers in terms of price, quality, and delivery schedules. For instance, manufacturers of laptop computers almost always have multiple competing suppliers of key components, such as keyboards, solid-state, and display screens.

Information Systems Strategies for Dealing with Competitive Forces

What is a firm to do when it is faced with all these competitive forces? And how can the firm use information systems to counteract some of these forces? How do you prevent substitutes and inhibit new market entrants? There are four generic strategies, each of which is often enabled by using information technology and systems: low-cost leadership, product differentiation, focus on market niche, and strengthening customer and supplier intimacy.



Supermarkets and large retail stores such as Walmart use sales data captured at the checkout counter to determine which items have sold and need to be reordered. Walmart's continuous replenishment system transmits orders to restock directly to its suppliers. The system enables Walmart to keep costs low while fine-tuning its merchandise to meet customer demands.

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Low-Cost Leadership

Use information systems to achieve the lowest operational costs and the lowest prices. The classic example is Walmart. By keeping prices low and shelves well stocked using a legendary inventory replenishment system, Walmart became the leading retail business in the United States. Walmart's continuous replenishment system sends orders for new merchandise directly to suppliers as soon as consumers pay for their purchases at the cash register. Point-of-sale terminals record the bar code of each item passing the checkout counter and send a purchase transaction directly to a central information system at Walmart headquarters. The information system collects the orders from all Walmart stores and transmits them to suppliers. Suppliers can also access Walmart's sales and inventory data using web technology.

Because the system replenishes inventory with lightning speed, Walmart does not need to spend much money on maintaining large inventories of goods in its own warehouses. The system also enables Walmart to adjust purchases of store items to meet customer demands. By using systems to keep operating costs low, Walmart pays a lower percentage of sales revenue for overhead than most firms in the retail industry.

Walmart's continuous replenishment system is also an example of an efficient customer response system. An **efficient customer response system** directly links consumer behavior to distribution and production and supply chains. Walmart's continuous replenishment system provides such an efficient customer response.

Efficient customer response system

Information system that directly links consumer behavior back to distribution and production, and supply chains.

Product Differentiation

Use information systems to enable new products and services or improve customer convenience in using your existing products and services. Big Tech firms like Google, Meta (Facebook), Amazon, Apple, and others are pouring billions of dollars into research and deployment of new services, and enhancements to their most valuable services and products in order to differentiate them from potential competitors. For instance, Apple has continued to differentiate its mobile devices with nearly annual introductions of new iPhones and iPad models. Crayola is creating new digital products and services (described in Section 3.9) that enable children to use digital tools for drawing, designing, and other creative tasks that used to require hand-held crayons. The continual stream of innovations flowing from these and other technology-driven companies ensures their products are unique and difficult to copy.

Manufacturers and retailers are using information systems to create products and services that are customized and personalized to fit the precise specifications of individual customers. For example, Baume, an offshoot of the luxury brand Baume et Mercier, uses recycled, upcycled, and sustainable materials to craft distinctive watches. With Baume's mobile-optimized website, which is built on Salesforce Commerce Cloud, customers can pick and mix different watch elements, including dial style, casing finish, strap material, and engraved messages. Customers can choose from 2,160 variations to create their own unique product. This ability to offer individually tailored products or services is called **mass customization**.

More and more companies are differentiating their products not just by the features of the products themselves, but also by the entire experience of buying and using the product. This is called the "customer experience," and **customer experience management**, which involves management of all the interactions between a customer and a company throughout their entire business relationship, has become incorporated into competitive strategy. The chapter-opening case on how PepsiCo used AI to improve the experience of eating Doritos is an example.

Table 3.3 lists a number of companies that developed IT-based products and services that other firms found difficult to copy—or at least took a long time to copy.

Focus on Market Niche

Use information systems to enable a specific market focus and serve this narrow target market better than competitors. Information systems support this strategy by producing and analyzing data for finely tuned sales and marketing techniques. Information systems enable companies to analyze customer buying patterns, tastes, and preferences closely so that they efficiently pitch advertising and marketing campaigns to smaller and smaller target markets.

The data come from a range of sources—credit card transactions, demographic data, purchase data from checkout counter scanners at supermarkets and retail stores, and data collected when people access and interact with websites. Sophisticated software tools find patterns in these large pools of data and infer rules from them to guide decision making. This has become one of the most popular applications today for AI/ML. Analysis of such data drives one-to-one marketing that creates personal messages based on individualized preferences. For example, the Charlotte Hornets of the NBA maintain detailed records on their millions of fans that include real-time data on buying tickets online and at games, purchasing team gear in-store and online, and eating and drinking at games. The team combines these fan records to create a single profile for each fan. The Hornets use these profiles to build a more personal relationship with fans, which can lead to a better fan experience, more targeted marketing, and more sales. Contemporary customer relationship management (CRM) systems augmented by AI tools feature analytical capabilities for this type of intensive data analysis.

Credit card companies are able to use this strategy to predict their most profitable cardholders. The companies gather vast quantities of data about consumer purchases and other behaviors and mine these data—often using AI/ML—to construct detailed profiles that identify cardholders who might be good or bad credit risks. We discuss the tools and technologies for data analysis in detail in Chapters 6, 11, and 12, and there are other real-world examples throughout the text.

Mass customization

The capacity to offer individually tailored products or services on a large scale.

Customer experience management

Management of all the interactions between a customer and a company throughout their entire business relationship.

Table 3.3 IT-Enabled New Products and Services Providing Competitive Advantage

Product or Service	Description
One-click shopping: Amazon	Amazon's patent on one-click shopping gave it a significant competitive advantage over other online retailers in the early years of e-commerce.
Online music: Apple iTunes Store	Apple was the first firm to develop a legal online marketplace for digital music from all five major record labels, and quickly became the largest music vendor in the United States.
Golf club customization: Ping	Customers can have golf clubs custom-fit and built to their individual specifications.
Online person-to-person payment: PayPal	PayPal enables the transfer of money between individual bank accounts and between bank accounts and credit card accounts.

Table 3.4
Four Basic Competitive Strategies

Strategy	Description	Example
Low-cost leadership	Use information systems to produce products and services at a lower price than competitors while enhancing quality and level of service	Walmart
Product differentiation	Use information systems to differentiate products, and enable new services and products	Uber, Nike, Apple
Focus on market niche	Use information systems to enable a focused strategy on a single market niche; specialize	Hilton Hotels, Harrah's
Customer and supplier intimacy	Use information systems to develop strong ties and loyalty with customers and suppliers	Toyota Corporation, Amazon

Strengthen Customer and Supplier Intimacy

Use information systems to tighten linkages with suppliers and develop intimacy with customers. Toyota, Ford, and other automobile manufacturers use information systems to facilitate direct access by suppliers to production schedules and even permit suppliers to decide how and when to ship supplies to their factories. This allows suppliers more lead time in producing goods. On the customer side, Amazon keeps track of user preferences and can recommend products purchased by others to its customers. Strong linkages to customers and suppliers increase **switching costs** (the expense incurred in lost time and expenditure of resources when changing from one supplier or system to a competing supplier or system) and loyalty to your firm.

Table 3.4 summarizes the competitive strategies we have just described. Some companies focus on one of these strategies, but you will often see companies pursuing several of them simultaneously. For example, Walmart has traditionally pursued a low-cost leadership strategy, but it also offers unique new services such as drone package delivery and more targeted marketing.

Switching costs

The expense incurred in lost time and expenditure of resources when changing from one supplier or system to a competing supplier or system.

The Internet’s Impact on Competitive Advantage

Because of the Internet, the traditional competitive forces are still at work, but competitive rivalry has become much more intense (Porter, 2001). Internet technology is based on universal standards that any company can use, making it easy for rivals to compete on price alone and for new competitors to enter the market. Because information is available to everyone, the Internet raises the bargaining power of customers, who can quickly find the lowest-cost provider on the web. Profits have been dampened. Table 3.5 summarizes some of the potentially negative impacts identified by Porter of the Internet on business firms.

The Internet has nearly destroyed some industries and has severely threatened more. For instance, the printed encyclopedia industry and the travel agency industry have been nearly decimated by the availability of substitutes over the Internet. Likewise, the Internet has had a significant impact on the retail, music, book, retail brokerage, software, telecommunications, and newspaper industries.

Table 3.5
Impact of the Internet on Competitive Forces and Industry Structure

Competitive Force	Impact of the Internet
Substitute products or services	Enables substitutes to emerge with new approaches to meeting needs and performing functions
Customers’ bargaining power	Availability of global price and product information shifts bargaining power to customers
Suppliers’ bargaining power	Procurement over the Internet tends to raise bargaining power over suppliers; suppliers can also benefit from reduced barriers to entry and from the elimination of distributors and other intermediaries standing between them and their users
Threat of new entrants	Reduces barriers to entry, such as the need for a sales force, access to channels, and physical assets; provides a technology for driving business processes that makes other things easier to do
Positioning and rivalry among existing competitors	Widens the geographic market, increasing the number of competitors and reducing differences among competitors; makes it more difficult to sustain operational advantages; puts pressure to compete on price

However, the Internet has also created entirely new markets; formed the basis for thousands of new products, services, and business models; and provided new opportunities for building brands with very large and loyal customer bases. Amazon, eBay, YouTube, Facebook, Expedia, and Google are examples. In this sense, the Internet has “transformed” entire industries, forcing firms to change how they do business.

Smart Products and the Internet of Things

The growing use of sensors in industrial and consumer products, often called the Internet of Things (IoT), is an excellent example of how the Internet is changing competition within industries and creating new products and services. Under Armour and other sports and fitness companies are pouring money into wearable health trackers and fitness equipment that use sensors to report users’ activities to remote corporate computing centers where the data can be analyzed. John Deere tractors are loaded with field radar, GPS transceivers, and hundreds of sensors that keep track of the equipment. The result is what are referred to as “smart products”—products that are a part of a larger set of information-intensive services sold by firms.

Smart, Internet-connected products offer new functionality, greater reliability, and more intense use of products while providing detailed information that can be used to improve both the products and the customer experience. They expand opportunities for product and service differentiation. When you buy a wearable digital health product, you not only get the product itself, you also get a host of services available from the manufacturer’s cloud servers. Smart products increase rivalry among firms that will either innovate or lose customers to competitors. Smart products generally raise switching costs and inhibit new entrants to a market because existing customers are trapped in the dominant firm’s software environment. Finally, smart products may decrease the power of suppliers of industrial components if, as many believe, the physical product becomes less important than the software and hardware that make it run.

3.5 Discuss the Value Chain Model

Although the Porter model is very helpful for identifying competitive forces and suggesting generic strategies, it is not very specific about what exactly to do, and it does not provide a methodology to follow for achieving competitive advantages. If your goal is to achieve operational excellence, where do you start? Here’s where the business value chain model is helpful.

The **value chain model** highlights specific activities in the business where competitive strategies can best be applied, and where information systems are most likely to have a strategic impact (Porter, 1985). This model identifies specific, critical leverage points where a firm can use information technology most effectively to enhance its competitive position. The value chain model views the firm as a series or chain of basic activities that add a margin of value to a firm’s products and services. Examine Figure 3.9, which illustrates how these activities can be categorized as either primary activities or support activities.

Primary activities are most directly related to the production and distribution of the firm’s products and services, which create value for the customer. Primary activities include inbound logistics, operations, outbound logistics, sales and marketing, and service. Inbound logistics includes receiving and storing materials for distribution to production. Operations transforms inputs into finished products. Outbound logistics entails storing and distributing finished products. Sales and marketing includes promoting and selling the firm’s products. The service activity includes maintenance and repair of the firm’s goods and services.

Support activities make the delivery of the primary activities possible and consist of activities related to the firm’s organizational infrastructure (administration and management), human resources (employee recruiting, hiring, and training), technology (improving products and the production process), and procurement (purchasing input).

Value chain model

Model that views the firm as a series or chain of basic activities that add a margin of value to a firm’s products or services and identifies where information systems can best be applied to achieve a competitive advantage.

Primary activities

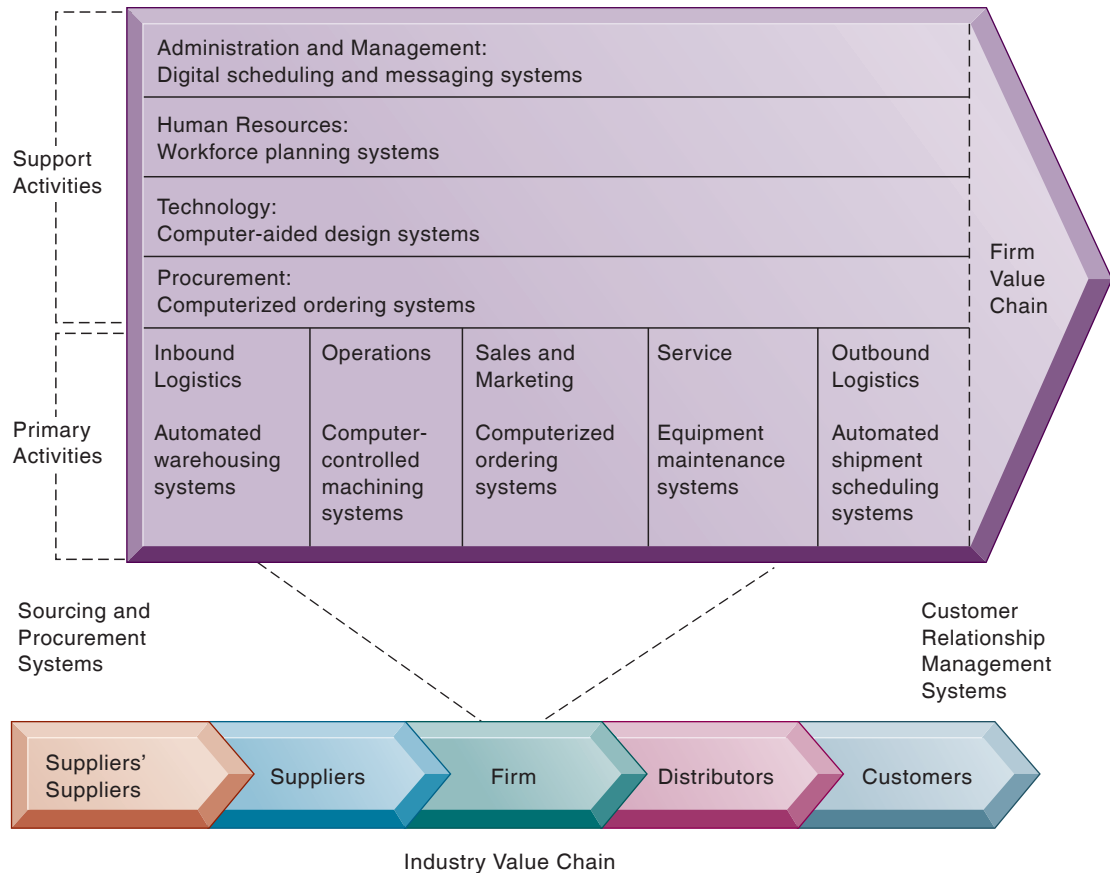
Activities most directly related to the production and distribution of a firm’s products or services.

Support activities

Activities that make the delivery of a firm’s primary activities possible. Consist of activities related to the firm’s organizational infrastructure, human resources, technology, and procurement.

Figure 3.9 The Value Chain Model

This figure provides examples of systems for both primary and support activities of a firm and of its value partners that can add a margin of value to a firm's products or services.



Now you can ask at each stage of the value chain, “How can we use information systems to improve operational efficiency and improve customer and supplier intimacy?” This will force you to critically examine how you perform value-adding activities at each stage and how the business processes might be improved. For example, value chain analysis indicated that PepsiCo, described in the Opening Case Study, could use technology to improve its processes for sales, marketing, customer service, and differentiating its products.

You can also begin to ask how information systems can be used to improve the relationship with customers and with suppliers who lie outside the firm's value chain but belong to the firm's extended value chain, where they are absolutely critical to your success. Here, supply chain management systems that coordinate the flow of resources into your firm and customer relationship management systems that coordinate your sales and support employees with customers are two of the most common system applications that result from a business value chain analysis. We discuss these enterprise applications in detail later in Chapter 9. We also discuss how AI technology can be used throughout the value chain in Section 3.7 of this chapter.

Using the business value chain model will also cause you to consider benchmarking your business processes against your competitors or others in related industries and identifying industry best practices. **Benchmarking** involves comparing the efficiency and effectiveness of your business processes to strict standards and then measuring performance against those standards. Industry **best practices** are usually identified by consulting companies, research organizations, government agencies, and industry associations as the most successful solutions or problem-solving methods for consistently and effectively achieving a business objective.

Once you have analyzed the various stages in the value chain at your business, you can come up with candidate applications of information systems. Then, once you have a list of candidate applications, you can decide which to develop first. By making improvements in

Benchmarking

Involves comparing efficiency and effectiveness of business processes to strict standards and then measuring organizational performance against those standards.

Best practices

The most successful solutions or problem-solving methods for consistently and effectively achieving a business objective that have been developed by a specific organization or industry.

your own business value chain that your competitors might miss, you can achieve competitive advantage by attaining operational excellence, lowering costs, improving profit margins, and forging a closer relationship with customers and suppliers. If your competitors are making similar improvements, then at least you will not be at a competitive disadvantage—the worst of all cases!

Extending the Value Chain: The Value Web

Figure 3.9 also shows that a firm's value chain is linked to the value chains of its suppliers, distributors, and customers. After all, the performance of most firms depends not only on what goes on inside a firm but also on how well the firm coordinates with direct and indirect suppliers, delivery firms (logistics partners, such as FedEx or UPS), and, of course, customers.

How can information systems be used to achieve strategic advantage at the industry level? By working with other firms, industry participants can use information technology to develop industrywide standards for exchanging information or business transactions electronically, which force all market participants to subscribe to similar standards. Such efforts increase efficiency, making product substitution less likely and perhaps raising entry costs—thus discouraging new entrants. Also, industry members can build industrywide, IT-supported consortia, symposia, and communications networks to coordinate activities concerning government agencies, foreign competition, and competing industries.

Looking at the industry value chain encourages you to think about how to use information systems to link up more efficiently with your suppliers, strategic partners, and customers. Strategic advantage derives from your ability to relate your value chain to the value chains of other partners in the process. For instance, if you are Amazon.com, you want to build systems that:

- Make it easy for suppliers to display goods and open stores on the Amazon site.
- Make it easy for customers to pay for goods.
- Develop systems that coordinate the shipment of goods to customers.
- Develop shipment tracking systems for customers.

Internet technology has made it possible to create highly synchronized industry value chains called value webs. A **value web** is a collection of independent firms that use information technology to coordinate their value chains to produce a product or service for a market collectively. It is more customer-driven and operates in a less linear fashion than the traditional value chain.

Examine Figure 3.10, which illustrates the value web as a networked system that operates in a less linear fashion than the traditional value chain displayed in Figure 3.9. It also shows that this value web synchronizes the business processes of customers, suppliers, and trading partners among different companies in an industry or in related industries. These value webs are more customer-driven, flexible, and adaptive to changes in supply and demand. Relationships can be bundled or unbundled in response to changing market conditions. Firms will accelerate time to market and to customers by optimizing their value web relationships to make quick decisions on who can deliver the required products or services at the right price and location.

Value web

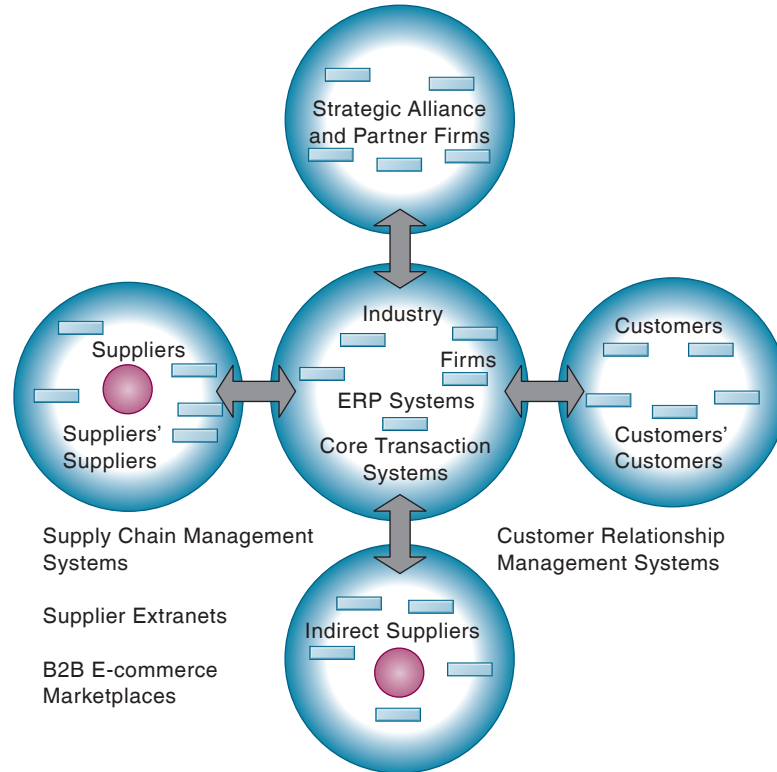
Collection of independent firms that use information technology to coordinate their value chains to collectively produce a product or service for a market.

3.6 Discuss Synergies, Core Competencies, and Network Economics

A large corporation is typically a collection of businesses. Often, the firm is organized financially as a collection of strategic business units, and the returns to the firm are directly tied to the performance of all the strategic business units. Information systems can improve the overall performance of these business units by promoting synergies and core competencies.

Figure 3.10 The Value Web

The value web is a networked system that can synchronize the value chains of business partners within an industry to respond rapidly to changes in supply and demand.



Synergies

Synergies develop when the output of some units can be used as inputs to other units, or when two organizations pool markets and expertise. These relationships lower costs and generate profits. Bank and financial firm mergers, such as the merger of JPMorgan Chase and Bank of New York, as well as Bank of America and Countrywide Financial Corporation, occurred precisely for this purpose.

One use of information technology in these synergy situations is to tie together the operations of disparate business units so that they can act as a whole. For example, acquiring MGM in 2022 enabled Amazon to bolster its position in the streaming entertainment industry. Information systems helped Amazon and MGM leverage their content, consolidate operations, lower costs, and increase cross-marketing of entertainment products.

Enhancing Core Competencies

Yet another way to use information systems for competitive advantage is to think about ways that systems can enhance core competencies. The argument is that the performance of all business units will increase insofar as these business units develop, or create, a central core of competencies. A **core competency** is an activity for which a firm is an industry leader. Core competencies may involve being the world's best miniature parts designer, the best package delivery service, or the best thin-film manufacturer. In general, a core competency relies on knowledge that is gained over many years of practical field experience with a technology. This practical knowledge is typically supplemented with a long-term research effort and committed employees.

Any information system that encourages the sharing of knowledge across business units enhances competency. Such systems might encourage or enhance existing competencies and help employees become aware of new external knowledge; such systems might also help a business leverage existing competencies to related markets. For example, Procter & Gamble (P&G), a world leader in brand management and consumer product innovation, uses a series of

Core competency

Activity at which a firm excels as an industry leader.

systems to enhance its core competencies by helping people working on similar problems share ideas and expertise. Employees working in research and development (R&D), engineering, purchasing, marketing, legal affairs, and business information systems around the world can share documents, reports, charts, videos, and other data from various sources online and locate employees with special expertise. P&G systems also can link to research scientists and entrepreneurs outside the company who are searching for new, innovative products worldwide.

Network-Based Strategies

Internet and networking technology have inspired strategies that take advantage of firms' abilities to create networks or network with each other. Network-based strategies include the use of network economics, a virtual company model, and business ecosystems.

Network Economics

Network economics refers to market situations where the economic value being produced depends, in large part, on the number of people using a product. For certain products and markets, the real economic value comes from the fact that other people use the product. In these situations, "network effects" are at work. For instance, what's the value of a telephone if it is not connected to millions of others? Email has value because it allows us to communicate with millions of others. Business models that are based on network effects have been highly successful on the Internet, including social networks, software, messaging apps, and on-demand companies like Uber and Airbnb.

In traditional economics—the economics of factories and agriculture—production experiences diminishing returns. The more any given resource is applied to production, the lower the marginal gain in output, until a point is reached where the additional inputs produce no additional outputs. This is the law of diminishing returns, and it is the foundation for most of modern economics.

In some situations, the law of diminishing returns does not work. For instance, in a network, the marginal costs of adding another participant are about zero, whereas the marginal gain is much larger. The larger the number of subscribers in a telephone system or the Internet, the greater the value to all participants because each user can interact with more people. It is not much more expensive to operate a television station with 1,000 subscribers than with 10 million subscribers. The value of a community of people grows with size, whereas the cost of adding new members is inconsequential. This is referred to as a "network effect." The value of Facebook to users increases greatly as more people use the social network.

From this network economics perspective, information technology can be strategically useful. Websites can be used by firms to build communities of users—like-minded customers who want to share their experiences. This builds customer loyalty and enjoyment and builds unique ties to customers. eBay, the giant online auction site, is an example. This business is based on a network of millions of users and has built an online community by using the Internet. The more people offering products on eBay, the more valuable eBay is to everyone because more products are listed, and more competition among suppliers lowers prices. Network economics also provides strategic benefits to commercial software vendors. The value of their software and complementary software products increases as more people use them, and there is a larger installed base to justify continued use of the product and vendor support.

Virtual Company Model

Another network-based strategy uses the model of a virtual company to create a competitive business. A **virtual company** uses networks to link people, assets, and ideas, enabling it to ally with other companies to create and distribute products and services without being limited by traditional organizational boundaries or physical locations. One company can use the capabilities of another company without being organizationally tied to that company. The virtual company model is useful when a company finds it cheaper to acquire products, services, or capabilities from an external vendor or when it needs to move quickly to exploit new market opportunities and lacks the time and resources to respond on its own.

Fashion companies such as GUESS, Ann Taylor, Levi Strauss, and Reebok enlist Hong Kong-based Li & Fung to manage production and shipment of their garments. Li & Fung handles product development, raw material sourcing, production planning, quality assurance,

Network economics

Refers to market situations where the economic value being produced depends in large part on the number of people using a product, and where the marginal costs of adding another participant are about zero but can create much larger marginal gains.

Virtual company

Organization using networks to link people, assets, and ideas, enabling it to ally with other companies to create and distribute products and services without being limited to traditional organizational boundaries or physical location.

and shipping. Li & Fung does not own any fabric, factories, or machines, outsourcing all of its work to a network of more than 15,000 suppliers in 40 countries all over the world. Customers place orders with Li & Fung over its private extranet. Li & Fung then sends instructions to appropriate raw material suppliers and factories where the clothing is produced. The Li & Fung extranet tracks the entire production process for each order. Working as a virtual company keeps Li & Fung flexible and adaptable so that it can design and produce the products ordered by its clients in short order to keep pace with rapidly changing fashion trends.

Business Ecosystems and Platforms

The Internet and the emergence of digital firms call for some modification of the industry competitive forces model. The traditional Porter model assumes a relatively static industry environment; relatively clear-cut industry boundaries; and a relatively stable set of suppliers, substitutes, and customers. Instead of participating in a single industry, some of today's firms participate in industry sets—collections of industries that provide related services and products that deliver value to the customer. Examine Figure 3.11, which illustrates these loosely coupled but interdependent networks of suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers. This is called a **business ecosystem**.

An example of a business ecosystem is the mobile Internet platform. In this ecosystem, there are four industries: device makers (Apple iPhone, Samsung, LG, and others), wireless telecommunication firms (AT&T, Verizon, T-Mobile, and others), independent software applications providers (generally small firms selling games, applications, and ring tones), and Internet service providers (who participate as providers of Internet service to the mobile platform). Each of these industries has its own history, interests, and driving forces. But these elements come together in a sometimes cooperative and sometimes competitive new mobile digital platform ecosystem, creating value for consumers that none of them could achieve acting alone.

Business ecosystems typically have one or only a few keystone firms that dominate the ecosystem and create a **platform** composed of information systems, technologies, and services that thousands of other firms in different industries use to enhance their own capabilities. For instance, Facebook is a platform used by billions of people and millions of businesses to interact and share information, as well as to buy, market, and sell numerous products and services. Microsoft is another example. More firms are trying to use information systems to develop into keystone firms by building IT-based platforms that other firms can use. Alternatively, firms should consider how their information systems will enable them to become profitable niche players in the larger ecosystems created by keystone firms (Zhou, 2024; Cutolo, 2021).

Business ecosystem

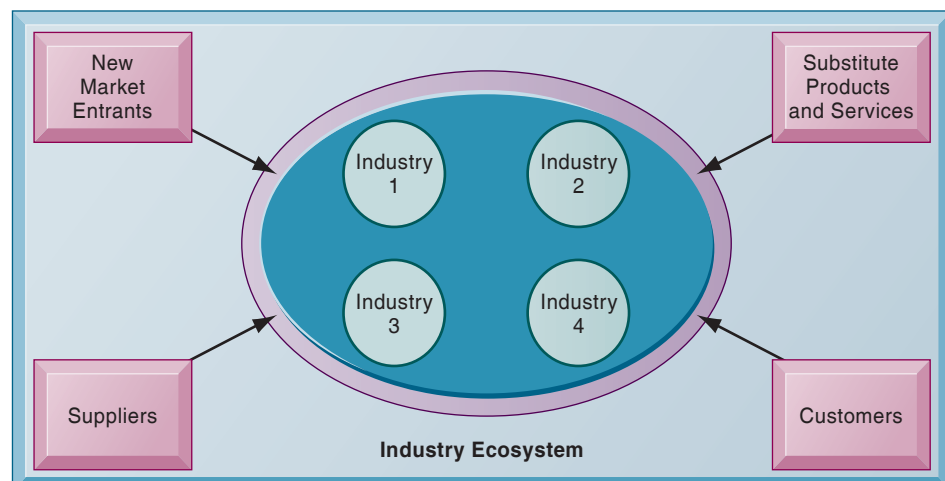
Loosely coupled but interdependent networks of suppliers, distributors, outsourcing firms, transportation service firms, and technology manufacturers.

Platform

Information systems, technologies, and services provided by a keystone firm in a business ecosystem that thousands of other firms in different industries use to enhance their own capabilities.

Figure 3.11 An Ecosystem Strategic Model

The digital firm era requires a more dynamic view of the boundaries among industries, firms, customers, and suppliers, with competition occurring among industry sets in a business ecosystem. In the ecosystem model, multiple industries work together to deliver value to the customer. IT plays an important role in enabling a dense network of interactions among the participating firms.



3.7 Understand How AI Can Be Used in Competitive Strategy

AI is transforming the way information systems operate and deliver value to organizations. Systems using AI technology can automate and enhance decision-making by analyzing patterns and trends in vast amounts of information, improve user experiences, open new revenue streams, optimize resource allocation and business processes, and create new products, services, and business models. You will find numerous examples of information systems that incorporate AI in every chapter of this text.

AI and the Value Chain

AI is making it easier for companies to improve performance based on their existing business models, but it is also spawning new business models and reshaping the competitive landscape. It has given birth to new products, services, and sources of value. Decision making has become much more data-driven, accurate, and precise for both operational and higher-level decisions. AI cannot make strategic decisions on its own, but it can be used to advise senior-level executives on how to pilot the firm more strategically. AI cannot fully substitute for human creativity and wisdom, but it can be a useful tool for ideation and problem-solving. Table 3.6 provides examples of how AI can add value at each step in the value chain, creating opportunities to boost competitive advantage.

AI and Porter's Competitive Forces Model

AI technology has impacted the various competitive forces identified in Porter's model, but not entirely in the same way as the Internet. There is intense pressure on firms to build products that will supply an unassailable competitive advantage. Some firms will succeed,

Table 3.6 Examples of AI Applications in the Value Chain

Value Chain Activity	AI Application
Inbound Logistics	AI in automated warehousing can maintain real-time, error-free inventory records, identify defects, and assign incoming shipments from suppliers to optimal warehouse locations.
Operations	AI identifies defects in large manufacturing runs, increasing quality while lowering costs.
Sales and Marketing	AI analyzes customer behavior to personalize marketing campaigns.
Service	AI chatbots provide quick answers to common queries from customers, sales reps, and service technicians.
Outbound Logistics	AI optimizes outbound shipment schedules, delivery routes, and route planning, reducing transportation costs.
Administration and Management	Predictive analytics can help managers anticipate demand fluctuations, optimize inventory levels, and allocate resources effectively.
Human Resources	AI algorithms predict which sourcing channels would most likely attract the right candidates for a specific job, reducing firm spending on identifying and recruiting new employees.
Technology	Firms can use AI technology to develop new products, such as noise cancelling technology for consumer products, hearing aids, earphones.
Procurement	AI can improve supplier selection by analyzing historical data, supplier performance, and market trends.

Table 3.7
AI and Porter’s Competitive Forces Model

Competitive Force	Impact of AI
Substitute products or services	AI can help companies develop unique products or services that are difficult to substitute. For example, a company that uses AI to offer heavily personalized services such as fashion recommendations for individual customers may face less threat from substitutes than a company offering generic services.
Customers’ bargaining power	AI can enhance the buying experience, building customer loyalty and reducing the bargaining power of buyers. For example, AI-powered shipping and delivery services or customer service systems make the experiences of purchasing and using a product more efficient and seamless, making customers less likely to switch to competitors.
Suppliers’ bargaining power	AI can reduce the bargaining power of suppliers. For instance, AI-driven supply chain management systems can help companies find alternative suppliers, reducing the bargaining power of existing suppliers. However, AI can also increase supplier bargaining power by enabling suppliers to provide customers with more customized and affordable goods and services to meet their specific requirements.
Threat of new entrants	AI applications that are expensive to build and run or that require highly specialized knowledge can deter new market entrants, as can AI applications that create a loyal customer base or a unique user experience. However, the proliferation of commercially available AI platforms and open-source software has also made it easier to implement some form of AI. Some new market entrants will be able to compete with existing firms in their market space. Established players will need to devote more of their resources to developing their own AI technology and datasets.
Positioning and rivalry among existing competitors	AI can intensify competition among existing firms. Companies that successfully use AI may gain a competitive edge over other firms in their industry, forcing these firms to adopt AI or risk becoming uncompetitive.

but many won’t because they lack the financial resources and know-how to work with AI tools. Equally essential are the enormous datasets required to train and build meaningful AI applications. The right data must be available, accurate, complete, error-free, and well-organized, and it must be appropriate for the problem to be solved by AI. Building successful AI applications also requires expertise with technology implementations and skillful change management (see Chapter 14). Companies that have AI expertise and the resources to pay for the extensive computing power and energy consumption of AI applications will be at an advantage. However, this will not entirely deter new market entrants who are able to take advantage of open-source software and AI platforms. AI can either increase or decrease the bargaining power of customers and suppliers, and it can be the foundation for new products and services. Table 3.7 shows how AI is impacting all aspects of Porter’s competitive forces model, recasting industry structures and the rules of competition.

3.8 Describe the Challenges Posed by Strategic Information Systems

Strategic information systems often change the organization as well as its products, services, and operating procedures, driving the organization into new behavioral patterns. Successfully using information systems to achieve a competitive advantage is challenging

and requires precise coordination of technology, organizations, and management. However, having a good strategy doesn't mean it's always the right strategy or that it is being properly executed, as the Spotlight on Management Case Study, *Spotify's Podcasting Strategy Stumbles*, illustrates.

Sustaining Competitive Advantage

The competitive advantages that strategic systems confer do not necessarily last long enough to ensure long-term profitability. Because competitors can retaliate and copy strategic systems, competitive advantage is not always sustainable. Markets, customer expectations, and technology change; globalization has made these changes even more rapid and unpredictable. The Internet can make competitive advantages disappear very quickly because virtually

Spotlight on Management Case Study Spotify's Podcasting Strategy Stumbles

Spotify is the world's largest on-demand music service, with more than 574 million active users in 180 countries. The company is known for its skillful use of technology to create ultra-personalized recommendations for songs and other audio content for its users.

Spotify doesn't have a monopoly on music streaming. Users can also find their favorite music, audiobooks, and podcasts on many other platforms. What makes Spotify stand out is its superior way of precisely serving the right audio (or other content) at the right time to each individual using its service. This ability to hyper-personalize recommendations for each Spotify user would not be possible without AI and is a major source of its competitive advantage.

Spotify uses AI in myriad ways to drive this personalization strategy. For example, Spotify's most popular feature is AI DJ, an AI-powered disc jockey that chooses what to play for users based on their specific music tastes and listening behavior. The AI DJ curates tracks for the individual user based on the data it has acquired about them, using generative AI to narrate its selections in a hyper-realistic voice. AI DJ will switch songs based on each user's real-time feedback. It can discover music you haven't listened to recently or recommend new selections based on what it thinks you'd like.

Spotify's Discover Weekly uses Spotify AI algorithms to craft a series of custom-made playlists of 30 songs each Monday for every user. Playlist selection criteria include each user's streaming history, likes and dislikes, and songs they added to the playlist. The playlist includes tracks that the user might not have heard before, but these recommendations are based on each user's search history and preferences. This capability helps introduce listeners to new tunes.

Spotify uses AI audio models to analyze data from raw audio tracks and categorize songs. This helps Spotify evaluate all songs to create recommendations, whether or not they are covered online. For example, if a new

song is released by a new artist on the Spotify platform, AI natural language processing (NLP) models might not discover it if coverage online and in social media is low. However, by using song data from audio models, the platform will still be able to analyze the track and recommend it to similar users along with other more popular songs.

Spotify's business strategy is multi-faceted and includes optimizing revenue from its free/freemium business model, forming partnerships with artists, music labels, and content providers, and expanding its platform to include podcasts, audiobooks, and other audio content, in addition to personalization. This approach has diversified Spotify revenue streams, amplified user engagement, and solidified its market presence. Spotify's profitability has fluctuated, but experts believe the company is positioned to grow its earnings going forward.

This multipronged business strategy had served Spotify well, but it didn't prevent the company from stumbling when it decided to also become a content creator, with podcasts delivering that content. Spotify's podcast strategy focused on original content from its in-house studios Parcast and Gimlet. These studios had excellent staff and built-in audiences. However, as part of Spotify, the studios failed to attract large audiences or produce new hit shows. Spotify did not give their shows the right marketing or support. Keeping certain podcasts exclusive to Spotify caused their audience to drop.

Spotify's preoccupation with original content also included making expensive deals with celebrities and celebrity studios as a means of attracting new listeners. Some deals worked out, including the *Batman Unburied* podcast by David S. Goyer and the *Case 63* scripted drama with Julianne Moore and Oscar Isaac. But many have not. Spotify's troubled celebrity deal with Prince Harry and Meghan Markle's Archewell Audio production company fell apart when

Archewell failed to deliver all of the podcasts specified in its \$20 million Spotify contract. Similar deals with high-profile studios such as Barack and Michele Obama's Higher Ground also fizzled out.

All told, Spotify spent more than \$1 billion to build its podcasting empire, far more than any of its rivals. It paid \$286 million for the podcast studios and spent \$250,000 or more per episode on exclusive shows to lure new listeners. These investments have not paid off. No one in the podcasting business is making much money. The amount of Spotify's investment relative to the size of the podcasting market was way out of line.

Spotify's management did not understand that podcasts have a fundamentally different formula for success from film, video games, books, and music. Franchises, intellectual property (IP), and name recognition can be enough to deliver a hit across many different mediums. For podcasts, however, a series from an acclaimed filmmaker, best-selling author, celebrity, or former president may barely register on podcast channels.

Spotify has now modified its podcast strategy to capitalize on its ability to sell ads to deliver more value for content creators and users. It cut podcasting staff and programming and it is upgrading its advertising tools for podcasters. The company is using listener preferences, based on music choices and listening habits, to target listeners with ads. It introduced a marketplace called Spotify Audience Network that enables advertisers to reach particular types of listeners, rather than buying

spots on specific shows. To date, Spotify hasn't come up with anything unique in podcasting to make its platform stand out from competitors such as Apple, Stitcher, SiriusXM, and others. The most popular platform for podcasts is YouTube.

SOURCES: Ipshita Sen, "How AI Helps Spotify Win in the Music Streaming World," *Outsideinsight.com*, accessed August 17, 2024; Mike Kaput, "How Spotify Uses AI (And What You Can Learn from It)," *Marketingaiinstitute.com*, January 26, 2024; Anne Steele and Sarah Krouse, "Spotify's \$1 Billion Podcast Bet Turns into a Serial Drama," *Wall Street Journal*, September 5, 2023; Amrita Khalid, "How Spotify's Podcast Plan Is Going Off the Rails," *Theverge.com*, June 21, 2023.

Case Study Questions

1. Analyze Spotify's business strategy using the value chain and competitive forces models.
2. What is the role of information technology and AI in Spotify's business strategy?
3. Why did Spotify's focus on podcasting initially stumble? What management, organization, and technology factors were involved?
4. Spotify Chief Executive Daniel Ek has said he wants Spotify to evolve from a music streaming company to being the world's largest audio company, with content including audiobooks, education, sports and news. According to Ek, podcasting has helped make the company a major podcasting platform despite its rocky start. Do you think Spotify will be able to achieve this goal? Why or why not?

all companies can use this technology. Classic strategic systems, such as American Airlines' SABRE computerized reservation system, Citibank's ATM system, and FedEx's package tracking system, benefited by being the first in their industries. Then rival systems emerged. Amazon was an e-commerce leader but now faces serious competition from Walmart, eBay, and Google. Information systems alone cannot provide an enduring business advantage. Systems originally intended to be strategic frequently become tools for survival, required by every firm to stay in business, or they may inhibit organizations from making the strategic changes essential for future success.

Aligning IT with Business Objectives

Research on IT and business performance has found that firms that can successfully align information technology with their business objectives are more likely to be profitable. When alignment is missing, the technology might work but it might not deliver an outcome the business needs (Hind et. al., 2023; Luftman et. al., 2017).

Most businesses get it wrong: Information technology takes on a life of its own and does not serve management and shareholder interests very well. Instead of businesspeople taking an active role in shaping IT to the enterprise, they ignore it or come up with unrealistic expectations of what IT can and cannot do. This is especially true for firms pursuing benefits from AI. Such firms pay a hefty price in poor performance and business results. Successful firms and managers understand what IT can do and how it works, take an active role in shaping its use, and measure its impact on revenues and profits.

Management Checklist: Performing a Strategic Systems Analysis

To align IT with the business and use information systems effectively for competitive advantage, managers need to perform a strategic systems analysis. To identify the types of systems that provide a strategic advantage to their firms, managers should ask the following questions:

1. What is the structure of the industry in which the firm is located?
 - What are some of the competitive forces at work in the industry? Are there new entrants to the industry? What is the relative power of suppliers, customers, and substitute products and services over prices?
 - Is the basis of competition quality, price, or brand?
 - What is the direction and nature of change within the industry? From where is the momentum and change coming? How much of this change is being driven by AI technology?
 - How is the industry currently using information technology? Is the organization behind or ahead of the industry in its application of information systems?
2. What are the business and industry value chains for this particular firm?
 - How is the company creating value for the customer—through lower prices and transaction costs or higher quality? Are there any places in the value chain where the business could create more value for the customer and additional profit for the company?
 - Does the firm understand and manage its business processes using the best practices available? Is it taking maximum advantage of supply chain management, customer relationship management, and enterprise systems?
 - Does the firm leverage its core competencies?
 - Is the industry supply chain and customer base changing in ways that benefit or harm the firm?
 - Can the firm benefit from strategic partnerships, value webs, ecosystems, or platforms?
 - Where in the value chain will information systems provide the greatest value to the firm?
3. Have we aligned IT with our business strategy and goals? Have we correctly articulated our business strategy and goals?
 - Is IT improving the right business processes and activities to promote this strategy?
 - Are we using the right metrics to measure progress toward those goals?
 - Are we using AI in ways that genuinely make our firm more competitive?

3.9 Understand How the Information in This Chapter Can Help Your Career

The information you have learned in Chapter 3 can help you advance your career by helping you understand how information technology impacts organizations and how you and your company should take this into account. The chapter also provides many examples of how information technology can support the various competitive strategies used in your job. Finding ways of using IT to improve your company's competitive position will benefit you throughout your career.

Real-World Scenario

Crayola Develops Its Business with Digital Technology

Crayola, based in Easton, Pennsylvania, has been a cherished brand for generations, producing high-quality, non-toxic crayons, markers, pencils, modeling clay, creative toys, and innovative art tools. Crayola’s popularity dipped with the advent of computers and online learning, as digital tools started to supplant hand-held art supplies. Crayola was able to meet this challenge by transforming its business model from a manufacturer of crayons and art tools into a trusted source of tools and experiences for creative play. The company still produces crayons and

traditional art tools, but it also has created a digital product line, which includes tools such as iMarker, an all-in-one digital pen, crayon, and pencil, designed for use with the Color Studio HD iPad app. Crayola also invested more in digital marketing, including online advertising, promotions, social media, and the use of business development representatives, to connect with parents and educators. Crayola closely tracks activity on its online channels through Google Analytics to make sure it is getting the most out of its marketing and ad campaigns.

Job Description and Career Path: Business Development Representative

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Business Development Representative	Sales professional who specializes in seeking new business opportunities by contacting and developing relationships with potential customers Job responsibilities include: - Lead Generation: Creating quality sales leads by employing methods such as cold email, cold calling, social selling, and networking - Qualifying leads and engaging with prospective buyers, including working with existing business accounts to identify potential opportunities - Booking appointments with prospective customers that can turn into legitimate sales opportunities - Developing sales and marketing strategies to tap into untapped markets	- Knowledge of Microsoft 365 tools: Excel, Word, PowerPoint, Teams - Experience with Salesforce CRM software - Experience with data analytics and business intelligence platforms such as Tableau	- Business Development Representative: \$50,000–\$60,000 - Business Development Manager: \$75,000–\$100,000 - Director of Business Development \$125,000–\$160,000 - VP Business Development: \$175,000–\$250,000 Salaries can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

AI Redesigns the Fashion Industry

Artificial intelligence is finding many applications across industries. In the fashion industry, it is becoming a game-changer. The ways clothing and fashion accessories are being designed, produced, marketed, and sold are undergoing profound transformation. AI is also reshaping the customer experience.

For starters, AI is revolutionizing how brands anticipate and capitalize on fashion trends. Fashion is all about keeping up with trends—what’s hot, whether clothing designs capture the trends of the moment, and where the hottest fashion trends can be found. Fashion forecasters traditionally made trend predictions by observing street fashion, designer runway shows, and pop culture. These predictions were not always accurate and were highly subjective.

Today, buying and merchandising can use AI predictive analytics and trend forecasting to make smarter buying decisions that will help fashion retailers increase sales and reduce overstock. AI tools can sift through enormous amounts of data, glean insights from various sources such as social media posts, fashion blogs, online retail sites, retail sale transactions, and global fashion events. This data includes styles, color schemes, fabric choices, and regional fashion preferences. AI can spot patterns and predict trends with impressive accuracy.

For example, Heuritech, a Paris-based AI platform, uses machine learning to analyze more than 3 million images from social media and the web each day for unbiased trend analysis. Heuritech AI can detect 2,000 different fashion attributes, such as macro prints, colors and fabrics, and granular shapes to forecast market shifts. The system can forecast and compare the current visibility of a color in China versus what it is likely to be in the United States in ten months, identify the consumer segments that will adopt it, and compare the ideal month to release items with that color. The system allows fashion brands such as Louis Vuitton and Dior to align their collections based on actual customer preferences and needs.

AI is also being used to optimize fashion supply chains. The supply chain for the fashion industry is one of the most complex and challenging to manage. Each step, from sourcing raw materials, to managing inventory, to ensuring timely delivery, is riddled with inefficiency and waste. AI can help streamline and optimize these processes. AI algorithms can analyze historical data to predict demand more accurately, allowing better inventory management. For instance, fast fashion retail chain H&M utilizes AI to analyze store receipts and returns, helping H&M adjust the quantities of each garment for each store. This kind of data analysis reduces the chances of having too much unsold stock, which enables companies to cut their costs, while promoting sustainability. Additionally, AI can also help in managing the logistics of the supply chain.

For instance, AI systems can track and analyze shipping data, ensuring that products are delivered in the most efficient manner. This also saves time and reduces carbon emissions.

Zara, another fast fashion leader noted for skillful supply chain management, has been using AI to make its fast fashion business processes even faster and more “intelligent.” Zara has a vertical supply chain where it controls everything from design to display to shipping, allowing it to gather immense quantities of data at every step. The data are analyzed to identify inefficiencies, pinpoint areas of success, and create more accurate demand forecasts and predictions. Armed with all of this information, Zara can develop and deliver new items to stores in as little as one week, a significantly faster turnaround time compared to the industry average of three to six months.

Sustainability is another area in which AI is having an impact on the fashion industry. The fashion industry is believed to be the second largest polluter worldwide, generating 186 billion pounds of textile waste annually. Companies create far more products than they can sell, and the fashion industry emits about the same quantity of greenhouse gases per year as the economies of France, Germany, and the United Kingdom combined. Most clothing and other fashion apparel are generally non-biodegradable and rarely recycled, so 87 percent of all material used ends up in landfills or incinerators. Fashion companies are starting to respond to consumer demands for more sustainable fabrics, but disposing of unsold clothing remains a huge problem.

Information technology, including AI, has proved very effective for curbing waste. Companies such as H&M and Zara are integrating AI into RFID technology and clothing tags with microchips, providing full visibility of their inventory, facilitating more informed forecasting decisions, and minimizing waste from unsold clothing. Zara also has an initiative with Jetlore, a consumer behavior prediction platform that uses AI to map consumer behavior into structured predictive attributes, such as size, color, fit, or style preferences, which help the company produce and sell only goods that customers actually want.

Lenzing Group, a world-leading provider of specialty fibers for the textile and nonwoven industries, is using AI to help it work with its customers and business partners to reduce carbon emissions. It uses machine learning and predictive analytics, along with a real-time ocean shipment tracker, to provide customers with access to carbon emission data at the shipment and container level. Lenzing and its customers use this real-time data to collaboratively establish goals and adjust shipment plans to reduce carbon emissions.

Refiberd has developed AI-based sorting technology for textile-to-textile re-sorting. The system redirects up to 70 percent of textile waste toward high-value recyclers. AI-powered hyperspectral imaging identifies specific fiber composition and contaminants present in textile waste. (Hyperspectral imaging collects and processes information across the electromagnetic spectrum to obtain the spectrum for each pixel in an image. This allows for the identification of objects and materials by analyzing their unique spectral signatures.)

One of the most significant applications of AI in fashion is its ability to personalize the fashion customer experience. For example, Thread.com, a UK-based startup, uses AI in combination with human stylists to provide personalized style advice and outfit recommendations for its customers. After customers supply their sizes, style preferences, and price ranges, AI algorithms, in collaboration with human stylists, curate a personalized shopping experience for each individual customer. StitchFix provides a similar service in the United States.

Small and medium-sized fashion retailers are also using AI to personalize their offerings. Apps like EasySize help online fashion shops offer personalized size recommendations to their customers, reducing returns and increasing customer satisfaction. A machine learning algorithm analyzes shopper feedback and calibrates size and fit recommendations based on the expectations and experiences of shoppers all over the world rather than body measurements. The EasySize app is available for e-commerce stores of all sizes through integration with the Shopify e-commerce platform. EasySize often works with small and fast-growing, direct-to-consumer fashion brands who have small teams and need things to be simple, reliable, and to work as fast as possible.

It gets even better: 3DLOOK is revolutionizing the way brands understand and cater to their customers' unique body shapes and sizes. Using mobile body scanning and AI, the technology provides highly accurate body measurements. 3DLOOK collects brand-specific data on core body measurements, fitting forms, avatars, and live models, along with specific grading rules. This provides a comprehensive understanding of the brand's unique sizing and fit standards. Utilizing this data, size recommendation algorithms map the brand-specific garment data to the customer's unique body data and fit preferences. The results are a highly personalized size suggestion for a perfect fit from each garment. By using 3DLOOK's YourFit solution, brands can potentially achieve a 13 to 16 percent increase in conversion rates and a 20 to 25 percent increase in their average order value.

AI is also transforming the creative process in the fashion industry, providing new design ideas and also helping fashion designers become more creative themselves. Designers can use generative AI tools such as DALL-E for clothing and jewelry design ideas in response to text prompts, or more powerful specialized design tools to create designs based on a designer's specific style.

Hong Kong fashion technology innovator Calvin Wong introduced a designer-led AI system called AiDA that generates blueprints from initial sketches and color ideas, preserving the designer's creativity while offering suggestions for improvement. Using AI image recognition, detection, and generation, the system provides a digital mood board for storing the sketches and color ideas. From this, AiDA's algorithms generate blueprints, offering designers a canvas for adjustments and personal touches. AiDA is a supportive tool and not a replacement for designer creativity or roles. Nike is another firm using AI to design products. Nike used AI to design its ISPA Universal shoe by feeding data on three previous shoe models to an algorithm, producing an eco-friendly shoe made from materials like Bio-EVA foam derived from the renewable raw material sugarcane.

Although AI has the potential to revolutionize the fashion industry, it also poses new challenges that businesses will need to address. With AI algorithms capable of doing the jobs of designers, marketers, and other fashion professionals, some jobs will be lost. This could have a significant impact on the entire fashion industry. There is also a risk that fashion will become less individual and less creative if AI is allowed to drive design and marketing decisions. This could lead to a decline in the quality and popularity of fashion products.

SOURCES: 3dlook.com, accessed August 26, 2024; Refiberd.com, accessed August 26, 2024; Easysize.com, accessed August 26, 2024; Euritech.com, accessed August 26, 2024; Stephanie Nikolopoulos and Melissa Epifano "H&M, Zara, Fast Fashion Turn to Artificial Intelligence to Transform the Supply Chain," Thomasnet.com, October 25, 2023; "Calvin Wong Launches AI Guiding Designers from Sketch to Runway," Apparelresources.com, September 18, 2023; Lenzing AG, "Lenzing Takes Supply Chain Transparency to Next Level by Combining Real-Time Shipment Tracking and Carbon Visibility," Lenzing.com, September 18, 2023.

Case Study Questions

1. Identify the points in the fashion industry value chain that could benefit from AI.
2. How has AI changed operations and decision making at the organizations discussed in this case study?
3. What strategic business objectives is AI helping the companies in this case study pursue? Explain your answer.

Review Summary

3.1 Discuss technical and behavioral views of organizations.

Information systems and organizations influence one another. The technical view of organizations sees the organization as a stable formal social structure that takes resources from the environment and processes them to produce outputs (products and services). The behavioral view of organizations emphasizes group relationships, values, and structures. Both views are required in order to fully understand the organizational impact of information technology.

3.2 Discuss features of organizations that affect information systems.

Modern organizations are hierarchical, specialized, and impartial, using explicit routines to maximize efficiency. Organizations have their own cultures and politics arising from differences in interest groups, and they are affected by their surrounding environment. Organizations differ in goals, groups served, social roles, leadership styles, incentives, types of tasks performed, and type of structure. These features help explain differences in organizations' use of information systems. Information systems and the organizations in which they are used interact with and influence each other.

3.3 Describe the impact of information systems on organizations.

The introduction of a new information system will affect organizational structure, goals, work design, values, competition between interest groups, decision making, and day-to-day behavior. At the same time, information systems must be designed to serve the needs of important organizational groups and will be shaped by the organization's structure, business processes, goals, culture, politics, and management. Information technology can reduce transaction and agency costs, and such changes have been accentuated in organizations using the Internet. New systems disrupt established patterns of work and power relationships, so there is often considerable resistance to them when they are introduced.

3.4 Discuss Porter's competitive forces model.

In Porter's competitive forces model, the strategic position of the firm and its strategies are determined by competition with its traditional direct competitors, but they are also greatly affected by new market entrants, substitute products and services, suppliers, and customers. Information systems help companies compete by maintaining low costs, differentiating products or services, focusing on market niche, strengthening ties with customers and suppliers, and increasing barriers to market entry with high levels of operational excellence.

3.5 Discuss the value chain model.

The value chain model highlights specific activities in the business where competitive strategies and information

systems will have the greatest impact. The model views the firm as a series of primary and support activities that add value to a firm's products or services. Primary activities are directly related to production and distribution, whereas support activities make the delivery of primary activities possible. A firm's value chain can be linked to the value chains of its suppliers, distributors, and customers. A value web consists of information systems that enhance competitiveness at the industry level by promoting the use of standards and industrywide consortia and by enabling businesses to work more efficiently with their value partners.

3.6 Discuss synergies, core competencies, and network economics.

Information systems achieve additional efficiencies or enhance services by tying together the operations of disparate business units. Information systems help businesses leverage their core competencies by promoting the sharing of knowledge across business units. Information systems facilitate business models based on large networks of users or subscribers that take advantage of network economics. A virtual company strategy uses networks to link to other firms so that a company can use the capabilities of other companies to build, market, and distribute products and services. In business ecosystems, multiple industries work together to deliver value to the customer. Information systems support a dense network of interactions among the participating firms. Business ecosystems typically have one or more keystone firms that dominate the ecosystem and create a platform that can be used by other firms.

3.7 Understand how AI can be used in competitive strategy.

AI is making it easier for companies to improve performance based on their existing business models, but it is also spawning new business models and reshaping the competitive landscape. It has given birth to new products, services, and sources of value. Decision making has become much more data-driven, accurate, and precise for both operational and higher-level decisions. AI cannot make strategic decisions on its own, but it can be used to advise senior-level executives on how to pilot the firm more strategically. AI cannot fully substitute for human creativity and wisdom, but it can be a useful tool for ideation and problem-solving.

3.8 Describe the challenges posed by strategic information systems.

Implementing strategic systems often requires extensive organizational change and a transition from one sociotechnical level to another. Such changes are often difficult and painful to achieve. Moreover, not all strategic systems are profitable, and they can be expensive to build. Many strategic information systems are easily copied by other firms, so that strategic advantage is not always sustainable.

Key Terms

Agency theory, 81	Efficient customer response system, 86	Support activities, 89
Benchmarking, 90	Mass customization, 87	Switching costs, 88
Best practices, 90	Network economics, 93	Transaction costs, 81
Business ecosystem, 94	Organization, 74	Transaction cost theory, 81
Competitive forces model, 84	Platform, 94	Value chain model, 89
Core competency, 92	Primary activities, 89	Value web, 91
Customer experience management, 87	Product differentiation, 85	Virtual company, 93
Disruptive technologies, 79	Routines, 76	

Review Questions

3.1 Discuss technical and behavioral views of organizations.

- Define organization and compare the technical and behavioral views of organizations.

3.2 Discuss the features of organizations that affect information systems.

- Identify and describe the features of organizations that help explain differences in organizations' use of information systems.
- Define and describe disruptive technologies and how they affect organizations.

3.3 Describe the impact of information systems on organizations.

- Describe the major economic theories that help explain how information systems affect organizations.
- Describe the major behavioral theories that help explain how information systems affect organizations.
- Explain why there is considerable organizational resistance to the introduction of information systems.

3.4 Discuss Porter's competitive forces model.

- Define Porter's competitive forces model and explain how it works.
- Describe what the competitive forces model explains about competitive advantage.
- Describe four competitive strategies enabled by information systems that firms can pursue.
- Describe how information systems can support each of these competitive strategies and give examples.
- Describe how the Internet has changed competitive forces and competitive advantage.

3.5 Discuss the value chain model.

- Define and describe the value chain model.
- Explain how the value chain model can be used to identify opportunities for information systems.
- Define the value web and show how it is related to the value chain.
- Explain how the value web helps businesses identify opportunities for strategic information systems.

3.6 Discuss synergies, core competencies, and network economics.

- Describe how promoting synergies and core competencies enhance competitive advantage.
- Explain how businesses benefit from using network economics and business ecosystems.
- Define and describe a virtual company and the benefits of pursuing a virtual company strategy.

3.7 Understand how AI can be used in competitive strategy.

- Describe how AI has impacted organizational performance and decision making.
- Describe how AI can add value at various steps in the value chain.
- Describe how AI has impacted the competitive forces identified in Porter's model.

3.8 Describe the challenges posed by strategic information systems.

- List and describe the management challenges posed by strategic information systems.
- Explain how to perform a strategic systems analysis.

Discussion Questions

3.9 It has been said that there is no such thing as a sustainable strategic advantage. Do you agree? Why or why not?

3.10 It has been said that the advantage that leading-edge retailers such as Amazon and Walmart have over their

competition isn't technology; it's their management. Do you agree? Why or why not?

3.11 What are some of the issues to consider in determining whether the Internet would provide your business with a competitive advantage?

Using AI Project

3.12 AI Project: Assessing Impact of Geographic and Demographic Factors

You have been hired as a consultant by two friends who want to open a retail store selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to help them understand the most important geographic and demographic factors that might affect the success of their prospective business,

such as proximity to good roads and parking, or the average age and income of the population near the store location. Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt that you can use to help advise them as to the most important geographic and demographic factors for a small retail pet store business featuring organic products. Save a copy of your prompt and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing the impact of business processes on business performance; using a database to improve sales performance; using web tools to configure and price a product; and collaborate in a team to identify strategic information systems.

For student files please visit www.pearsonhighered.com/laudon or MyLab MIS.

3.13 Management Decision Problem

Entrust Corporation, formerly Datacard Group, produces hardware and software for establishing trusted identities and the issuance of personalized financial cards, passports, driver's licenses, identification cards, and mobile devices. Entrust is one of the world's leading suppliers of payment cards. As a part of doing business with Entrust, customers—which include financial institutions and credit card bureaus, government agencies, and major corporations—supply the Shakopee, Minnesota-based company with large volumes of business data. Entrust had a predominantly manual process for distribution pricing for its secure identification and card issuance products to about 250 partners quarterly by emailing a price book of multiple spreadsheets. Creating the book involved hours of manual input by people in marketing and sales. The process of changing a price was complex, time-consuming, and people-dependent. People who received the price books had to manually type them into their own systems and check the data—a process that could take up to two days. Assess the impact of this process on Entrust's business performance, and suggest some management, organization, and technology issues that would need to be addressed by a solution.

3.14 Using Data Analytics to Analyze Sales Performance

Software skills: Database querying and reporting; database design

Business skills: Sales trend analysis

In this exercise, you will start out with raw transactional sales data and then use Microsoft Access database software to develop queries that help managers make better decisions about product pricing, sales promotions, and inventory replenishment. The MIS18_Project3.14_Student

File contains a Store and Regional Sales Database developed in Microsoft Access. The database provides raw data on weekly store sales of computer equipment in various sales regions. The database includes fields for store identification number, sales region, item number, item description, unit price, units sold, and the weekly sales period when the sales were made. Use Access to develop some queries to make this information more useful for running the business. This information includes the following:

- All 24-inch monitor sales by Sales Region for Week Ending December 29, 2024, sorted within region from highest to lowest number of units sold.
- All 24-inch monitor sales by store for the East Sales Region, sorted by Store No. and Week Ending in ascending order.
- All sales of all products for the East Sales Region, sorted from highest to lowest total number of units sold by item number.

Then describe how the information in these queries could be used for management decision making.

3.15 Getting to Know Tableau

In this project, you will learn how you can use Tableau to develop a competitive strategy based on identifying market niches. Access the Superstore data set, which is a sample data set available on the Tableau Public website and also within Tableau Desktop. The data set includes data about sales, profits, and other metrics from an online retailer. Use Tableau to create a dashboard with at least four data visualizations, such as bar charts, line charts, scatter plots, and/or maps, to analyze sales and profit data across the following dimensions: Segment; Product category and subcategory; Region, State, and Zip code. Use the dashboard to help you identify one or more specific market niches that the retailer should focus on. Write a report or create a presentation summarizing your findings and explaining how the visualizations helped you to reach your conclusions. If you don't have access to Tableau Desktop, ask your instructor, or alternatively, you can create a free Tableau Public account.

3.16 Using Online Tools to Configure and Price a Product

Software skills: Online research tools

Business skills: Researching product information and pricing

In this exercise, you will use online research tools to find product information about a car of your choice and use that information to make a purchase decision. You will also evaluate two of these tools.

You are interested in purchasing a lightly used low mileage Honda CR-V (or some other car of your choice). Go to the website of CarsDirect.com and begin your investigation. Research the various used CR-V models, and choose one you prefer in terms of price, features, and safety ratings. Locate and read at least two reviews. Then visit the Carvana.com website or app. Compare the information on pricing and available inventory for the car at the Carvana website or app with that at CarsDirect. Try to locate the lowest price for the car you want in inventory along with financing and vehicle pickup and delivery.

Compare the Carsdirect.com customer experience to that provided by Carvana. Which do you prefer for this type of purchase? Explain your answer.

3.17 Collaboration and Teamwork Project: Identifying Opportunities for Strategic Information Systems

With a team of two or more other students, select a company described in *The Wall Street Journal*, *Fortune*, *Forbes*, or another business publication available on the web. Visit the company's website to find additional information about that company and to see how the firm is using the web. On the basis of this information, analyze the business. Include a description of the organization's features, such as important business processes, culture, structure, and environment as well as its business strategy. Suggest strategic information systems appropriate for that particular business, including those based on Internet technology, if appropriate. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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Ethical and Social Issues in Information Systems



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Learning Objectives

After completing this chapter, you will be able to:

- | | |
|--|---|
| 4.1 Describe IT/IS-related ethical, social, and political issues. | 4.5 Discuss system quality issues. |
| 4.2 Describe principles for ethical conduct. | 4.6 Discuss accountability and control issues. |
| 4.3 Discuss information rights and privacy issues. | 4.7 Discuss quality of life issues. |
| 4.4 Discuss intellectual property issues. | 4.8 Understand how the information in this chapter can help your career. |

Chapter Case Studies

- **Opening Case Study:** Apps That Track: A Double-Edged Sword
- **Spotlight on Organizations Case Study:** Section 230 Under Fire
- **Chapter Case Study:** Meta's Many Ethical Challenges

Opening Case Study

Apps That Track: A Double-Edged Sword

How often do you check your local weather forecast? If you are like many people, you might do so multiple times a day, probably by opening a weather app on a mobile device. Few realize that this simple act can open the door to the widespread sale of data about your exact physical location to third parties, who can then use it for purposes far removed from the function of the app that originally gathered it. And it's not just weather apps that collect and sell this kind of data. Location data forms the basis for a multibillion-dollar industry, populated by largely unregulated companies that operate behind the scenes.

To figure out where you are, mobile devices use data from a variety of technologies embedded within the device, such as GPS sensors, Bluetooth, and connections to local Wi-Fi and cellular networks. Mobile apps often make use of a third-party SDK (software development kit) such as Foursquare's Pilgrim SDK, a free software tool that enables an app to collect data, or may be paid by an SDK developer to include its code within an app.

Apple has taken some steps to make privacy issues related to apps more visible. If an app uses an SDK, the developer must describe what data the SDK collects and how the data may be used. Apps on the iPhone platform must ask for your permission before tracking your activity (including your physical location). Google instituted similar policies in 2022. However, many users are so conditioned to clicking "Yes" that they don't stop to consider the ramifications. In addition, some app developers employ "dark patterns," a design tactic that encourages you to make a choice detrimental to your own interests, for example, by prompting you to enable location tracking while simultaneously suggesting that the app might not work as intended if location tracking is not enabled. As a case in point, contrary to what many people assume, weather apps do not actually "need" to take constant readings of location data to provide a local weather forecast, but these apps do not necessarily make that fact apparent.



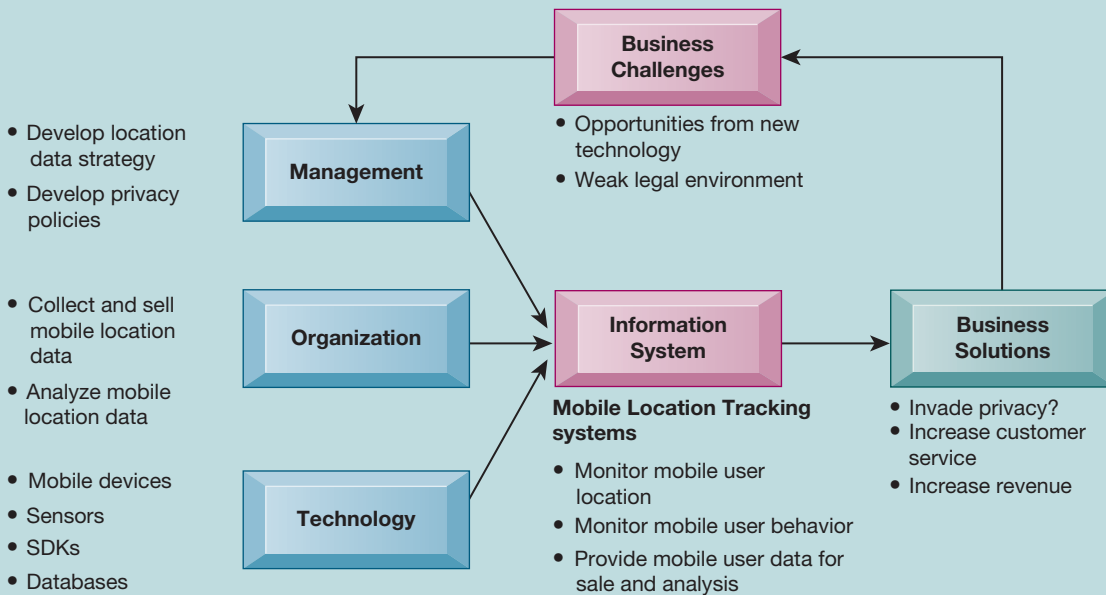
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Once you agree to tracking, the app will typically refer you to a privacy policy that details how that data can be used. Research has shown that few users ever actually read such policies or understand their ramifications. Companies that sell user data often justify that doing so provides them with a necessary revenue stream. For example, AccuWeather's privacy policy notes that its ability to provide its app for free is supported by its sale of user information to third parties. Companies that sell user data also note that the information collected is usually anonymized (although studies have shown that it can be relatively easy to tie such data to a particular person). Companies that buy data may also be doing so for a variety of legitimate reasons such as for analytics, fraud detection, and advertising and marketing purposes.

You might not object to your data being used to provide you with relevant advertising. However, it's what happens to that data once it is in the hands of a data broker that is of concern. Currently, there are few laws or regulations in the United States that restrict what a data broker can do with that data. Once it has been collected, it can be sold over and over again. For instance, although the U.S. Supreme Court has held that the government needs a warrant to compel companies such as cell phone carriers to produce location data, no such restriction applies to the government's purchase of such data from data brokers.

That might be starting to change, however. Federal government agencies have begun to take action against location data brokers and telecommunications providers. For instance, in November 2023, the Federal Trade Commission (FTC) filed an amended complaint in U.S. District Court against data broker Kochava, alleging that Kochava was collecting, using, and disclosing massive amounts of precise geolocation data without user notification or consent. In February 2024, the Court ruled that the suit could proceed, endorsing the FTC's position that the unfettered data collection of sensitive geolocation data could be considered a violation of Section 5 of the FTC Act, which prohibits unfair or deceptive acts or practices in or affecting commerce. In April 2024, the FTC finalized an order prohibiting data broker Outlogic (formerly known as X-Mode) from sharing or selling to third parties, including U.S. government agencies, raw location data that Outlogic gathered from hundreds of different mobile apps and which could be used to track a person's visits to sensitive locations such as medical clinics and places of worship. In May 2024, the FTC also finalized a similar settlement with digital marketing and data broker InMarket Media.

The Federal Communications Commission (FCC) is also taking action. In April 2024, it fined AT&T, Verizon, T-Mobile, and Sprint (now owned by T-Mobile) nearly \$200 million for selling real-time location information to data aggregators without their customers'



consent, albeit more than five years after those practices first came to light.

However, critics charge that these actions do not go nearly far enough and that relying on concepts like notification and user consent are a totally inadequate method of protecting privacy. There has been a new push for the enactment of a comprehensive federal data privacy law with the introduction of the proposed American Privacy Rights Act, but the prospects for its passage, although perhaps better than in the past, still remain uncertain.

SOURCES: "Collecting, Using, and Sharing Location Data," Reuters.com, June 2024; Forrester, "Proposed Federal Privacy Law Gains Momentum," Forbes.com, May 15, 2024; Joseph Jerome, "The FCC's Big Fines Over Location Data Aren't a Privacy Success Story," Techpolicy.com, May 3, 2024; Wendy Davis, "Kochava Must Face FTC Location Privacy Charges," Mediapost.com, February 5, 2024; Dell Cameron, "The Sad Truth of the FTC's 'Historic' Privacy Win," Wired.com, January 16, 2024; Alfred Ng and Jon Keegan, "Who Is Policing the Location Data Industry," Themarkup.org, February 24, 2022; Jon Keegan and Alfred Ng, "There's a Multibillion-Dollar Market for Your Phone's Location Data," Themarkup.com, September 30, 2021.

The challenges that location tracking poses to privacy, described in the Opening Case Study, show that technology can be a double-edged sword. It can be the source of benefits, including the ability to use mobile devices to access maps and driving directions or local weather and business news. At the same time, digital technology creates new opportunities for invading privacy and using information that could cause harm.

The Opening Case Study diagram above calls attention to important points that this case and chapter raise. Advances in mobile device technologies, mobile communications, and data management and analytics have created opportunities for organizations to track the

locations of individuals who are using mobile devices at any given time. The organizations described in the opening case are benefiting by collecting and using location-based data to monitor user behavior so that this information can be sold for marketing and other purposes. The use of location data is also taking benefits away from individuals because such use can invade their privacy. Individuals might be subject to job discrimination or special scrutiny based on behavior patterns revealed by location tracking. There are very few privacy protections that apply to location tracking.

This case also illustrates an ethical dilemma because it shows two sets of interests at work: the interests of organizations that want to use location data to enhance profit and the interests of those who fervently believe that businesses and public organizations should not use personal data because doing so invades privacy or may harm individuals. The case further illustrates the importance of being sensitive to both the positive and negative impacts of information systems and the importance of learning how to resolve ethical dilemmas involving information systems.

Case Study Questions

1. Does analyzing location data create an ethical dilemma? Why or why not?
2. Should there be new privacy laws to protect personal data collected from mobile device users? Why or why not?

CAREER BENEFIT

Awareness of the ethical, social, and legal issues surrounding the use of information technology and information systems is a critical skill for everyone who uses them in an organization. Recognizing ethical dilemmas and understanding how to best resolve them is also a critical skill.

4.1 Describe IT/IS-related Ethical, Social, and Political Issues

In this chapter, we examine the ethical, social, and political issues raised by the use of information technology and information systems. **Ethics** refers to the principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behavior. Information systems raise ethical questions for both individuals and societies because these systems create opportunities for intense social change and, thus, threaten existing distributions of power, money, rights, and obligations. Like other technologies, such as steam engines, electricity, and the telephone, information technology can be used to achieve social progress, but it can also be used to commit crimes and threaten cherished social values. The development of information technology will produce benefits for many and costs for others.

Ethical, social, and political issues in information systems have been given additional urgency by the rise of information technologies, the Internet, and most recently, the increased use of artificial intelligence, unleashing concerns about the appropriate use of customer information, the protection of personal privacy, and the protection of intellectual property.

Other pressing ethical issues that information systems raise include setting standards to safeguard system quality that protect the safety of the individual and society, establishing accountability for the consequences of information systems, and preserving values and institutions considered essential to the quality of life in today's information society. When using information systems, it is essential to ask, "What is the ethical and socially responsible course of action?"

Ethics

Principles of right and wrong that can be used by individuals acting as free moral agents to make choices to guide their behavior.

Ethical Challenges in Conducting Business

Before delving into the ethical, social, and political issues surrounding the use of information systems, it is important to consider the general topic of ethics in business. Organizations and their employees face numerous ethical challenges in conducting business. Table 4.1 provides a small sample of recent examples involving failures in ethical business judgment. These lapses in judgment occurred across a broad spectrum of industries.

In today's environment, employees who are convicted of violating the law face the risk of time in prison. The U.S. Department of Justice has indicated that prosecuting corporate executives for "white-collar crime" is one of the Department's highest priorities. U.S. federal sentencing guidelines mandate that federal judges impose stiff sentences based on the monetary value of the crime, the presence of a conspiracy to prevent discovery of the crime, the use of structured financial transactions to hide the crime, and failure to cooperate with prosecutors.

Although business firms in the past often paid for the legal defense of employees enmeshed in civil charges and criminal investigations, firms are now encouraged to cooperate with prosecutors to reduce charges against the firm itself. More than ever, as a manager or employee, you will have to decide for yourself what constitutes proper legal and ethical conduct.

Failures in ethical business judgment are not usually masterminded by employees of information systems departments, but information technology and information systems are often instrumental in many of these crimes. In many cases, the perpetrators of these crimes artfully used information systems to bury their actions from public scrutiny. We deal with the issue of information systems controls in Chapter 8.

Table 4.1 Recent Examples of Failures in Ethical Business Judgment

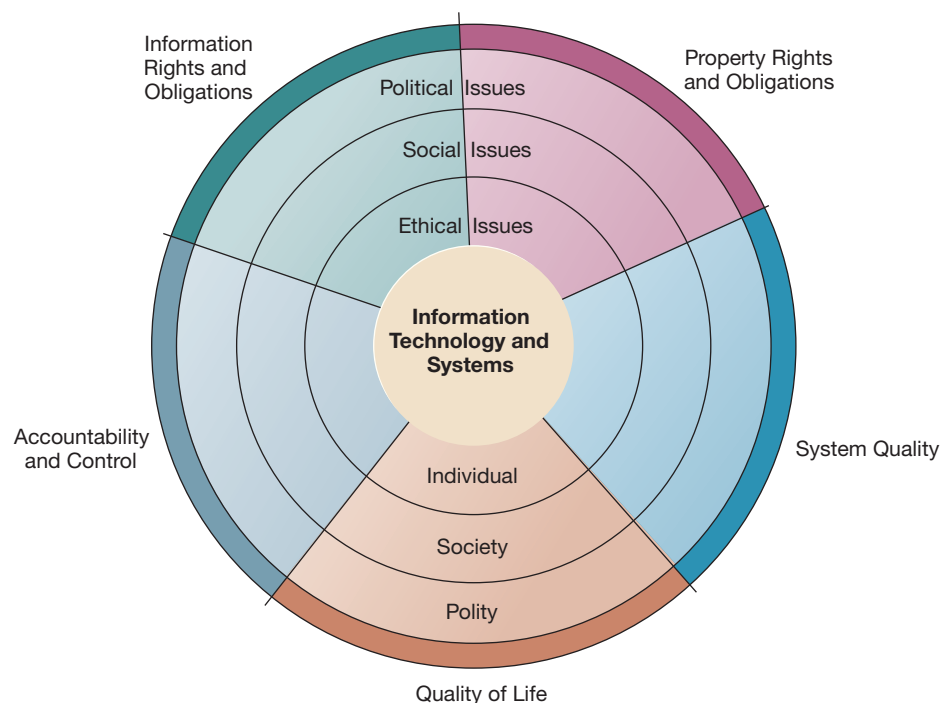
Company	Failure in Ethical Business Judgment
FX	Sam Bankman-Friedman, head of the cryptocurrency trading firm FX, was sentenced to 25 years in prison for his role in what has been described as one of the biggest financial frauds in U.S. history, leading to the collapse of FX. Bankman-Friedman's crimes were enabled by defects in internal controls, regulatory compliance, human resources, and system integrity.
Robinhood	Fintech brokerage firm Robinhood was fined a record \$70 million by the Financial Industry Regulatory Authority (FINRA) for systemic supervisory failures in several critical parts of its business, including distributing false and misleading information to customers of the firm, as well as systems outages that prevented customers from placing timely trades.
Volkswagen AG	Installed "defeat-device" emissions software on more than 500,000 diesel cars in the United States and on roughly 10.5 million more worldwide that allowed the cars to meet U.S. emissions standards during regulatory testing while actually spewing unlawful levels of pollutants into the air in real-world driving. Penalties paid internationally have totalled \$35 billion, with \$9.5 billion going to U.S. car owners. Volkswagen sued top company executives for being complicit in the fraud and obtained settlements totalling \$365 million. Criminal charges were also brought against a number of VW executives, including U.S.-based Oliver Schmidt, who was sentenced to seven years in prison and fined \$400,000.
Wells Fargo	Wells Fargo employees admitted to opening millions of false accounts, manipulating terms of mortgages, and forcing auto loan customers to purchase unneeded insurance. Wells Fargo was fined \$2.5 billion by the US government.

A Model for Thinking about Ethical, Social, and Political Issues

Ethical, social, and political issues are closely linked. The ethical dilemma you may face as someone who is involved in the use of information systems by your organization typically is reflected in social and political debate. Examine Figure 4.1, which illustrates one way to think about these relationships. Imagine society as a more or less calm pond on a summer day, a delicate ecosystem in partial equilibrium with individuals and with social and

Figure 4.1 The Relationship Among Ethical, Social, and Political Issues in an Information Society

The introduction of new information technology has a ripple effect, raising new ethical, social, and political issues that must be dealt with on the individual, social, and political levels. These issues have five moral dimensions: information rights and obligations, property rights and obligations, system quality, quality of life, and accountability and control.



political institutions. Individuals know how to act in this pond because social institutions (family, education, organizations) have developed well-honed rules of behavior, and these rules are supported by laws developed in the political sector that prescribe behavior and promise sanctions for violations. Now toss a rock into the center of the pond. What happens? Ripples, of course.

Imagine instead that the disturbing force is a powerful shock of new information technology and systems hitting a society more or less at rest. Suddenly, individual actors are confronted with new situations often not covered by the old rules. Social institutions cannot respond overnight to these ripples—it may take years to develop etiquette, expectations, social responsibility, or approved rules. Political institutions also require time to develop new laws and often require the demonstration of real harm before they act. In the meantime, you may have to act. You may be forced to act in a legal gray area.

We can use this model to illustrate the dynamics that connect ethical, social, and political issues. This model is also useful for identifying the main moral dimensions of the information society, which cut across various levels of action—individual, social, and political.

Five Moral Dimensions of the Information Age

The major ethical, social, and political issues that information systems raise include the following moral dimensions.

- *Information rights and obligations* What information rights do individuals and organizations possess with respect to themselves? What can these rights protect?
- *Property rights and obligations* How will traditional intellectual property rights be protected in a digital society in which tracking and accounting for ownership are difficult while ignoring such property rights is so easy?
- *System quality* What standards of data quality and system quality should we demand to protect individual rights and the safety of society?
- *Accountability and control* Who can and will be held accountable and liable for the harm done to individual and collective rights?
- *Quality of life* What values should be preserved in an information- and knowledge-based society? Which institutions should we protect from violation? Which cultural values and practices does information technology support?

We explore these moral dimensions in detail in Sections 4.3 through 4.7.

Key IT Trends That Raise Ethical Issues

Ethical issues long preceded information technology. Nevertheless, information technology has heightened ethical concerns, taxed existing social arrangements, and made some laws obsolete or insufficient. Technological trends that are responsible for many of these ethical stresses are summarized in Table 4.2.

The ever-increasing rise in computing power over the past several decades has made it possible for most organizations to use information systems for their core production processes. As a result, our dependence on systems and our vulnerability to system errors and poor data quality have increased. Social rules and laws have not yet fully adjusted to this dependence. In addition, standards for ensuring the accuracy and reliability of information systems (see Chapter 8) are not universally accepted or enforced.

Advances in data storage techniques and rapidly declining storage costs have enabled the collection of big data and have been responsible for the proliferation of databases on individuals—employees, customers, and potential customers—maintained by private and public organizations. These advances in data storage have made the routine violation of individual privacy both inexpensive and effective. Data storage systems for terabytes and petabytes of data are now available onsite or as online services for firms of all sizes to use in identifying customers.

Advances in data analysis techniques for large pools of data are another technological trend that heightens ethical concerns because organizations can find out highly detailed personal information about individuals. With contemporary data management tools (see Chapter 6), companies can assemble and combine myriad pieces of data about you stored in information systems much more easily than they could in the past.

Table 4.2 IT Trends that Raise Ethical Issues

Trend	Impact
Ever-increasing rise in computing power	More organizations depend on computer systems for critical operations and become more vulnerable to system failures.
Rapid decline of data storage costs	The steep reduction in cost enables the collection and storage of big data. Organizations can easily maintain detailed databases on individuals. There are no limits on the data collected about you.
Advances in data analysis	Companies can analyze vast quantities of data gathered on individuals to develop detailed profiles of individual behavior. Large-scale population surveillance is enabled.
Advances in networking	The cost of making data accessible from anywhere falls exponentially. Access to data becomes more difficult to control.
Proliferation of mobile devices	Mobile devices may be tracked without user consent or knowledge. The always-on device becomes a tether.
Artificial Intelligence (AI)	Increased reliance on various forms of AI in decision making substitutes data-driven calculations for human judgment. AI also poses challenges to privacy, further facilitating large-scale population surveillance, and to intellectual property rights.

Think of all the ways you generate digital information about yourself—browsing online; using mobile apps; making online and in-person retail and service purchases; interacting with the financial services industry; participating in social networks; making mobile phone calls; using a smart TV, smart speaker, and other smart home devices; driving an Internet-enabled car; as well as interacting with local, state, and federal government institutions (including courts and the police). Each one of these actions generates digital data. Put together and mined properly, this information can reveal not only your credit information but also your tastes, your associations, what you read and watch, your political interests, your driving habits, and more.

Companies purchase personal information from a variety of sources to help them more finely target their marketing campaigns. Chapters 6 and 12 describe how companies can analyze large pools of data from multiple sources to rapidly identify buying patterns of customers and make individualized recommendations. The use of computers to combine data from multiple sources and create digital dossiers of detailed information on individuals is called **profiling**. For example, Google Marketing Platform tracks the activities of visitors of participating websites and uses this information to create a profile of each online visitor, adding more detail to the profile as the visitor accesses an associated Google Marketing Platform site.

Profiling

The use of computers to combine data from multiple sources and create electronic dossiers of detailed information on individuals.



Credit card purchases can make personal information available to a wide variety of organizations. Advances in information technology facilitate the invasion of privacy.

Over time, Google Marketing Platform can create a detailed dossier of a person's spending and online behavioral habits that is sold to companies to help them target their online ads more precisely. Advertisers can combine online consumer information with offline consumer information such as credit card purchases at stores.

LexisNexis Risk Solutions gathers data from police, criminal, and motor vehicle records; credit and employment histories; current and previous addresses; professional licenses; and insurance claims to assemble and maintain dossiers on almost every adult in the United States. The company sells this personal information to businesses and government agencies. Demand for personal data is so enormous that data broker businesses, such as Acxiom, Epsilon Data Management, CoreLogic, and LexisNexis Risk Solutions, are flourishing. The three major credit monitoring services, TransUnion, Equifax, and Experian, also maintain huge databases of personal data.

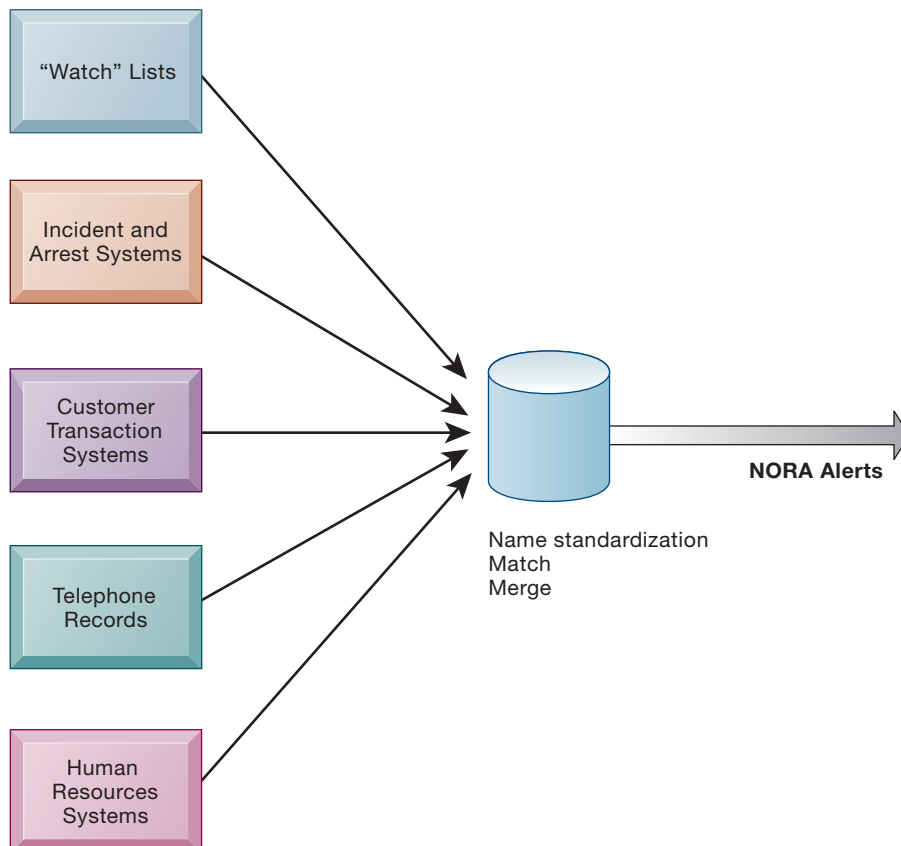
A data analysis technology called **nonobvious relationship awareness (NORA)** has given both the government and the private sector even more powerful profiling capabilities. Examine Figure 4.2, which illustrates how NORA works. NORA can take information about people from many disparate sources, such as “watch” lists, incident and arrest systems, consumer transaction systems, telephone records, and human resources systems, and correlate relationships to find obscure connections that might help identify criminals or terrorists. NORA technology processes the data and extracts information as the data are being generated so that, for example, it can—before a person gets on an airplane—instantly discover that the person at the airline ticket counter has the same phone number as a known terrorist. NORA is also often used by the casino industry as an anti-fraud technology. Although NORA technology is considered a valuable security tool, it does have privacy implications because it can provide such a detailed picture of the activities and associations of a particular individual. In 2005, IBM purchased the company that originally developed NORA technology, and now offers it under the name IBM InfoSphere Identity Insight.

Nonobvious relationship awareness (NORA)

Data analysis technology that can find obscure hidden connections among people and entities by analyzing information from many different sources to correlate relationships.

Figure 4.2 Nonobvious Relationship Awareness (NORA)

NORA technology can take information about people from disparate sources and find obscure, nonobvious relationships. It might discover, for example, that an applicant for a job at a casino shares a telephone number with a known criminal and then issue an alert to the hiring manager.



Advances in networking, including the Internet, have greatly reduced the costs of accessing large quantities of data and have enabled the mining of large pools of data using desktop/laptop computers, mobile devices, and cloud servers, permitting an invasion of privacy on a scale and with a precision heretofore unimaginable. The proliferation of mobile devices, such as smartphones, that users carry everywhere means that users can potentially be tracked without their knowledge or consent.

Finally, companies and individuals are increasingly relying on AI technologies as a substitute for human judgment in decision making. However, doing so raises serious ethical issues. Human judgment relies not only on reasoning but also on other human capabilities, such as empathy, and also typically takes into account factors that inform moral and ethical behavior. AI, on the other hand, may make decisions without regard to consequences that might violate ethical and moral standards. Another issue with AI is that it reduces multidimensional reality into a more simplistic representation of the world, one that appears on the surface to be precise and undeniably correct, thus encouraging acceptance of its decisions without questioning them. AI also poses challenges to privacy and intellectual property rights. We discuss these issues further later in this chapter and in Chapter 11 (Moser, den Hond, and Lindebaum, 2022).

4.2 Describe Principles for Ethical Conduct

Ethics is a concern of humans who have freedom of choice. Ethics is about individual choice: When faced with alternative courses of action, what is the correct moral choice? What are the main features of ethical choice?

Basic Concepts: Responsibility, Accountability, Liability, and Due Process

Responsibility

Accepting the potential costs, duties, and obligations for the decisions one makes.

Accountability

Feature of systems and social institutions; mechanisms for assessing responsibility for decisions made and actions taken.

Liability

Body of laws that permit individuals to recover the damages done to them by other actors, systems, or organizations.

Due process

A process in which laws are known and understood and there is an ability to appeal to higher authorities to ensure that laws are applied correctly.

Ethical choices are decisions made by individuals who are responsible for the consequences of their actions. **Responsibility** is a key element of ethical action. Responsibility means that you accept the potential costs, duties, and obligations for the decisions you make. **Accountability** is a feature of systems and social institutions; it means that mechanisms are in place to determine who took action and who is responsible. Systems and institutions in which it is impossible to find out who took what action are inherently incapable of ethical analysis or ethical action. **Liability** extends the concept of responsibility further to the area of laws. Liability is a feature of political systems in which a body of laws is in place that permits individuals to recover the damages done to them by other actors, systems, or organizations. **Due process** is a related feature of law-governed societies and is a process in which laws are known and understood, and the ability exists to appeal to higher authorities to ensure that laws are applied correctly.

These basic concepts form the underpinning of an ethical analysis of information systems and those who manage them. First, information technologies are filtered through social institutions, organizations, and individuals. Systems do not have impacts by themselves. Whatever information system effects exist are products of institutional, organizational, and individual actions and behaviors. Second, responsibility for the consequences of technology falls clearly on the institutions, organizations, and individuals who choose to use the technology. Using information technology in a socially responsible manner means that you can and will be held accountable for the consequences of your actions. Third, in an ethical political society, individuals and others can recover damages done to them through a set of laws characterized by due process.

Ethical Analysis

When confronted with a situation that seems to present ethical issues, how should you analyze it? The following five-step process should help:

1. *Identify and describe the facts clearly.* Find out who did what to whom and where, when, and how. In many instances, you will be surprised at the errors in the initially reported

facts, and often you will find that simply getting the facts straight helps define the solution. It also helps to get the opposing parties involved in an ethical dilemma to agree on the facts.

2. *Define the conflict or dilemma and identify the higher-order values involved.* Ethical, social, and political issues always reference higher values. The parties to a dispute all claim to be pursuing higher values (e.g., freedom, privacy, protection of property, or the free enterprise system). Typically, an ethical issue involves a dilemma: two diametrically opposed courses of action that support worthwhile values. For example, the Opening Case Study, *Apps That Track: A Double-Edged Sword*, illustrates two competing values: organizational efficiency and profit versus respect for individual privacy.
3. *Identify the stakeholders.* Every ethical, social, and political issue has stakeholders: players in the game who have an interest in the outcome, who have invested in the situation, and who usually have vocal opinions. Find out the identity of these groups and what they want. This will be useful later when designing a solution.
4. *Identify the options that you can reasonably take.* You may find that none of the options satisfy all the interests involved, but that some options do a better job than others. Sometimes arriving at a good or ethical solution may not always be a balancing of consequences to stakeholders.
5. *Identify the potential consequences of your options.* Some options may be ethically correct but disastrous from other points of view. Other options may work in one instance but not in similar instances. Always ask yourself, “What if I choose this option consistently over time?”

Ethical Principles

Once your analysis is complete, what ethical principles or rules should you use to make a decision? What higher-order values should inform your judgment? Although you are the only one who can decide which among many ethical principles you will follow, and how you will prioritize them, it is helpful to consider some ethical principles with deep roots in many cultures that have survived throughout recorded history:

1. The **Golden Rule**: do unto others as you would have them do unto you. Putting yourself in the place of others and thinking of yourself as the object of the decision can help you think about fairness in decision making.
2. **Immanuel Kant’s categorical imperative**: if an action is not right for everyone to take, it is not right for anyone. Ask yourself, “If everyone did this, could the organization, or society, survive?”
3. The **slippery slope rule**: if an action cannot be taken repeatedly, it is not right to take at all. An action may bring about a small change now that is acceptable, but if it is repeated, it would bring unacceptable changes in the long run. In the vernacular, it might be stated as “once started down a slippery path, you may not be able to stop.”
4. The **utilitarian principle**: take the action that achieves the higher or greater value. This rule assumes you can prioritize values in a rank order and understand the consequences of various courses of action.
5. The **risk aversion principle**: take the action that produces the least harm or the least potential cost. Some actions have extremely high failure costs of very low probability (e.g., building a nuclear generating facility in an urban area) or extremely high failure costs of moderate probability (speeding and automobile accidents). Avoid actions that have extremely high failure costs; focus on reducing the probability of accidents occurring.
6. The **ethical no-free-lunch rule**: assume that virtually all tangible and intangible objects are owned by someone else unless there is a specific declaration otherwise. If something someone else has created is useful to you, it has value, and you should assume the creator wants compensation for this work.

Golden Rule

Ethical principle that involves putting oneself in the place of others and thinking of yourself as the object of a decision.

Immanuel Kant’s categorical imperative

Ethical principle that states that if an action is not right for everyone to take it is not right for anyone.

Slippery slope rule

Ethical principle that if an action cannot be taken repeatedly, it is not right to take at all.

Utilitarian principle

Ethical principle that assumes one can put values in rank order of utility and understand the consequences of various courses of action.

Risk aversion principle

Ethical principle that one should take the action that produces the least harm or the least potential cost.

Ethical no-free-lunch rule

Assumption that virtually all tangible and intangible objects are owned by someone else, unless there is a specific declaration otherwise, and that the creator wants compensation for this work.

Actions that do not easily pass these rules deserve close attention and a great deal of caution. The appearance of unethical behavior may do as much harm to you and your company as actual unethical behavior.

Professional Codes of Conduct

When groups of people claim to be professionals, they take on special rights and obligations because of their special claims to knowledge, wisdom, and respect. Professional codes of conduct are promulgated by associations of professionals such as the American Medical Association (AMA), the American Bar Association (ABA), the Association of Information Technology Professionals (AITP), and the Association for Computing Machinery (ACM). These professional groups take responsibility for the partial regulation of their professions by determining entrance qualifications and competence. Codes of ethics are promises by professions to regulate themselves in the general interest of society. For example, avoiding harm to others, honoring property rights (including intellectual property), and respecting privacy are among the General Moral Imperatives of the ACM's Code of Ethics and Professional Conduct.

Governance Codes

One tool that organizations are using to help ensure ethical decision making is to create and implement governance codes, which establish guidelines and policies, as well as implementing technical guardrails with respect to conduct. For example, global consulting firm Accenture has adopted a Responsible AI Compliance program that rests on a set of core principles, such as “human by design” (focus on AI's impact on humans), fairness, transparency, safety, accountability, compliance, and sustainability. To implement these principles, they established an AI governance structure, conducted an AI risk assessment, created an AI compliance program, and established ongoing monitoring of AI (Accenture, 2024).

To help enterprises develop their own policies, the Responsible AI Institute, whose members include Amazon Web Services, Boston Consulting Group, ATB Financial, and Shell has recently introduced an AI policy template. The template is described as an “industry-agnostic, plug-and-play policy document” that allows organizations to develop policies that are aligned with both business needs and risks (Barker, 2024). We discuss Responsible AI further in Chapter 11.

Some Real-World Ethical Dilemmas

Information systems have created new ethical dilemmas in which one set of interests is pitted against another. For example, many companies use voice recognition software to reduce the size of their customer support staff by enabling computers to recognize a customer's responses to a series of automated questions. Many companies monitor what their employees are doing on the Internet to prevent them from wasting company resources on nonbusiness activities.

In each instance, you can find competing values at work, with groups lined up on either side of the debate. A company may argue, for example, that it has the right to use information systems to increase productivity and reduce the size of its workforce to lower costs and stay in business. Employees displaced by information systems may argue that employers have some responsibility for their welfare. Business owners might feel obligated to monitor employee email and Internet use to minimize drains on productivity. Employees might believe they should be able to use the Internet for short personal tasks. A close analysis of the facts can sometimes produce solutions that give each side “half a loaf.” Try to apply some of the principles of ethical analysis described to each of these cases. What is the right thing to do?

4.3 Discuss Information Rights and Privacy Issues

In this section, we take a closer look at information rights, one of the moral dimensions of information systems first described in Figure 4.1. We identify ethical, social, and political issues and use real-world examples to illustrate the values involved, the stakeholders, and the options chosen.

Information Rights: Privacy and Freedom in the Internet Age

Information rights are the rights that individuals and organizations have with respect to the information that pertains to them. **Privacy** is a component of information rights and is the claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations, including the state. Claims to privacy are also involved at the workplace. Millions of employees are subject to digital and other forms of high-tech surveillance. Information technology and systems threaten individual claims to privacy by making the invasion of privacy cheap, profitable, and effective.

Information rights

The rights that individuals and organizations have with respect to information that pertains to them.

Privacy

Claim of individuals to be left alone, free from surveillance or interference from other individuals or organizations, including the state; component of information rights.

Legal Protections for Privacy in the United States

The claim to privacy is protected in the United States Constitution in a variety of ways and also through various statutes. The same holds true for many other countries. In the United States, the claim to privacy is protected primarily by the First Amendment's guarantees of freedom of speech and association, the Fourth Amendment's protections against unreasonable search and seizure of one's personal documents or home, and the Fifth and Fourteenth Amendments' guarantee of due process.

Table 4.3 describes the major U.S. federal statutes that set forth the conditions for handling information about individuals in such areas as credit reporting, education, financial records, newspaper records, and electronic and digital communications. The Privacy Act of 1974 was one of the first, and arguably, the most important of these laws, regulating the federal government's collection, use, and disclosure of information.

Most U.S. and European privacy law is based on a set of principles called **Fair Information Practices (FIP)**, first set forth in a 1973 report by a federal government advisory committee (U.S. Department of Health, Education, and Welfare, 1973). FIP is a set of principles governing the collection and use of information about individuals. FIP principles are based on the notion of a mutuality of interest between the record holder and the individual. The individual has an interest in engaging in a transaction, and the record keeper—usually a business or government agency—requires information about the individual to support the transaction. After information

Fair Information Practices (FIP)

A set of principles originally set forth in 1973 that governs the collection and use of information about individuals and forms the basis of most U.S. and European privacy laws.

Table 4.3 Federal Privacy Laws in the United States

Privacy Laws That Apply to the U.S. Federal Government	Privacy Laws That Apply to Private Institutions
Privacy Act	Fair Credit Reporting Act
Privacy Protection Act	Family Educational Rights and Privacy Act
Electronic Communications Privacy Act	Right to Financial Privacy Act
Computer Matching and Privacy Protection Act	Cable Communications Policy Act
Computer Security Act	Video Privacy Protection Act
Federal Managers Financial Integrity Act	Telephone Consumer Protection Act (TCPA)
Driver's Privacy Protection Act	Health Insurance Portability and Accountability Act (HIPAA)
E-Government Act	Children's Online Privacy Protection Act (COPPA)
USA Freedom Act	Financial Services Modernization Act (Gramm-Leach-Bliley Act)

Table 4.4 Federal Trade Commission’s Fair Information Practice Principles

Principle	Description
Notice/awareness (core principle)	Websites must disclose their information practices before collecting data. Includes identification of collector; uses of data; other recipients of data; nature of collection (active/inactive); voluntary or required status; consequences of refusal; and steps taken to protect confidentiality, integrity, and quality of the data.
Choice/consent (core principle)	A process enabling choice must be in place, allowing consumers to choose how their information will be used for purposes other than supporting the transaction, including internal use and transfer to third parties.
Access/participation	Consumers should be able to review and contest the accuracy and completeness of data collected about them in a timely, inexpensive process.
Security	Data collectors must take responsible steps to ensure that consumer information is accurate and secure from unauthorized use.
Enforcement	A mechanism must be in place to enforce FIP principles. This can involve self-regulation, legislation giving consumers legal remedies for violations, or federal statutes and regulations.

is gathered, the individual maintains an interest in the record, and the record may not be used to support other activities without the individual’s consent. In 1998, the Federal Trade Commission (FTC) restated and extended the original FIP principles to provide guidelines for protecting online privacy and updated them further in 2010 to take into account new privacy-invading technology. Table 4.4 describes the FTC’s FIP principles.

The FTC’s FIP principles have been used throughout the years as guidelines to drive changes in privacy legislation. For instance, the Children’s Online Privacy Protection Act (COPPA) requires websites and mobile apps to obtain parental permission before collecting information on children under the age of 13. The FTC has also added three principles to its framework for privacy. Firms should adopt privacy by design, building products and services that protect privacy; firms should increase the transparency of their data practices; and firms should require consumer consent and provide clear options to opt out of data-collection schemes. The FTC has extended its privacy policies and enforcement actions to a wide range of privacy issues across multiple industries, including behavioral targeting, geolocation tracking, social media, the Internet of Things (IoT), the mobile app ecosystem, and AI, among others (Federal Trade Commission, 2024).

Public opinion polls show an ongoing distrust of online marketers. Although there are many studies of privacy issues, there has been no significant federal legislation in recent years. A recent survey by the Pew Research Center found that more than 80 percent of those surveyed were concerned about their online privacy; most believe they have little to no control over what companies do with their data; and most also say they have little to no understanding about what companies do with their data (Pew Research, 2023).

Privacy protections have also been added to laws deregulating financial services and safeguarding the maintenance and transmission of health information about individuals. The Gramm-Leach-Bliley Act, which repeals earlier restrictions on affiliations among banks, securities firms, and insurance companies, includes some privacy protection for consumers of financial services. All financial institutions are required to disclose their policies and practices for protecting the privacy of nonpublic personal information and to allow customers to opt out of information-sharing arrangements with nonaffiliated third parties.

The Health Insurance Portability and Accountability Act (HIPAA) includes privacy protection for medical records. The law gives patients access to their personal medical records that healthcare providers, hospitals, and health insurers maintain and the right to authorize how protected information about themselves can be used or disclosed. Doctors, hospitals, and other healthcare providers must limit the disclosure of personal information about patients to the minimum amount necessary to achieve a given purpose.

General Data Protection Regulation (GDPR)

Updated framework for protecting personally identifiable information; unifies data privacy laws across the European Union, focusing on making businesses more transparent and expanding the privacy rights of data subjects.

The European General Data Protection Regulation

In 2018, the European Commission implemented the European Union’s **General Data Protection Regulation (GDPR)**, which is arguably the most important privacy legislation since the FTC’s FIP principles. It applies to all firms and organizations that collect, store, or

process personal information of EU citizens, and these protections apply worldwide regardless of where the processing takes place (European Commission, 2018).

The GDPR is an updated framework for protecting personally identifiable information (PII) and replaces an earlier Data Protection Directive. In Europe, privacy protection is historically much stronger than it is in the United States. In the United States, there is no federal agency charged with enforcing privacy laws, and there is no single privacy statute governing the use of PII by private organizations. Instead, privacy laws are piecemeal, sector by sector (e.g., medical privacy, educational privacy, and financial privacy laws). These are enforced by the FTC, through self-regulation by businesses, and by individuals, who must sue companies in court to recover damages.

In the EU, data protection laws are comprehensive, apply to all organizations, and are enforced by data protection agencies in each country to pursue complaints brought by citizens and to actively enforce privacy laws. The GDPR protects a wide variety of PII: basic identity information such as name, address, and ID numbers; web data such as location, IP address, cookie data, and RFID tags; health and genetic data; mobile phone number; driver's license and passport number; biometric and facial data; racial and ethnic data; political opinions; and sexual orientation.

The main objective of the GDPR is to strengthen the rights of citizens to their own personal information and to strengthen oversight of firms to ensure they implement these individual rights. A second thrust was to harmonize conflicting data protection standards among the EU member nations and create a single EU agency to implement and enforce the regulation. And third, to enforce these conditions worldwide for all organizations that operate in the EU, or process data pertaining to EU citizens, regardless of where the organization is located.

For individuals, the GDPR requires organizations to allow consumers to access all their personal information without charge within one month; delete personal data (the right to be forgotten); ensure data portability so consumers are not locked into a particular service; and guarantee the right to sue providers for damages or abuse of PII, including class action lawsuits.

Organizational requirements have been strengthened. Organizations must have a data protection officer that reports to senior management. Explicit consent must be obtained from users before the organization can collect data (positive opt-in), eliminating default opt-in processes. Organizations must publish the rationale for data collection and the length of time data are retained, and report any breaches or hacks within 72 hours. Organizations must also build privacy protections into all new systems (privacy by design); limit targeting and retargeting of individuals to audience-level anonymized data (rather than targeting based on intimate, personal profiles); and limit the collection of personal data to only that which is needed to support a task or a transaction, and then delete it shortly thereafter. Abuse of PII can be fined up to \$20 million or 4 percent of the organization's global revenue, whichever is greater. Finally, the EU will enforce the GDPR requirements with non-EU countries like the United States using intergovernmental privacy shield agreements that ensure that EU data processed in non-EU nations meets GDPR standards. Privacy shield agreements are a more enforceable version of earlier safe harbor agreements. A **safe harbor** is a private self-regulating policy and enforcement mechanism that meets the objectives of government regulators and legislation but does not involve government regulation or enforcement.

The GDPR is primarily aimed at Meta (owner of Facebook, Instagram, and WhatsApp), Google, and other online ad-based businesses that track and collect personal data from users in order to create comprehensive digital profiles and target the users with ads. Google and Meta dominate their markets in Europe, but at the same time, they are widely criticized for invading privacy and not protecting PII (see the Chapter Case Study, *Meta's Many Ethical Challenges*).

The GDPR and AI

While AI is not explicitly mentioned in the GDPR, the broad definition of "processing" under the GDPR makes it clear that the GDPR does apply to AI systems to the extent such systems include personal data at some point in their lifecycle of either development or operation. The GDPR also includes provisions with respect to automated decision-making, which are also likely to apply, given that AI systems are frequently used to make automated decisions that impact individuals. Finally, as you will learn later in the chapter, the European Parliament has also recently adopted a comprehensive act (the EU AI Act) regulating artificial intelligence.

Safe harbor

Private self-regulating policy and enforcement mechanism that meets the objectives of government regulations but does not involve government regulation or enforcement.

The relationship between AI and data protection is expressly recognized in the text of the EU AI Act, which states that it is without prejudice to the GDPR (meaning that it does not override any protections the GDPR may also offer).

Challenges Posed by the Internet

The Internet poses numerous challenges for the protection of individual privacy. Companies can track a user’s online search history, the websites and web pages visited, the online content accessed, and the items inspected or purchased online. Mobile apps can also track users (see the Opening Case Study, *Apps that Track: A Double-Edged Sword*). This monitoring and tracking occurs in the background without the user’s knowledge. It is conducted not just by individual websites (known as first-party tracking) but by advertising networks such as Microsoft Advertising, Google’s Marketing Platform, and Meta’s Audience Network that are capable of tracking personal browsing behavior across thousands of websites (known as third-party tracking). In the past, website publishers and the online advertising industry have defended tracking of individuals across the web because doing so allows more relevant ads to be targeted to users, and this helps pay for the cost of providing online content. The commercial demand for this personal information is virtually insatiable. However, these practices also impinge on individual privacy.

Cookies

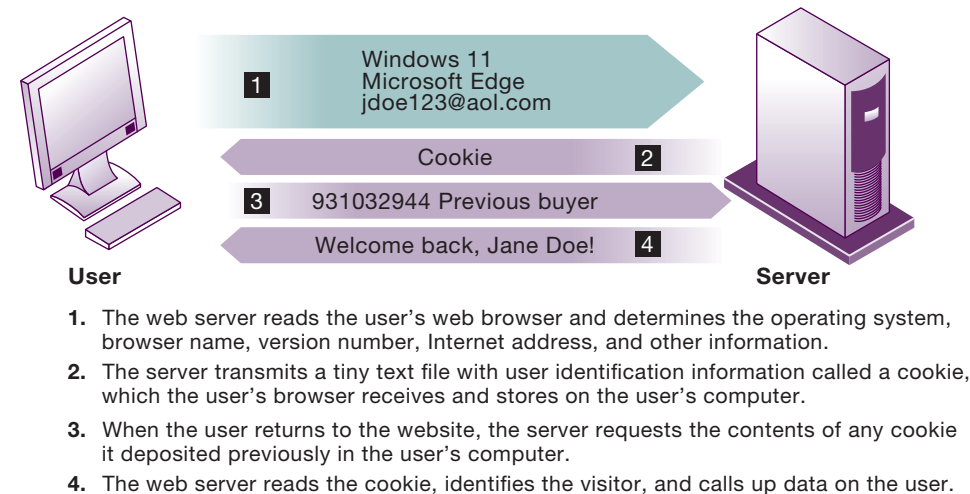
Small text file placed on a user’s computer or mobile device when the user visits a website; used to identify the user and track visits to the website.

Cookies are one method used to monitor and track online users. Examine Figure 4.3, which illustrates how cookies work. When a user visits a website, the website’s web server places a small text file (a “cookie”) on the user’s computer or mobile device. Cookies identify the visitor’s web browser software, as well as other information, and track visits to the website. When the visitor returns to a site that has stored a cookie, the website software searches the visitor’s computer or mobile device, finds the cookie, and knows what that person has done in the past. It may also update the cookie, depending on the activity during the visit. In this way, the site can customize its content for each visitor’s interests. For example, if you purchase a book on Amazon.com and return later from the same browser, the site will welcome you by name and recommend other books of interest based on your past purchases. Cookies also make “quick checkout” options possible by allowing a site to keep track of users as they add items to a shopping cart. This type of cookie is known as a first-party cookie. First-party cookies are typically considered to be “good” cookies because they help enhance the user experience.

Websites using cookie technology cannot directly obtain visitors’ names and addresses. However, if a person has registered at a site, that information can be combined with cookie data to identify the visitor. Website owners can also combine the data they have gathered from cookies and other website monitoring tools with personal data from other sources, such as

Figure 4.3 How First-Party Cookies Identify Web Visitors

Cookies are placed by a website on a visitor’s computer. When the visitor returns to that website, the web server requests the ID number from the cookie and uses it to access the data stored by that server on that visitor. The website can then use these data to display personalized information.



offline data collected from surveys or paper catalog purchases, to develop very detailed profiles of their visitors. First-party cookies can be blocked and/or deleted by users, although the majority of users do not do so.

Third-party cookies operate similarly to first-party cookies but enable advertising platforms to track user behavior across websites and devices. In the past, third-party cookies were supported by all the major web browsers, but that has begun to change. Apple's Safari and Mozilla's Firefox browsers now block third-party cookies by default. Google, which had previously stated that it would also follow suit, delayed implementation several times and recently announced that it had scrapped that plan, and would retain third-party cookies after all. Instead, it said it would introduce new methods that would enable users to choose how they wanted to be tracked while using Google search products. We discuss other technology-based efforts to enhance privacy in the following section on technological solutions.

In addition to cookies, there are other tools for the surveillance of Internet users. A **web beacon**, also called a *web bug* (or simply tracking file), is a tiny image that keeps a record of users' online clickstreams. It reports data back to whoever owns the tracking file, which can be invisibly embedded in an email message or web page to monitor the behavior of the user visiting the website or receiving the email. Web beacons are placed on popular websites by third-party firms who pay the websites a fee for access to their audience. So how common is web tracking? A recent analysis by *Wired* of the top 10,000 most popular websites found that thousands of websites are sharing data with hundreds of different advertising firms, with dozens data sharing with more than 1,000 companies. For instance, publisher Dotdash Meredith, which operates popular websites such as Investopedia.com, Allrecipes.com, and People.com, share data with more than 1,600 firms. The biggest trackers were Google, Microsoft, and Meta, all of whom are in the business of selling ads to advertising firms and marketers (Burgess, 2024).

Adware can secretly install itself on an Internet user's computer by piggybacking on larger applications. Once installed, adware calls out to websites to send ads and other unsolicited material to the user. A more malicious version of adware, known as spyware, can also track the user's browsing habits. More information is available about intrusive software in Chapter 8.

More than 90 percent of global Internet users use Google Search and other Google services, making Google the world's largest collector of online user data. Whatever Google does with its data has an enormous impact on online privacy. Most experts believe that Google possesses the largest collection of personal information in the world—more data on more people than any government agency. Google uses behavioral targeting (also known as interest-based advertising) to target individuals as they move from one site to another to show them relevant ads based on their search activities and other data that Google has collected about them. For instance, one of its programs enables advertisers to target ads based on the search histories of Google users, along with any other information the user submits to Google such as age, demographics, region, and web activities (such as blogging). Google's AdSense program enables Google to help advertisers select keywords and design ads for various market segments based on search histories such as helping a clothing website create and test ads targeted at teenagers. Google displays targeted ads on YouTube and Google mobile applications, and its Google Marketing Platform ad network serves targeted ads. We further discuss behavioral targeting in Chapter 10.

In the United States, businesses are allowed to gather transaction information generated in the marketplace and then use that information for other marketing purposes without obtaining the **informed consent** (consent given with knowledge of all the facts needed to make a rational decision) of the individual whose information is being used. Firms that do so argue that when users agree to the sites' terms of service, they are also agreeing to allow the site to collect information about their online activities. An **opt-out** model of informed consent permits the collection of personal information until the consumer specifically requests the data not to be collected. Privacy advocates would like to see wider use of an **opt-in** model of informed consent in which a business is prohibited from collecting any personal information unless the consumer specifically takes action to approve information collection and use. Here, the default option is no collection of user information. Many websites now employ "cookie banners"—a pop-up window that allows users to accept or reject cookies from the site. Although intended as a form of opt-in informed consent, most privacy advocates feel that cookie banners are ineffective, as few people actually read the disclosures about what data are being collected before clicking the Accept button.

Web beacon

Tiny image invisibly embedded in email messages and web pages that is designed to monitor the behavior of the user visiting a website or sending email; also called a web bug or tracking file.

Informed consent

Consent given with knowledge of all the facts needed to make a rational decision.

Opt-out

Model of informed consent permitting the collection of personal information until the consumer specifically requests that the data not be collected.

Opt-in

Model of informed consent permitting prohibiting an organization from collecting any personal information unless the individual specifically takes action to approve information collection and use.

The online industry has preferred self-regulation to privacy legislation for protecting consumers. Members of the online advertising industry have created an industry association called the Network Advertising Initiative (NAI) to develop privacy policies to help consumers opt out of advertising network programs and to provide consumers redress from abuses. Individual firms such as Apple, Google, and Microsoft have also launched privacy initiatives in an effort to address public concern about tracking people online.

In general, most businesses do little to protect the privacy of their customers, and consumers do not do as much as they should to protect themselves. For websites and apps that depend on advertising for support, most revenue derives from selling access to customer information. The vast majority of people claim to be concerned about online privacy, but most do not read the privacy statements on websites. In addition, website privacy policies are often ambiguous about key terms and too complicated for the average consumer to understand (Laudon and Traver, 2024). What firms call a privacy policy is in fact a data use policy. The concept of privacy is associated with consumer rights, which many firms do not wish to recognize. A data use policy simply tells customers how the information will be used and does not mention rights.

Challenges Posed by AI

AI is the latest information technology that poses a threat to privacy. Many of the issues already noted above are amplified by the use of AI. Google, OpenAI, and Microsoft have also been accused of training their generative AI tools on the personal data of millions of people that was scraped from the Internet without the users' knowledge or consent.

The development of more specifically focused AI tools on a firm-wide level also raises concerns. For instance, AI applications developed by firms in the health and financial services industries may require sensitive data for training purposes. AI applications developed for use in online advertising can create very detailed profiles of individuals based on their online behavior, which also raises privacy concerns.

The collection of data for use by facial recognition systems (see Chapter 11) has also been the subject of much concern. Massive databases of people's faces collected from social networks, photo websites, dating services, and cameras placed in public spaces have been compiled by a number of companies as well as by academic researchers. Georgetown University has estimated that photos of nearly half of all U.S. adults have been entered into at least one facial recognition database. Both Meta and Google have faced lawsuits charging that their actions in amassing facial databases violate the Illinois Biometric Information Privacy Act, one of a growing number of state laws that address this issue, and which require firms to obtain user consent before collecting biometric and facial information. In another recent case, Prisma Labs, the developer of the image-generating app Lensa, has also been accused of collecting and storing the facial information of users without their consent in violation of Illinois' law.

Technological Solutions

In addition to legislation, a number of technological solutions have been developed to help protect user privacy (sometimes referred to as privacy-enhancing techniques (PETs)). Table 4.5 lists some of the most common tools that are available. For the most part, technical solutions thus far have failed to provide effective protection for online privacy, in part because most of them require users to be proactive in implementing them.

4.4 Discuss Intellectual Property Issues

Intellectual property

Products of the mind created by individuals or corporations and which may be protected by copyright, patent, trademark and trade secret law.

Today's information technologies and systems have severely challenged existing laws and social practices that protect intellectual property. **Intellectual property** refers to products of the mind created by individuals or corporations. Information technology has made it difficult to protect intellectual property because digital information can be so easily copied or distributed. Intellectual property is subject to a variety of protections under four legal traditions: copyright, patents, trademarks, and trade secrets.

Table 4.5 Technological Protections for Online Privacy

Technology	Protection
Apple App Tracking Transparency (ATT)	Requires any app that wants to track user activity and share it with other apps or websites to ask user for permission
Apple Intelligent Tracking Prevention (ITP) for Safari web browser	Monitors and disables cross-site tracking cookies and blocks trackers' ability to identify users by IP address
Google Privacy Sandbox	Originally developed by Google to replace third-party cookies; now being focused on technologies that will allow users to choose how they want to be tracked while using Google search products
Differential privacy software	Reduces the ability to merge different files and de-anonymize consumer data. In AI, this means adding "statistical noise" to the raw data to protect individual privacy.
Privacy default browsers	Eliminates tracking cookies and prevents IP tracking
Message encryption apps	Encrypts text and other data transferred using smartphones
Spyware blockers	Detects and removes spyware, adware, keyloggers, and other malware
Pop-up and ad blockers	Prevents calls to ad servers; restricts downloading of images at user request
Secure email	Email and document encryption
Anonymous remailers	Enhanced privacy protection for email
Anonymous surfing	Enhanced privacy protection for web browsing
Cookie managers	Blocks third-party cookies
Public key encryption	Enables encryption of email and documents

Copyright

Copyright protects creators of intellectual property from having their work copied by others for a certain period of time, depending on several factors. As a general rule, copyright protection lasts for the life of the author, plus an additional 70 years after the author's death. For corporate-owned works, copyright protection generally lasts for 95 years after initial creation. Copyright protects literary, dramatic, musical, and artistic works, such as novels, poetry, songs, plays, maps, drawings, and artwork. The intent behind copyright laws has been to encourage creativity and authorship by ensuring that creative people receive the financial and other benefits of their work. Most industrial nations have their own copyright laws, and there are several international conventions and bilateral agreements through which nations coordinate and enforce their copyright laws.

In the mid-1960s, the Copyright Office began registering software programs, and in 1980, Congress passed the Computer Software Copyright Act, which provides protection for software program code and copies of the original code sold in commerce; it sets forth the rights of the purchaser to use the software, while the creator retains legal title.

Copyright protects against copying entire programs or their parts. Damages and relief are readily obtained for infringement. The drawback to copyright protection is that the underlying ideas behind a work are not protected; only their manifestation in a work is protected. A competitor can use your software, understand how it works, and build new software that follows the same concepts without infringing on a copyright.

Look-and-feel copyright infringement lawsuits are precisely about the distinction between an idea and its expression. For instance, in the early 1990s, Apple sued Microsoft and Hewlett-Packard, claiming that the defendants copied the expression of overlapping windows in Apple's Macintosh interface. The defendants countered that the idea of overlapping windows can be expressed only in a single way and, therefore, was not protectable. When ideas and their expression merge, the expression cannot be copyrighted. Ultimately, the courts agreed with the defendants.

In general, courts follow the reasoning of a 1989 case—*Brown Bag Software v. Symantec Corp.*—in which the court dissected the elements of software alleged to be infringing. The court found that similar concepts, functions, general functional features (e.g., drop-down menus), and colors are not protectable by copyright law (*Brown Bag Software v. Symantec Corp.*, 1992).

Copyright

Protects creators of certain types of intellectual property against copying by others for a certain period of time, depending on several factors.

Patents

Patent

Grants the owner an exclusive monopoly on the ideas behind an invention for a certain period of time, typically 20 years; designed to ensure that inventors of new machines or methods are rewarded for their labor while enabling the widespread use of their inventions.

A **patent** grants the owner an exclusive monopoly on the ideas behind an invention for a certain period of time, typically 20 years. The congressional intent behind patent law was to ensure that inventors of new machines, devices, or methods receive the full financial and other rewards of their labor, and yet to make widespread use of the invention possible by providing detailed diagrams for those wishing to use the idea under license from the patent's owner. The granting of a patent is determined by the United States Patent and Trademark Office and relies on court rulings.

The key concepts in patent law are originality, novelty, and invention. The Patent Office did not accept applications for software patents routinely until a 1981 Supreme Court decision held that computer programs could be part of a patentable process.

The strength of patent protection is that it grants a monopoly on the underlying concepts and ideas of software. The difficulty is passing stringent criteria of nonobviousness (e.g., the work must reflect some special understanding and contribution), originality, and novelty as well as years of waiting to receive protection.

Trademarks

Trademarks

Marks, symbols, or images used to distinguish products in the marketplace.

Trademarks are the marks, symbols, and images used to distinguish products in the marketplace. Trademark laws protect consumers by ensuring that they receive what they paid for. These laws also protect the investments that firms have made to bring products to market. Typical trademark infringement violations occur when one firm appropriates or pirates the marks of a competing firm. Infringement also occurs when firms dilute the value of another firm's marks by weakening the connection between a mark and the product. For instance, if a search engine firm copies the trademarked Google icon, colors, and images, it would be infringing on Google's trademarks. It would also be diluting the connection between the Google search service and its trademarks, potentially creating confusion in the marketplace.

Trade Secrets

Trade secret

Any intellectual work product used for a business purpose, provided it is not based on information in the public domain.

Any intellectual work product—a formula, device, pattern, or compilation of data—used for a business purpose can be classified as a **trade secret**, provided that it is not based on information in the public domain. Protections for trade secrets vary from state to state. In general, trade secret laws grant a monopoly on the ideas behind a work product, but the monopoly can be very tenuous.

Software that contains novel or unique elements, procedures, or compilations can be considered a trade secret. Trade secret law protects the actual ideas in a work product, not only their manifestation. To make a claim of trade secret infringement, the creator or owner must take care to bind employees and customers with nondisclosure agreements and prevent the secret from falling into the public domain.

The limitation of trade secret protection is that although virtually all software programs of any complexity contain unique elements of some sort, it is difficult to prevent the ideas in the work from falling into the public domain when the software is widely distributed.

Challenges to Intellectual Property Rights Posed by Today's Information Technologies and Systems

Today's information technologies and systems pose severe challenges to existing methods of protecting intellectual property and, therefore, create significant ethical, social, and political issues. Digital media differ from books, periodicals, and other media in terms of the former's ease of replication; ease of transmission; ease of alteration; ease of theft because of compactness; and difficulties in establishing uniqueness.

Challenges Posed by the Internet

The proliferation of digital networks, including the Internet, has made it even more difficult to protect intellectual property. Before the widespread use of digital networks, copies of software, books, magazine articles, or films had to be stored on physical media such as paper, computer

disks, or videotape, creating some hurdles to distribution. Using digital networks, information can be more widely reproduced and distributed. Software and media piracy remains an ongoing issue, with the number of visits to piracy websites across all industries, from movies, television, and music, to publishing and software, increasing to 230 billion worldwide in 2023, an increase of 6.7 percent from 2022 (MUSO, 2024).

The Internet was designed to transmit information, including copyrighted information, freely around the world. You can easily copy and distribute virtually anything to millions of people worldwide, even if they are using different types of computer systems. Information can also be illicitly copied from one place and distributed throughout other systems and networks, even though these parties do not willingly participate in the infringement.

Individuals have been illegally copying and distributing digitized music files on the Internet for several decades. File-sharing services such as Napster and, later, Grokster, Kazaa, Morpheus, Megaupload, and The Pirate Bay sprang up in the early years of the web to enable users to locate and swap digital music and video files, including those protected by copyright. Illegal file sharing became so widespread that it threatened the viability of the music recording industry and, at one point, consumed 20 percent of Internet bandwidth. The recording industry won several legal battles for shutting these services down, but it has not been able to halt illegal file sharing entirely. The motion picture and cable television industries continue to wage similar battles. Several European nations have worked with U.S. authorities to shut down illegal sharing sites, with mixed results.

However, as legitimate online music stores such as iTunes and streaming services such as Spotify, Apple Music, Amazon Prime Music, and Pandora have expanded, illegal file sharing has significantly declined. The Apple iTunes Store legitimized paying for music and entertainment and created a closed environment from which music and videos could not be easily copied and widely distributed. Amazon's Kindle also protects the rights of publishers and writers because its books cannot be copied to the Internet and distributed. Music and video streaming services also inhibit piracy because the streams cannot be easily recorded on separate devices. Since 2015, music industry revenues have more than doubled after a precipitous drop of more than 50 percent between 1999 and 2014.

The **Digital Millennium Copyright Act (DMCA)** also provides some copyright protection. The DMCA implements a World Intellectual Property Organization Treaty that makes it illegal to circumvent technology-based protections of copyrighted materials. Internet service providers (ISPs) are required to take down sites of copyright infringers they are hosting when the ISPs are notified of the infringement. Microsoft and other major software and information content firms are represented by the Software and Information Industry Association (SIIA), which lobbies for new laws and enforcement of existing laws to protect intellectual property around the world. The SIIA runs an antipiracy hotline for individuals to report piracy activities, offers educational programs to help organizations combat software piracy, and has published guidelines for employee use of software.

Digital Millennium Copyright Act (DMCA)

Makes it illegal to make, distribute, or use devices that circumvent technology-based protections of copyrighted materials.

Challenges Posed by AI

AI raises numerous issues with respect to the protection of various forms of intellectual property. We provide an overview here and also discuss these issues further in Chapter 11.

Copyright

One of the primary issues that has arisen thus far has to do with the use, without permission, by companies such as OpenAI, Google, Microsoft, and Meta of copyrighted works to train their AI models. For instance, it has been alleged that upwards of 170,000 copyrighted works were used to train the Meta LLAMA large language model. In the first major test with respect to this issue, the New York Times has filed a high-profile lawsuit against OpenAI, the developer of ChatGPT, and its partner Microsoft, charging that OpenAI scraped millions of the New York Times articles, along with those of other copyright owners, from the web without permission. OpenAI and Microsoft argue that its action constituted "fair use," a doctrine that allows the use of copyrighted material without first obtaining permission under specified circumstances. Other similar lawsuits have also been filed. For example, Getty Images, a stock photo company, has sued Stability AI, alleging that Stability AI infringed on its copyrights by using Getty's images to train its AI models.

Copyright issues also arise with respect to output generated by AI. If AI produces content that “looks like” or is in the style of a creator, such as a writer, artist, or singer, does that infringement that party’s copyright? There are also instances when AI generates content that is a verbatim copy. For example, in its lawsuit against OpenAI and Microsoft, the New York Times claims that ChatGPT sometimes generates verbatim portions of New York Times articles in response to user queries. OpenAI, Microsoft, and its subsidiary GitHub are currently being sued on the grounds that GitHub Copilot, a code-generating AI application, was replicating long sections of licensed open-source code without providing credit, as required by the licenses under which they were originally published.

There is also an issue as to whether works generated by AI are eligible for copyright protection under U.S. law. In March 2023, the U.S. Copyright Office issued a policy statement declaring that works created solely by AI engines were not eligible for copyright protection. Only works that include human-authored elements are considered copyrightable. In August 2023, in the first court ruling on this issue, the court affirmed this policy, stating that human authorship is a bedrock requirement of copyright.

Patents

Questions have also arisen with respect to the use of AI in the creation of inventions. In February 2024, the USPTO issued guidance that makes clear that while an AI system cannot be listed as an inventor on a patent application, that it would consider issuing patents for inventions that were created with the assistance of an AI system, so long as the human inventor significantly contributed to the invention. The guidance includes instructions on how to determine whether the human contribution is sufficient to qualify (USPTO, 2024).

Trademarks

As with copyright, the use of generative AI raises trademark concerns, particularly in terms of the output generated. Given that AI algorithms are trained using preexisting materials, there is an increased likelihood that an AI algorithm may generate logos or designs that closely resemble existing trademarks, which if actually used in commerce, could potentially lead to consumer confusion or dilution of established brands. AI chatbots or virtual assistants might also unknowingly infringe on trademarks when responding to customer inquiries or providing product recommendations. Whether AI-generated images can be trademarked remains an open question. Thus far, the USPTO has not issued any guidance indicating that AI-generated trademarks are subject to different rules than traditionally developed trademarks.

Trade Secrets

AI poses a potential threat to the protection of trade secrets. According to research firm Cyberhaven Labs, although many companies have prohibited the use of unlicensed, non-corporate AI chatbots such as ChatGPT, Google Gemini, and Microsoft Copilot for work-related tasks, many employees are using them and sharing company legal documents, source code, and employee information in the process, potentially leading to invalidation of trade secret protection. Most users do not understand what happens to their companies’ data once they share it with an unlicensed AI. So far, there have not been any high-profile reports about major company trade secrets revealed by large public AIs, but it is quite possible that this may happen in the future.

Trade secret laws are also being used by companies and public authorities to avoid having to disclose how their AI engines work, leading to what is referred to as a “black box” issue. In a world where AI increasingly is playing a pivotal role, how AI actually makes those decisions is not apparent, in part because those details constitute proprietary trade secrets. We discuss this issue further in Section 4.6.



4.5 Discuss System Quality Issues

In this and the following two sections, we take a closer look at the issues that information technologies and systems raise with respect to system quality, accountability and control, and the quality of everyday life, which, along with information rights and obligations and property rights and obligations, are part of the moral dimensions of information systems first described in Figure 4.1.

Given the centrality of information systems to the functioning of businesses and society, what is an acceptable level of system quality? If a person is injured by a machine controlled, in part, by software, who should be held accountable and, therefore, held liable? Should a social network be held accountable and liable for the posting of offensive material? What kinds of acts should be considered computer crimes? How should individuals, society, and governments deal with the potentially negative impacts of new information technologies on the quality of everyday life?

System Quality: Data Quality and System Errors

In 2023, many of the largest cloud providers experienced significant outages, which took out sites and services throughout the United States and Europe. For example, in June 2023, an Amazon Web Services (AWS) outage impacted the websites of the New York Metropolitan Transportation Authority, the Boston Globe, Associated Press, the Securities and Exchange Commission's EDGAR system, and Southwest Airlines, among others. The root cause of the issue appeared to be with a subsystem responsible for capacity management for AWS's Lambda service. In April 2023, Microsoft users faced a nearly six-hour-long issue with Microsoft 365 online applications, including Teams, SharePoint Online, and Outlook. And in February 2023, Oracle's global cloud infrastructure (OCI) suffered a multi-day outage affecting customers in numerous regions around the world arising from a performance issue in the back-end infrastructure that supports the OCI public DNS API.

Despite these high-profile incidents, outages at cloud computing services are infrequent. But as more and more firms rely on cloud providers and centralize their data and operations with a small group of cloud providers, these outages have called into question the reliability and quality of cloud services. Are these outages acceptable?

The debate raises a related but independent moral dimension: What is an acceptable, technologically feasible level of system quality? At what point should system managers say, "Stop testing, we've done all we can to perfect this software. Ship it!" Individuals and organizations may be held responsible for avoidable and foreseeable consequences, which they have a duty to recognize and correct. The gray area is that some system errors are foreseeable and correctable only at very great expense, an expense so great that pursuing this level of perfection is not feasible economically—no one could afford the product!

For example, although software companies try to debug their products before releasing them to the marketplace, they knowingly ship buggy products because the time and cost of fixing all the minor errors would prevent these products from ever being released. What if the product was not offered in the marketplace? Would social welfare as a whole falter and perhaps even decline? Carrying this further, just what is the responsibility of a producer of computer services—should the producer withdraw the product that can never be perfect, warn the user, or forget about the risk (let the buyer beware)?

Three principal sources of poor system performance are (1) software bugs and errors, (2) hardware or facility failures caused by natural or other causes, and (3) poor input data quality. Zero defects in software code of any complexity cannot be achieved, and the seriousness of the remaining bugs cannot be estimated (Choi, 2020). Hence, there is a technological barrier to perfect software, and users must be aware of the potential for catastrophic failure. The software industry has not yet arrived at testing standards that would result in software of acceptable but imperfect performance.

Although software bugs and facility catastrophes are likely to be widely reported in the press, by far the most common source of business system issues is poor data quality (see Chapter 6). Few companies routinely measure the quality of their data, but individual organizations report data error rates ranging from 0.5 to 30 percent.

4.6 Discuss Accountability and Control Issues

The use of information technology and systems may sometimes result in harm, creating questions as to who, if anyone, should be held accountable for that harm. What laws exist to control the misuse of information technology and systems?

Software and Internet Liability Issues

Poorly designed software can sometimes result in injury or economic loss. For instance, it has been reported that up to 60 percent of data breaches are caused by unpatched software vulnerabilities. Who, if anyone, is liable if a business or person is harmed by software that is defective? Should the company that developed the software be held legally liable for any harm that results from using the software? What about the individual software developers? The main concern is: What level of defects, if any, should be considered ordinary and customary and, therefore, acceptable?

Liability for software-related problems is a complicated issue from a legal standpoint. If someone suffers a personal injury or property damage because of a defective physical product, the seller, distributor, and/or manufacturer of that product can normally be held liable under the concept of strict product liability, whether any of those parties were negligent or not. Whether software is a “product” that is subject to strict product liability remains unsettled. Courts that have grappled with the question have generally found that software is not a product. This finding is important because it is typically very difficult to prove negligence, which is an alternative legal standard that might provide a basis for a lawsuit. However, when software is an essential part of a machine, such as a car, plane, or robot, it is more likely to be considered a product and potentially be subject to strict product liability laws or other types of liability. For example, Boeing has admitted to full responsibility for the 2019 crash of its 737 Max in Ethiopia because of a faulty software system.

Software vendors and software service providers typically use contracts and end-user license agreements to limit any potential liability they may have for defective software. Unfortunately, very few users actually read such agreements before using the software and, even when they do read them, most users do not understand that the agreements effectively exempt software vendors from almost all liability. Courts have typically upheld such agreements, severely impacting the ability of software users to recover for any damages they may have suffered.

The Internet has introduced additional questions surrounding the issue of liability. Should a website or social network such as Facebook or X (formerly Twitter) be held liable and accountable for the content a user posts when that content contains false, offensive, or potentially harmful material, or should it be held harmless from any liability with respect to that content? The Spotlight on Organizations Case Study, *Section 230 Under Fire* further examines this issue.

Spotlight on Organizations Case Study

Section 230 Under Fire

In 1996, the Internet and web as we know it today did not yet really exist. There was no Facebook, Twitter, Google, YouTube, or Wikipedia. Only about 20 million U.S. adults had access to the Internet, and they spent fewer than 30 minutes a month online. AOL was the world’s largest Internet service provider, along with Prodigy and CompuServe. But there were a few features that would be recognizable today, including bulletin boards (forums) and blogs (although they were not yet called that). The U.S. Congress at that time was very concerned about online pornography and in that year, passed the Communications Decency Act, a law designed primarily to regulate obscenity and indecency on the Internet. Most of the law was quickly found to be unconstitutional by the Supreme Court, but one provision survived: Section 230. Congress had no inkling at the time that Section 230

would later come to be viewed as making the Internet of today possible.

Section 230 has two major provisions: Section 230(c)(1) and Section 230(c)(2). Section 230(c)(1) provides that online companies that enable the posting of user content should not be treated as a publisher or speaker of that content and, therefore, should not be held legally responsible (liable) for that content. This section of the law likens such services to the telephone system or bookstores, which are not responsible for unlawful content that they may play a part in distributing. Congress believed that such a provision would give needed support for what was, at the time, a nascent industry. Without this provision, early Internet companies might have had to spend too much time and resources moderating content to avoid being subjected to potential liability.

Section 230(c)(2) provides that online companies shall not be held liable if they choose, in good faith, to remove user content they deem offensive. This provision resulted from a lawsuit involving user content on a Prodigy bulletin board that a company alleged was defamatory. A New York state court found Prodigy liable because Prodigy exercised editorial control over bulletin board content, at times removing content it believed objectionable. The court case presented Internet firms with a difficult choice—either moderate content (but be subject to the risk of potential legal liability) or don't moderate any content (but run the risk of objectionable content that might harm their business interests). Section 230(c)(2) aimed to remove that concern.

In the years that followed the passage of Section 230, courts interpreted it as creating broad immunity with respect to any claims based on user content, even if the platform knew the content was unlawful. Doing so enabled business models based on user content, such as social media, to flourish. The law has been widely characterized as laying the foundation for the Internet we know today.

However, Section 230 has become a lightning rod in recent years, particularly in relation to social media platforms such as Facebook and X (formerly Twitter) that host a flood of divisive content, including what many believe to be harmful disinformation. In the current heightened political climate, Section 230 has been attacked from both sides. Many Republicans have called for its repeal or reform because they believe that Big Tech companies are biased against conservative viewpoints and that, enabled by Section 230(c)(2), they act to censor certain voices. These critics want to prevent Big Tech platforms from being able to unilaterally decide what content they will remove, label, or restrict. Many Democrats also want reform, but for a different reason. They contend that Section 230(c)(1), by granting Internet platforms legal immunity with respect to user content, reduces their incentive to proactively remove content that causes social harm, especially when that content can be economically valuable to them. Seemingly no one is happy with Section 230 anymore, except the technology companies it was originally designed to protect. Most of these companies want Section 230 to remain as it has always been, with some companies viewing any kind of reform effort as an existential threat to the Internet.

Some Section 230 critics focus on the amplification caused by algorithms and suggest removing liability protection when social media algorithms promote harmful content. However, it is unclear whether it would be possible to design an algorithm that amplifies only “good” content, nor is it clear as to how “good” versus “bad” content should be determined. Others have suggested that a platform's safe harbor protections be tied to the use of reasonable content moderation policies, although this

raises First Amendment issues. Some lawmakers have proposed eliminating Section 230 protections for “market-dominant” Internet platforms or have said that eligibility for Section 230 safe harbors should be based on political neutrality. Still others warn that any Section 230 reform itself may have unintended and harmful side effects, pointing to a previous amendment aimed at fighting online sex trafficking that reportedly actually made prosecuting such crimes more difficult. The Internet Society warns that Section 230 impacts not just platforms such as Facebook and Google but also underlying infrastructure providers, such as cloud providers, that move content and data from one place to another.

To date, the U.S. Supreme Court has sidestepped making any rulings on the scope of Section 230, leaving the issue thus far to legislators to address. In May 2024, a new bipartisan effort was launched by lawmakers in the U.S. House of Representatives to “sunset” Section 230. Energy and Commerce Committee Chair Cathy Rodgers, a Republican, and Ranking Member Frank Pallone, a Democrat, have proposed giving Congress 18 months to create a new regulatory framework to replace Section 230. In addition to all of the existing issues with Section 230, lawmakers are particularly interested in how Section 230 might apply to content created by generative AI. However, the effort remains very contentious, and given the polarized views about Section 230, it's unclear if enough overall support for the proposal can be developed anytime soon. If and when that time comes, just as the initial passage of Section 230 helped create the Internet of today, its repeal or amendment will shape the Internet of the future.

SOURCES: Melissa Quinn, “Supreme Court Declines to Review Scope of Section 230 Liability Shield for Internet Companies,” *Cbsnews.com*, July 2, 2024; Lauren Feiner, “Lawmakers Debate Ending Section 230 in Order to Save It,” *Theverge.com*, May 22, 2024; Marty Swant, “WTF Is Section 230,” *Digiday.com*, March 14, 2024; Congressional Research Service, “Section 230: A Brief Overview,” *Crsreports.congress.gov*, February 2, 2024; Julie Pattison-Gordon, “Can Section 230 Reform Advocates Learn from Past Mistakes,” *Govtech.com*, March 15, 2022; Marguerite Reardon, “Section 230: How It Shields Facebook and Why Congress Wants Changes,” *Cnet.com*, October 6, 2021; Michael D. Smith and Marshall Van Alstyne, “It's Time to Update Section 230,” *Harvard Business Review*, August 12, 2021; Adi Robertson, “What Will Changing Section 230 Mean for the Internet,” *Theverge.com*, February 26, 2021; Stephen Engleberg, “Twenty-Six Words Created the Internet. What Will It Take to Save It,” *Propublica.com*, February 9, 2021; Farhad Manjoo, “Jurassic Web,” *Slate.com*, February 24, 2009.

Case Study Questions

1. What are the ethical, social, and political issues that Internet platforms face with respect to user content?
2. Do you think Section 230 should be repealed or amended? Why or why not?
3. In what ways does the controversy over Section 230 illustrate the difficulties of creating laws governing the use of new digital technologies?

AI Liability Issues

AI raises a number of issues with respect to the question of liability for harm. Most AI algorithms operate as a “black box,” with a lack of transparency into how they actually work and arrive at decisions. It makes it difficult to hold anyone accountable for incorrect output or decisions. This is a particular concern in areas such as healthcare, where AI tools are increasingly performing a wide range of health-related tasks (Duffourc and Gerke, 2024).

In Europe, the European Parliament formally adopted the EU Artificial Intelligence Act (the EU AI Act) in March 2024. The **EU AI Act** is the world’s first broad-based law governing the use of AI. At the same time, the European Parliament also formally adopted a revamped Product Liability Directive. The revisions to the EU Product Liability Directive make clear that the provider of an AI system is considered a manufacturer under the Directive, and thus can be held primarily liable for harm caused by AI systems. The EU Directive also simplifies the burden of proof on consumers in relation to complex or “black box” AI, i.e., where the consumer might have difficulties proving the product is defective because of the “complexity” of the product.

EU AI Act

World’s first broad-based law governing the use of AI.

Computer Crime and Abuse

New technologies, including computers, create new opportunities for committing crime by creating new, valuable items to steal, new ways to steal them, and new ways to harm others.

Computer crime can be generally defined as the commission of illegal acts through the use of a computer or against a computer system. Simply accessing a computer system without authorization or with intent to do harm, even accidentally, is a federal crime. The most frequent types of incidents comprise a greatest hits list of cybercrime: malware, phishing, network interruption, spyware, and denial-of-service attacks. The true cost of all computer crime is unknown, but it is estimated to be in the billions of dollars. You can find a more detailed discussion of computer crime in Chapter 8.

Computer abuse is the commission of acts involving a computer that may not be illegal but are considered unethical. The popularity of the Internet, email, and mobile devices has turned one form of computer abuse—spamming—into a serious problem for both individuals and businesses. Originally, **spam** was junk email that an organization or individual sent to a mass audience who had not expressed any interest in the product or service being marketed. Today, spam typically tries to entice users to select malicious links or may market fraudulent deals and services, outright scams, pornography, illegal or counterfeit drugs, or other objectionable products. Most spam originates from bot networks, which consist of thousands of captured devices that can initiate and relay spam messages.

Spam costs for businesses are high because of the computing and network resources and the time required to deal with a large volume of unwanted email messages, as well as the potential that they may contain malicious links. In 2023, the percentage of all email that is spam worldwide was estimated to be around 45 percent (Petrosyan, 2024).

Identity and financial-theft cybercriminals are also targeting mobile devices with spam. Mobile phone spam usually comes in the form of SMS text messages, but increasingly, users are receiving spam in their Facebook, Instagram, Messenger, and WhatsApp apps as well.

ISPs and individuals can combat spam by using spam filtering software to block suspicious email before they enter a recipient’s email inbox. However, spam filters may block legitimate messages. Spammers know how to skirt filters by continually changing their email accounts, by incorporating spam messages in images, by embedding spam in email attachments and digital greeting cards, and by using other people’s computers that have been hijacked by botnets (see Chapter 8). Many spam messages are sent from one country but hosted by another country.

Some countries have passed laws to outlaw spamming or restrict its use. For instance, spamming is more tightly regulated in Europe than in the United States. Unsolicited commercial messaging is banned in the European Union. Digital marketing can be targeted only to people who have given prior consent.

In the United States, the CAN-SPAM Act does not outlaw spamming but does ban deceptive email practices by requiring commercial email messages to display accurate subject lines, identify the true senders, and offer recipients an easy way to remove their names

Computer crime

The commission of illegal acts through the use of a computer or against a computer system.

Computer abuse

The commission of acts involving a computer that may not be illegal but are considered unethical.

Spam

Unsolicited commercial email.

from email lists. It also prohibits the use of fake return addresses. Although a few people have been prosecuted under this law, it has had a negligible impact on spamming, in large part because of the Internet's exceptionally poor security and the use of offshore servers and botnets.



4.7 Discuss Quality of Life Issues

As the power of information technologies has increased, the potential negative social and personal costs of information technologies and systems have also begun to mount. Many of these negative social consequences are not violations of individual rights or property rights. Nevertheless, they can be extremely harmful to individuals, societies, and political institutions. Information technologies potentially can destroy valuable elements of culture and society even as they bring benefits. They can also create serious personal health risks. If there is a balance of good and bad consequences of using information systems, whom do we hold responsible for the bad consequences? In this section, we briefly examine some of the negative social and personal consequences of systems, considering individual, social, and political responses.

Big Tech: Concentrating Economic and Political Power

In the United States, Amazon accounts for more than 40 percent of all e-commerce retail sales, with almost 95 percent of all book, music, and video sales occurring on its platform. Google accounts for around 90 percent of online searches. Almost 90 percent of all social network users use Facebook. More than 60 percent of all digital ad revenue goes to Google, Meta, and Amazon, collectively. More than 90 percent of online video viewers use YouTube (owned by Google), and more than 60 percent use Amazon Prime Video. In the office, Microsoft dominates, with around 75 percent of the world's desktop computers using Windows operating system software and Microsoft software products. Apple accounts for about 60 percent of the U.S. market in smartphones, while phones with an Android operating system (developed by Google) account for the remaining 40 percent. Regulators' attention in the United States has also turned to the artificial intelligence industry, where OpenAI, Microsoft, and Nvidia increasingly dominate with their software and AI chips.

Reviewing the above, it becomes clear that in today's world of "Big Tech" firms, oligopolies and monopolies dominate. The wealth created by these firms inevitably translates into political influence: These same firms have amassed an army of lobbyists in Washington, DC and state capitals to try to ensure that legislation, or legislative inquiries that might affect their market and tax concerns reflect their interests.

Concentrations of market power are not new in the United States or Europe. Beginning in 1890 with the Sherman Antitrust Act in the United States, and continuing through the 1960s, monopolies have been considered threats to competition and to smaller start-up businesses, generally restraining free trade. Monopolies typically achieve their size by purchasing smaller competitors, crushing competitors by developing similar products, or engaging in predatory pricing by dropping prices drastically for short periods of time to force smaller firms out of business. Big Tech firms have a well-documented history of these behaviors. But antitrust thinking changed in the 1970s to a different standard of harm: consumer welfare. In this view, bigness per se was not a danger, nor was anticompetitive behavior. Instead, price and consumer welfare became paramount. As long as consumers were not forced to pay higher prices, market power was not important and was not considered a social or economic harm. In this view, because the offerings of Meta, Google, and Amazon are either free or very low cost, there can be no harm.

Critics point out that consumer welfare is harmed in ways other than price, namely, by preventing new, innovative companies from market access or from surviving long enough to prosper as independent firms. Complaints and lawsuits originated by small start-up firms alleging anticompetitive and unfair practices, and concerns about the abuse of personal privacy by Big Tech firms, have led to a torrent of public criticism, governmental investigations and regulatory actions, and lawsuits. In August 2024, in a landmark decision, a federal district

court found Google guilty of violating antitrust law by illegally maintaining a monopoly in Internet search. Google is also facing another antitrust case with respect to its ad technology. Antitrust cases are also pending against Apple, Amazon, and Meta.

In Europe, the European Union's Digital Markets Act (DMA), which aims to prevent "gatekeeper" platforms (defined on the basis of market value and which include Alphabet [owner of Google and YouTube], Meta, Amazon, Apple, Microsoft, and ByteDance [owner of TikTok]), from exploiting their interlocking products and services to lock in their users and squash potential rivals, went into effect in March 2024. The law is expected to have a far-reaching impact on the operation of app stores, online advertising, e-commerce, and messaging services. U.S. critics of Big Tech hope the passage of the DMA will provide a roadmap for the passage of similar legislation in the United States (Peters, 2024; Satariano, 2022).

Digital Divide

Does everyone have an equal opportunity to participate in the digital age? Will the social, economic, and cultural gaps that exist in the United States and other societies be reduced by information systems technology? Or will the cleavages be increased?

These questions have not yet been fully answered because the impact of digital technologies on various groups in society has not been thoroughly studied. What is known is that there is a **"digital divide"**: a significant disparity in access to computers and the Internet, as well as many other information resources, among different racial, ethnic, and income-level groups and in rural locations. For instance, a number of studies have found that low-income groups in the United States are less likely to have computers or online Internet access even though computer ownership and Internet access have soared over the past decade. Although the gap in computer access is narrowing, higher-income families in each racial or ethnic group are still more likely to have home computers and broadband Internet access than are lower-income families in the same group. Moreover, the children of higher-income families are far more likely to use their Internet access to pursue educational goals, whereas lower-income children are much more likely to spend time on entertainment and games. This is called the **"time-wasting" gap**.

Left uncorrected, this digital divide could lead to a society of information haves, who are computer literate and skilled, versus a group of information have-nots, who are computer illiterate and unskilled. Narrowing this digital divide by making digital information services—including broadband Internet service—available to as many people as possible, just as basic telephone service is now, has become an important priority for federal, state, and local governments in the United States.

The use of AI can also lead to increased levels of bias. For example, if an AI system is trained on data that reflects existing societal biases, it might perpetuate or amplify these biases, leading to unfair treatment or discrimination. AI's capacity to analyze and draw patterns from large datasets can lead to the stereotyping of certain groups, leading to potential algorithmic discrimination and bias. For example, predictive policing algorithms employed by U.S. law enforcement have been shown to reinforce prevailing biases. Biased algorithms can also lead to discriminatory outcomes in loan applications and job applications. We discuss this issue further in Chapter 11.

Employment: Job Loss Due to Automation, Business Process Reengineering, and AI

The reengineering of work is typically hailed in the information systems community as a major benefit of new information technology. It is much less frequently noted that the redesign of business processes and automation have also caused millions of people to lose their jobs. On the other hand, some economists believe that new technologies create as many or more jobs than they destroy. For instance, the growth of e-commerce has led to a decline in retail sales jobs but an increase in jobs for warehouse workers, supervisors, and delivery work. These economists also believe that bright, educated workers who are displaced by technology will move to better jobs in fast-growth industries.

Digital divide

Significant disparity in access to computers and the Internet, as well as many other information resources, among different racial, ethnic, and income-level groups and in rural locations.

The increasing use of AI has raised tremendous concerns about the prospect of wide-scale loss of jobs. Experts are divided as to the ultimate impact, but it is clear that AI will have a significant impact on how business is conducted, how work is performed, and the types of jobs that will be available in the future. In the short term, companies appear to be reducing the number of people they are hiring for job roles that might ultimately be partially or totally replaced by AI, such as customer service roles. We examine issues related to AI and jobs further in Chapter 11.

Rapidity of Change: Reduced Response Time to Competition

Information systems have helped to create much more efficient national and international markets. Today's rapidly moving global marketplace has reduced the normal social buffers that permitted businesses many years ago to adjust to competition. Time-based competition has an ugly side: The business you work for may not have enough time to respond to global competitors and may be wiped out, along with your job. We run the risk of developing a just-in-time society with just-in-time jobs and just-in-time workplaces, families, and vacations. One impact of on-demand services firms such as Uber is to create just-in-time jobs with no benefits or insurance for employees.

Maintaining Boundaries: Family, Work, and Leisure

Although the idea of ubiquitous computing, remote work and telecommuting, nomad computing, mobile computing, and the do-anything-anywhere computing environment is alluring to many, it also raises a number of issues. One impact of such an environment is that it weakens the traditional boundaries that separate work from family and leisure.

The advent of information systems, coupled with the growth of knowledge-work occupations, means that more and more people are working when traditionally they would have been interacting with family and friends. The work umbrella now extends far beyond the eight-hour day into commuting time, vacation time, and leisure time. The explosive growth and use of mobile devices have only heightened the sense for many employees that they are never away from work.

Even leisure time spent on the computer threatens these close social relationships. Extensive Internet and mobile device use, even for entertainment or recreational purposes, takes people away from their family and friends. Among middle school and teenage children, it can lead to harmful antisocial behavior, such as cyberbullying.



Although some people enjoy the convenience of working at home, the do-anything-anywhere computing environment can blur the traditional boundaries between work and family time.

Weakening these institutions poses clear-cut risks. Family and friends historically have provided powerful support mechanisms for individuals, and they act as balance points in a society by preserving private life and providing a place for people to collect their thoughts, think in ways contrary to their employer, and dream.

Physical, Mental, and Cognitive Health Risks

Repetitive stress injury (RSI)

Injury that occurs when muscle groups are forced through repetitive actions with high-impact loads or thousands of repetitions with low-impact loads.

Carpal tunnel syndrome (CTS)

Type of RSI in which pressure on the median nerve through the wrist's bony carpal tunnel structure produces pain.

Computer vision syndrome (CVS)

Eyestrain condition related to display screen use; symptoms include headaches, blurred vision, and dry and irritated eyes.

Information technology can pose a number of physical, mental, and cognitive health risks. For instance, a common occupational injury today is **repetitive stress injury (RSI)**. RSI occurs when muscle groups are forced through repetitive actions that are often made with high-impact loads (such as playing tennis) or tens of thousands of repetitions under low-impact loads (such as working at a computer keyboard). The incidence of RSI is estimated to affect as much as one-third of the labor force and accounts for one-third of all disability cases.

The single-largest source of RSI is computer keyboards. The most common kind of computer-related RSI is **carpal tunnel syndrome (CTS)**, in which pressure on the median nerve through the wrist's bony structure, called a carpal tunnel, produces pain. The pressure is caused by the constant repetition of keystrokes. Symptoms of CTS include numbness, shooting pain, inability to grasp objects, and tingling. Millions of workers have been diagnosed with CTS. In the United States and most other developed countries, CTS affects an estimated 1 to 3 percent of the general population in any given year (Sevy and Varacallo, 2022).

However, RSI is avoidable. Designing workstations for a neutral wrist position (for example, using a wrist rest to support the wrist), proper monitor stands, and footrests all contribute to proper posture and reduced RSI. Ergonomically correct keyboards are also an option. These measures should be supported by frequent rest breaks and rotation of employees to different jobs.

RSI is not the only occupational illness computers cause. Back and neck pain, leg stress, and foot pain also result from poor ergonomic designs of workstations. **Computer vision syndrome (CVS)** refers to any eyestrain condition related to display screen use with desktop computers, laptops, e-readers, smartphones, and handheld video game players. CVS affects about 90 percent of people who spend three hours or more per day using a display screen. Its symptoms, which are usually temporary, include headaches, blurred vision, and dry, irritated eyes.

Much has also been written about the challenges that the Internet, mobile apps, and social media pose for mental health, particularly that of children and young adults. The impact of the increase in screen time and time spent with digital devices during the Covid-19 pandemic has further highlighted this issue. Potential negative impacts include



Repetitive stress injury (RSI) is a leading occupational disease today. The single-largest cause of RSI is computer keyboard work.

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increased exposure to negative online social interactions (cyberbullying, feelings of missing out, and social comparisons); reduced physical activity; sleep disruption; isolation from in-person contact; and the possibility of Internet/social media addiction. The topic remains controversial, however. One recent study of 430,000 U.S. and UK adolescents found little evidence for an increased association between adolescents' technology engagement and mental health problems over the past 30 years. However, the U.S. Surgeon General has released an advisory stating that there are ample indicators that social media can pose a risk of harm to the mental health and well-being of children and adolescents, and in June 2024 announced that he will push for a warning label on social media platforms similar to those that appear on tobacco and alcohol products (Barry and Kang, 2024; APS, 2021).

Information technology also may be harming our cognitive functions or at least changing how we think and solve problems. Although the Internet has made it much easier for people to access, create, and use information, some experts believe that it is also preventing people from focusing and thinking clearly on their own. They argue that excessive use of computers and smartphones reduces intelligence. Some scholars believe that exposure to computers encourages looking up answers rather than engaging in real problem-solving. People, in this view, don't learn much while surfing the web or answering email when compared to listening, drawing, arguing, looking, and exploring. In addition, increased reliance on various forms of AI (see Chapter 11) may be impacting our power to make decisions based on our own judgments (Moser, den Hond, and Lindebaum, 2022; Carr, 2015).

As it further develops, the metaverse may also present new physical and mental health challenges. The metaverse is a visual, three-dimensional (3D) virtual reality that some believe will be at the center of the future web. It aims to move the Internet beyond 2D screens and toward an immersive 3D experience.

Information technology has become part of our lives—personally as well as socially, culturally, and politically. It is unlikely that the issues and our choices will become easier as information technology continues to transform our world. The growth of the Internet, the information economy, and AI suggests that all the ethical, social, and political issues we have described will be heightened as we move further into the first fully digital century.

4.8 Understand How the Information in This Chapter Can Help Your Career

The information in this chapter can help your career by giving you an overview of the ethical, social, and political issues that you should be aware of as you interact with information technologies and information systems. Companies are becoming increasingly aware that such issues can have a significant impact on their bottom line, and awareness of these issues is necessary for everyone in a company.

Real-World Scenario

Cermentos Argos Uses Privacy Analysts to Cement Competitive Advantage

Cermentos Argos is a leading multinational producer and distributor of cement, concrete, and aggregates. Cermentos Argos operates in 16 countries and territories, including the United States, and Central and South America. Cermentos Argos decided that it needed to improve its decision making

in order to gain a competitive advantage. The company established a dedicated business analytics center, investing in skilled business analysts, data scientists, as well as privacy analysts. Privacy analysts played a crucial role in ensuring compliance with privacy laws and ethical data practices.

Job Description and Career Path: Privacy Analyst

Job	Job Description	IT Knowledge	Career Path and Average Salaries
Privacy analyst	A privacy analyst specializes in an organization’s data privacy practices. A privacy analyst is pivotal to maintaining consumer trust and upholding an organizations’ reputation. Job responsibilities include: • Acting as the first point of contact for internal and external privacy-related questions • Proactively informing and involving internal stakeholders (such as Customer Service, Legal, Technology, Finance) • Supporting various departments to make sure processes and controls are aligned with privacy business strategies as well as supporting requests from data subjects and authorities • Working as part of the team in creating documentation and encouraging and facilitating privacy best practices • Providing support on privacy-related projects, improvements, and innovations, including assisting in the development and implementation of privacy processes	• Strong grasp of IT and security systems, including familiarity with data encryption, data management, access controls, cybersecurity threats, and incident response strategies • Experience with data mapping and inventory tools such as OneTrust and Trust Arc • Experience with privacy impact assessment tools, such as Privacy Designer • Experience with consent management tools, such as OneTrust Preference Choice; Cookiebot	• Junior/Entry-level: \$60,000–\$80,000 • Mid-level: \$80,00–\$100,000 • Senior: \$110,000–\$125,000 • Executive/Chief Privacy Officer (CPO): \$250,000–\$350,000 Salary can vary based on factors like location, industry, and company size.

Additional Career Resources

In MyLab MIS Multimedia Library, you can find additional Career Resources to help you identify appropriate jobs requiring some knowledge of IT, create effective resumes, prepare for job interviews, and create a digital portfolio showcasing your business knowledge, application software skills, Internet skills, and analytical skills.

Chapter Case Study

Meta's Many Ethical Challenges

Meta is the parent company of some of the most “popular” social networks in the world, including Facebook and Instagram. At the same time, Meta (which changed its name from Facebook in 2021) is also one of the most reviled companies in the world. This seemingly contradictory situation stems in large part from Meta’s business practices and how it has responded to a variety of ethical challenges throughout the years.

Meta’s stated mission is to give people the power to build community and bring the world closer together. It provides various platforms that enable people to connect and share their opinions, photos, and videos with audiences ranging from just friends and family to the public at large, to find communities, and to grow businesses. As it has grown, its platforms have also become a source of information for its users about what is happening in the world.

From just about its earliest days, Meta has been hounded by controversy over its business practices. When it originally launched Facebook in 2004, it promised its users nearly complete control over who could see their personal profiles and information. But that did not last long. Shortly after it launched, it took its first steps toward monetizing the platform to help cover its costs. It quickly turned to online advertising to generate revenue. What the online advertising industry prizes is data—specifically, personal data about people as well as data that enables companies to predict how people will think and act. It soon became clear that Facebook was generating a trove of such data, setting the stage for one of Meta’s most enduring ethical challenges: the privacy of its users versus Meta’s desire to generate revenue based on that data. In that battle, the desire to maximize its revenue has usually won.

The Beacon program was an early indication of this challenge. Introduced in 2007, the Beacon program broadcast Facebook users’ activities on participating websites, such as their purchases, to their friends’ News Feeds, without user permission. Advertisers could then run ads next to those posts. Privacy advocates blistered with criticism, and more than 50,000 users signed a petition objecting to the program. Meta believed that users would accept Beacon once they had time to get used to it. But it was wrong, and it ultimately was forced to withdraw the Beacon program.

A look back over the past 15 years shows that Meta has been repeatedly embroiled in conflicts over user privacy. It has been slapped by the Federal Trade Commission more than once: first in 2012, when it agreed to stop deceiving users about their ability to control their personal data and to stop sharing data with third parties without informing users, and then again in 2019, when it was fined a record-breaking \$5 billion for clearly and intentionally violating the 2012 settlement. The 2019 settlement followed closely on the heels of what has become known as the Cambridge Analytica

scandal. A UK researcher had obtained permission to use Facebook for a personality quiz app. Unbeknownst to users, the app collected data not only from those who downloaded it but also from their friends and their friends’ friends. Ultimately, it created a database on more than 87 million Facebook users, which was then sold to political consulting and data analytics firm Cambridge Analytica, which used the data in an effort to influence the 2016 US presidential election. Not long thereafter, a *New York Times* exposé revealed that Meta had data-sharing partnerships with at least 60 device makers as well as selected app developers, which enabled those third parties to gain access not only to data about Facebook users but also to personal data about their Facebook friends, without their consent.

Meta has also been repeatedly sued for privacy-related violations and has paid out millions to settle those lawsuits, including \$650 million in 2021 to settle claims it violated an Illinois privacy law by storing facial recognition data without first obtaining users’ consent, and \$725 million in 2023 to settle a federal class action lawsuit that originated from the Cambridge Analytica scandal and then expanded to target Meta’s broader data-sharing practices, including granting numerous third parties access to users’ Facebook content and information without consent.

Meta has also continually run afoul of the European Union’s General Data Protection Regulation, which mandates more stringent privacy protections than the United States does, and in 2023 was fined €1.2 billion, a new record for financial penalties imposed under the GDPR, related to Meta’s transfer of EU users’ data to servers located in the United States. Meta had been previously fined €405 million for violating the GDPR by failing to protect children’s data on Instagram, €390 million for compelling users to consent to personalized advertisements, and €265 million for failing to prevent hackers from stealing the personal information of more than 530 million Facebook users.

Given Meta’s checkered history of privacy violations, why do so many people continue to use and share sensitive details of their lives on its social network platforms? Often, it’s because users do not realize all the data that are being collected about them and the purposes for which that data may be used. Many of Facebook’s features and services are enabled by default, and a study by Siegel+Gale found that Facebook’s privacy policy is more difficult to comprehend than government notices or typical bank credit card agreements, which are notoriously dense.

But issues surrounding user privacy are not Meta’s only ethical challenge: Within the last several years, there has been an increasing focus on Meta’s role in providing a platform for misinformation and other harmful content. For instance, in 2021, the *Wall Street Journal* published an exposé based on a review of Meta documents including research reports,

online employee discussions, and internal presentations. According to the *Wall Street Journal*, Meta knew that its platforms caused harm but did little, if anything, about it. For example, Meta researchers knew that Instagram could be harmful to mental health, particularly for teenage girls, but in public statements, Meta consistently downplayed its negative effect on teens. Meta employees warned that its platforms were being used to incite ethnic violence and to facilitate criminal activity such as human trafficking in certain countries, but Meta did little to respond to those concerns on a systemic basis. Meta also knew that a change to the Facebook News Feed algorithm, which it hoped would increase user engagement, rewarded posts oriented toward outrage and sensationalism, making the Facebook environment more and more toxic. A former employee known as the Facebook whistleblower painted a picture of a company unable to control the “monster” it had created. For instance, Meta’s internal documents revealed its efforts to use artificial intelligence to keep its platforms free from offensive or dangerous content have fallen far short of its expectations.

Meta’s business practices with respect to its competitors have also come under fire. The FTC has sued Meta for attempting to maintain its dominant position by engaging in a series of anti-competitive actions that violated U.S. antitrust law. For example, the FTC pointed to acquisitions of rivals Instagram and WhatsApp in 2014 as predatory moves meant to crush the competition. The FTC also argued that Meta harmed rival app developers by blocking them from accessing data and tools on its platform. As of September 2024, the lawsuit remains ongoing. Meta is facing similar charges in the European Union, as well as a new investigation seeking to determine whether Meta has violated the EU’s new Digital Markets Act.

Meta’s latest ethical challenge involves AI. Meta has been working on its own version of a generative AI large language model, which it calls LLaMA (Large Language Model Meta AI) and in April 2024, began rolling out AI assistant tools across all of its platforms, including Facebook, Instagram, and WhatsApp. The tools appear in feeds, chats, and search, without any option to disable its use. Meta’s AI-everywhere push quickly raised user hackles. For instance, as with other AI chatbots, Meta’s AI can make things up or spread misinformation. For instance, in one private Facebook group focused on parenting, the chatbot posted a comment in which it claimed to have a child enrolled in a prestigious gifted and talented program. Another issue involves the use of users’ public information to train its AI services (Meta claims that it does not use private posts and messages). It is not clear when Meta began scraping data from users based in the United States, nor is there an easy method of opting out. Whether Meta has also violated copyright laws in training its

AI system is another issue. It reportedly considered buying the publishing house Simon & Schuster in order to gain access to longer works and trying to negotiate licenses with publishers, artists, musicians, and the news industry, but rejected both those options as taking too long. Instead, according to published reports, it relied on summarized books, essays, and other works found on the Internet, without obtaining permission, despite warnings from a lawyer about ethical concerns.

Meta, not surprisingly, generally denies most of the claims that its detractors allege, although at times it has apologized for some of its missteps. But Meta’s future may rely on how it responds to its ethical challenges going forward. Over the last several years, there has been an increasing focus on ethical business practices and a growing awareness that good ethics is good business. The ESG (environmental, social, and governance) movement, which focuses on a company’s performance beyond just its profit and loss, is gaining increased traction and becoming a strategic business imperative. Meta says it is taking a number of steps to respond to these challenges and has published a long list of environmental, social, and governance resources explaining how it is doing so. However, given that Meta’s business model relies almost entirely on the largely unfettered use of users’ personal information, and given its past history and business practices, it is unclear whether Meta can be trusted.

SOURCES: Jesus Jimenez, “Can I Opt Out of Meta’s A.I. Scraping on Instagram and Facebook? Sort Of,” *New York Times*, June 7, 2024; “Meta Investor Relations: Environmental Social Governance Resources,” Investor.fb.com, accessed June 7, 2024; Brian X. Chen, “Meta’s A.I. Assistant Is Fun to Use, but It Can’t Be Trusted,” *New York Times*, April 24, 2024; “Meta’s New AI Assistant Wanted to Answer Users’ Questions and Now It’s Getting on Their Nerves,” Qz.com, April 23, 2024; Michael Cader, “Meta Stole Everyone’s Books Because It Was Cheap and Convenient; Considered Buying S&S, But That Was Unnecessary,” *Lunch.publishersmarketplace.com*, April 8, 2024; Foo Yun Chee and Bart H. Meijer, “Apple, Google, Meta Targeted in EU’s First Digital Markets Act Probes,” *Reuters.com*, March 25, 2024; “Meta Platforms Inc. Form 10-K for the fiscal year ended December 31, 2023,” *Sec.gov*, February 2, 2024; Dev Kundaliya, “Meta Fined More than €1 billion for GDPR Breach,” *Computing.co.uk*, May 22, 2023; “The Facebook Files,” *Wall Street Journal*, January 12, 2022; “FTC Alleges Facebook Resorted to Illegal Buy-or-Bury Scheme to Crush Competition after String of Failed Attempts to Innovate,” *Ftc.gov*, August 19, 2021; “A Timeline of Trouble: Facebook’s Privacy Record and Regulatory Fines,” *Guild.com*, August 4, 2021.

Case Study Questions

1. Choose an ethical dilemma discussed in this case and perform an ethical analysis of it.
2. Will Meta be able to have a successful business model without invading users’ privacy? Explain your answer.
3. What ethical issues are involved in Meta’s development and use of Meta AI?
4. Investigate the ESG movement. How might it impact Meta’s future?

Review Summary

4.1 Describe IT/IS-related ethical, social, and political issues.

Information technology and systems are introducing changes for which laws and rules of acceptable conduct have not yet been developed. Increasing computing power, storage, data analysis, and networking capabilities—including the Internet—expand the reach of individual and organizational actions and magnify their impacts. The increasing use of mobile devices and artificial intelligence also raises ethical issues. The ease and anonymity with which information is now communicated, copied, and manipulated in online environments pose new challenges to the protection of privacy and intellectual property. The main ethical, social, and political issues that information systems raise center on information rights and obligations, property rights and obligations, system quality, accountability and control, and quality of life.

4.2 Describe principles for ethical conduct.

Basic concepts underlying ethical decisions include responsibility, accountability, liability, and due process. When confronted with an ethical issue, a five-step ethical analysis process that involves identifying and describing the facts, defining the conflict or dilemma and identifying the higher-order values involved, identifying the stakeholders, identifying the options that can be reasonably taken, and identifying the potential consequences of the options should be used. Ethical principles that should be followed in conjunction with an ethical analysis include the Golden Rule, Immanuel Kant's categorical imperative, the slippery slope rule, the utilitarian principle, the risk aversion principle, and the ethical no-free-lunch rule. These principles should be used in conjunction with an ethical analysis.

4.3 Discuss information rights and privacy issues.

Today's information technologies and systems enable companies to gather personal data about individuals from many sources easily and analyze these data to create detailed digital profiles about individuals and their behaviors. Data flowing over the Internet can be monitored at many points. Cookies and other monitoring tools closely track the activities of website visitors and mobile app users. Not all websites and mobile apps have strong privacy protection policies, and they do not always allow for informed consent regarding the use of personal information. AI is the latest information technology to raise privacy concerns.

4.4 Discuss intellectual property issues.

Intellectual property is subject to a variety of protections under four legal traditions: copyright, patents, trademarks, and trade secrets. However, today's information technologies and systems have severely challenged existing laws and social practices that protect intellectual property.

Traditional copyright laws are insufficient to protect against piracy because digital material can be copied so easily and transmitted to many locations simultaneously over the Internet. AI is the latest information technology to create concerns about the protection of intellectual property.

4.5 Discuss system quality issues.

New information technologies are raising questions about what constitutes acceptable levels of system quality, given the centrality of information systems to the functioning of businesses and society.

4.6 Discuss accountability and control issues.

The use of information technology may sometimes result in harm, creating questions as to who, if anyone, should be held accountable for that harm. What laws exist to control the misuse of information technology? Who, if anyone, is liable if a business or person is harmed by software that is defective? Should a website or social network be held accountable for the content that a user posts? Widespread use of computers increases opportunities for computer crime and computer abuse: How should such occurrences be handled by the legal system?

4.7 Discuss quality of life issues.

Although today's information technologies and systems have been sources of efficiency and wealth, they also have some negative social consequences that impact the quality of life by, for example, enabling the concentration of economic and political power. Not everyone has an equal opportunity to participate in the digital economy, and this digital divide may be exacerbating socioeconomic disparities among different ethnic groups and social classes. The use of AI may also result in the development of bias. Jobs can be lost when computers and AI replace workers or when tasks become unnecessary in reengineered business processes. Information systems and technology can also create issues with maintaining appropriate boundaries separating family, leisure and work, as well as create physical, mental, and cognitive health risks.

Key Terms

Accountability, 116
Carpal tunnel syndrome (CTS), 136
Computer abuse, 132
Computer crime, 132
Computer vision syndrome (CVS), 136

Cookies, 122
Copyright, 125
Digital divide, 134
Digital Millennium Copyright Act (DMCA), 127

Due process, 116
Ethical no-free-lunch rule, 117
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Fair Information Practices (FIP), 119

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Liability, 116	Repetitive stress injury (RSI), 136	Utilitarian principle, 117
	Responsibility, 116	Web beacon, 123

Review Questions

4.1 Describe IT/IS-related ethical, social, and political issues

- Define ethics.
- Explain how ethical, social, and political issues are connected, and give some examples.
- Describe the key technological trends that heighten ethical concerns.

4.2 Describe principles for ethical conduct.

- Define responsibility, accountability, liability, and due process.
- Describe the five steps in an ethical analysis.
- Describe six ethical principles.

4.3 Discuss information rights and privacy issues.

- Define information rights, privacy, and Fair Information Practices.
- Discuss how the regulation of privacy in Europe differs from that in the United States.
- Discuss the challenges to the protection of individual privacy posed by the Internet.
- Discuss the challenges to the protection of individual privacy posed by AI.
- Explain how informed consent, legislation, industry self-regulation, and technology tools help protect the individual privacy of Internet users.

4.4 Discuss intellectual property issues.

- Define intellectual property.

- Describe the four types of laws that protect intellectual property rights.
- Discuss the challenges to intellectual property rights posed by the Internet.
- Discuss the challenges to intellectual property rights posed by AI.

4.5 Discuss system quality issues.

- Discuss the issues involved in determining an acceptable level of system quality.
- List and describe the principal causes of system quality problems.

4.6 Discuss accountability and control issues.

- Explain why it is so difficult to hold software vendors and service providers liable for failure or injury.
- Discuss the issues that AI raises with respect to the question of liability for harm.
- Describe the difference between computer crime and computer abuse.

4.7 Discuss quality of life issues.

- Name and describe four quality-of-life impacts of information technologies and systems.
- Explain how AI may potentially contribute to the development of bias.
- Describe RSI, CTS, and CVS, and explain their relationship to information technology.

Discussion Questions

4.8 Should cloud service providers be held liable for economic injuries suffered when their systems fail?

4.9 Should companies be responsible for unemployment that their information systems cause? Why or why not?

4.10 Discuss the pros and cons of allowing companies to amass personal data for behavioral targeting.

Using AI Project

4.11 AI Project: Business Use of Social Media

You have been hired as a consultant by two friends who want to open a retail pet shop selling organic pet food and treats, as well as pet toys and clothing made from organic materials. They have asked you to advise them about the ethical, social, and legal issues they need to consider if they establish an online presence on Facebook and Instagram.

Using Microsoft's Copilot, Google's Gemini, or ChatGPT, develop a prompt to help you advise them about these issues. Save a copy of your prompt and the output of the chat session. Submit to your instructor as requested.

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing the privacy implications of using online data brokers; using blog creation tools to create a simple blog; developing a corporate policy for employee web usage; and analyzing web browser privacy.

4.12 Management Decision Problem

InfoFree's website is linked to massive databases that consolidate personal data on millions of people. Users can purchase marketing lists of consumers broken down by location, age, gender, income level, home value, home ownership, and length of residence. Do data brokers such as InfoFree raise privacy issues? Why or why not? If your name and other personal information were in this database, what limitations on access would you want in order to preserve your privacy? Consider the following data users: government agencies, your employer, private business firms, and other individuals.

4.13 Creating a Simple Blog

Software skills: Blog creation

Business skills: Blog and web page design

In this project, you'll learn how to build a simple blog of your own design using the online blog creation software available at Blogger.com or Squarespace. Pick a sport, hobby, or topic of interest as the theme of your blog. Name the blog, give it a title, and choose a template for the blog. Post at least four entries to the blog, adding a label for each posting. Edit your posts if necessary. Upload an image, such as a photo from your computer or the web, to your blog. Add capabilities for other registered users, such as team members, to comment on your blog. Briefly describe how your blog could be useful to a company selling products or services related to the theme of your blog. List the tools available in Blogger that would

make your blog more useful for business, and describe the business uses of each. Save your blog and show it to your instructor.

4.14 Analyzing Web Browser Privacy

Software Skills: Web browser software

Business Skills: Analyzing web browser privacy protection features

This project will help develop your skills in using the privacy protection features of leading web browser software. Examine the privacy protection features and settings for two leading web browsers, such as Microsoft Edge, Mozilla Firefox, or Google Chrome. Make a table comparing the features of two of these browsers in terms of functions provided and ease of use.

- How do these privacy protection features protect individuals?
- How do these privacy protection features affect what businesses can do on the Internet?
- Which browser does a better job of protecting privacy? Why?

4.15 Collaboration and Teamwork Project: Developing a Corporate Code of Ethics

With two or more of your classmates, develop a corporate ethics code on privacy that addresses both employee privacy and the privacy of customers and users of the corporate website. Be sure to consider email privacy and employer monitoring of worksites as well as corporate use of information about employees concerning their off-the-job behavior (e.g., lifestyle, marital arrangements, and so forth). If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

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